Introduction to Nonparametric Statistics (M1)

24 mars 2026

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Introduction

These lecture notes follow on from the notes for the introductory course in nonparametric statistics by Vincent Rivoirard (which were inspired by a course by Catherine Mathias), and from the notes by Laëtitia Comminges and Gabriel Turinici.

Chapitre 1

Introduction and review

1.1 What is nonparametric statistics ?

Parametric statistics is the classical framework of statistics. The statistical model is defined by a parameter $\theta \in \mathbb{R}^k$ for some integer k.

Exemple 1.1.
- *Gaussian linear model. The law $\mathbb{P}_\theta$ of the observations satisfies $\mathbb{P}_\theta = \mathcal{N}(\mu, \sigma^2 I_n)$. The parameter $\theta = (\mu, \sigma^2) \in \mathbb{R}^n \times \mathbb{R}_+^*$ is enough to determine the law of the observations.*
- *Observing the number of arrivals at a service desk : $Y \sim P(\lambda)$ (Poisson).*

By contrast, in nonparametric statistics, the model is not described by a finite number of parameters (or equivalently, not by a finite-dimensional parameter).

Exemple 1.2. *A car manufacturer studies the purchasing behavior of its customers. The company believes that the amount they are willing to pay is a function of their income and the distance they travel daily, and from n observations collected by a survey, it posits the following statistical model :*

$$Y_i = f(X_i) + \epsilon_i,\ \ i = 1, \dots, n$$

where the ϵ_i are i.i.d. with distribution $\mathcal{N}(0, \sigma^2)$ and $X_i = (X_{i1}, X_{i2}) =$ (income, distance) and $Y_i =$ amount to pay.

One can make different prior assumptions on the function f (depending on experience, prior knowledge about the data, or after a graphical representation of the data) :

- *one may assume that f is an affine function of the explanatory variables ; one then obtains a linear model (Gaussian here since we assumed Gaussian errors) :*

$$f(X_i) = \theta_1 + \theta_2 X_{i1} + \theta_3 X_{i2}.$$

- *One may also make no assumption on the form of f, and only assume minimal regularity. One then obtains a nonparametric model.*

1.2 A few nonparametric statistics problems

1.2.1 Estimating the distribution function

We observe $X_1, \dots, X_n$, n real-valued random variables with law $\mathbb{P}$. We seek to estimate $\mathbb{P}$. Now $\mathbb{P}$ is entirely described by its distribution function

$$F : \begin{array}{ccc} \mathbb{R} & \to & [0,1] \\ x & \to & \mathbb{P}(]-\infty, x]) \end{array}$$

We construct an estimator $\hat{F}_n$ of F using the n observations $X_1, \dots, X_n$.

1.2.2 Density estimation

We still observe $X_1, \dots, X_n$, n real-valued random variables with law $\mathbb{P}$. But we also assume that $\mathbb{P}$ is absolutely continuous with respect to Lebesgue measure and we wish to estimate its density f. In general, differentiating $\hat{F}_n$ is not a good solution.

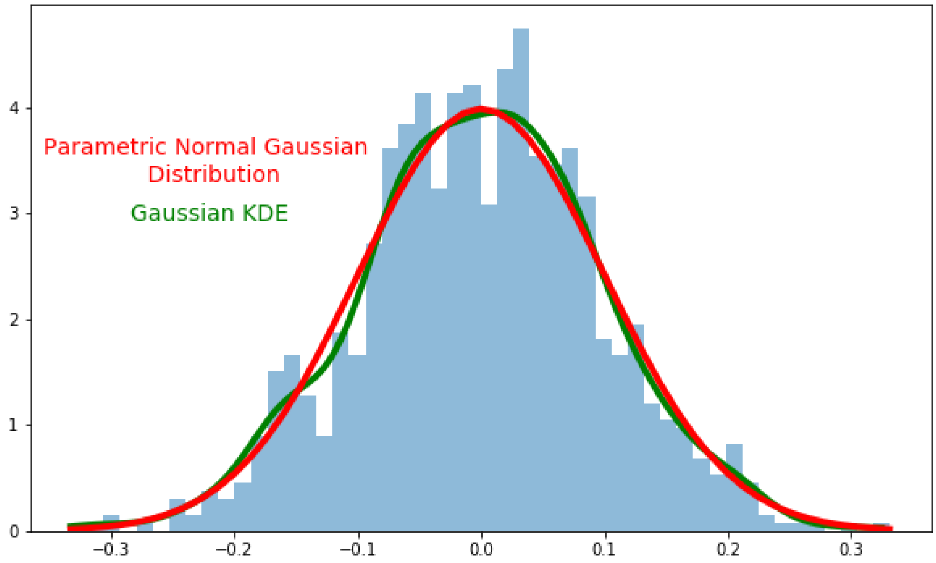


FIGURE 1.1 – Density estimation with Python, function "gausian_kde" from the package "scipy.stats.kde".

1.2.3 Nonparametric regression

We observe a sequence of pairs $\big((X_i, Y_i)\big)_{1 \le i \le n}$ following the model

$$Y_i = f(X_i) + \epsilon_i, \ i = 1, \dots, n.$$

We seek to estimate the regression function f.

One may also consider other nonparametric statistics problems that are not directly estimation problems.

1.2.4 Nonparametric tests

Two examples of possible problems :

— Let X be a random variable and $\mathbb{P}$ be a given distribution. Using $X_1, \dots, X_n$ i.i.d. with the same distribution as X, test :

$$H_0 : X \sim \mathbb{P}, \quad \text{versus} \quad X \nsim \mathbb{P}.$$

— Let X and Y be two random variables, and $(X_1, \dots, X_n)$ and $(Y_1, \dots, Y_m)$ be samples with the same laws as X and Y, respectively. Using the two samples, one can :

  — test whether they have the same distribution :

$$H_0 : X \sim Y \quad \text{versus} \quad H_1 : X \nsim Y;$$

  — test independence between X and Y :

$$H_0 : X \perp\!\!\!\perp Y \quad \text{versus} \quad H_1 : X \text{ and } Y \text{ are not independent.}$$

1.3 Review of classical inequalities

Markov's inequality

Let X be a nonnegative real-valued random variable such that $\mathbb{E}(X) < \infty$. Then for all $t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}(X)}{t}.$$

Bienaymé–Chebyshev inequality (B–C)

Let X be a real-valued random variable such that $\mathbb{E}(X^2) < \infty$. Then for all $t > 0$,

$$\mathbb{P}(|X - \mathbb{E}(X)| > t) \leq \frac{\mathbf{Var}(X)}{t^2}.$$

Hoeffding's inequality

Let $Y_1, \ldots, Y_n$ be independent centered real-valued random variables such that

$$a_i \leq Y_i \leq b_i \ \text{ a.s. for all } i.$$

Then

$$\forall t > 0, \quad \mathbb{P}\Big(\sum_{i=1}^{n} Y_i \geq t\Big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big).$$

Remarque 1.1

Under the same assumptions, we also have

$$\forall t > 0, \quad \mathbb{P}\Big(\big|\sum_{i=1}^{n} Y_i\big| \geq t\Big) \leq 2\exp\Big(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big).$$

Exemple 1.2

Comparison between Hoeffding and B–C : let $X_1, \ldots, X_n \overset{iid}{\sim} Be(p)$ with $p \in (0,1)$. We seek a two-sided confidence interval of level $1-\alpha$ for p using one of the inequalities above :

— Using the B–C inequality,

$$\mathbb{P}(|\bar{X} - p| > c) \leq \frac{p(1-p)}{nc^2} \leq \frac{1}{4c^2 n} := \alpha.$$

Therefore

$$\mathbb{P}\Big(p \in [\bar{X} - \tfrac{1}{2\sqrt{n\alpha}}, \bar{X} + \tfrac{1}{2\sqrt{n\alpha}}]\Big) \geq 1 - \alpha.$$

For $\alpha = 5\%$ and $n = 100$, the accuracy, i.e. the half-length, of this interval is $\frac{1}{\sqrt{n\alpha}} = 0.22$.

— Using Hoeffding's inequality,

$$\mathbb{P}(|\bar{X} - p| > c) \leq 2\exp(-2nc^2).$$

Therefore

$$\mathbb{P}\Big(p \in [\bar{X} - \sqrt{\tfrac{\log(\frac{2}{\alpha})}{2n}}, \bar{X} + \sqrt{\tfrac{\log(\frac{2}{\alpha})}{2n}}]\Big) \geq 1 - \alpha.$$

For $\alpha = 5\%$ and $n = 100$, the accuracy, i.e. the half-length, of this interval is

$$\sqrt{2\frac{\log(\frac{2}{\alpha})}{n}} = 0.14.$$

1.4 Classical convergence theorems

1.4.1 Slutsky's lemma

Théorème 1.3

Let $(X_n)_{n\geq 0}$ and $(Y_n)_{n\geq 0}$ be two sequences of random vectors such that

— $X_n \overset{loi}{\to} X$ where X is an arbitrary random vector ;

— $Y_n \overset{proba}{\to} c$ where c is a constant vector.

Then $(X_n, Y_n) \overset{loi}{\to} (X, c)$.

Consequence : $X_n + Y_n \overset{loi}{\to} X + c$, $X_n Y_n \overset{loi}{\to} cX$, and more generally, for any continuous function f (or continuous on the set where the variables take their values),

$$f(X_n, Y_n) \overset{loi}{\to} f(X, c).$$

1.4.2 Delta method

Théorème 1.4

Let $(U_n)_n$ be a sequence of random vectors in $\mathbb{R}^m$, let $(a_n)_n$ be a deterministic sequence, and let $\ell : \mathbb{R}^m \to \mathbb{R}^p$ be a map such that

— $a_n \to +\infty$;

— there exist a deterministic vector $U \in \mathbb{R}^m$ (= constant) and a random vector V such that

$$a_n(U_n - U) \overset{loi}{\to} V;$$

— ℓ is differentiable at U, with differential $D\ell(U) \in M_{pm}(\mathbb{R})$.

Then

$$a_n(\ell(U_n) - \ell(U)) \overset{loi}{\to} D\ell(U)V.$$

Exemple 1.5

Let $X_1, \ldots, X_n \overset{iid}{\sim} P(\lambda)$ with $\lambda > 0$. Then, by the central limit theorem (hereafter denoted CLT) we have

$$\sqrt{n}(\bar{X} - \lambda) \overset{loi}{\to} N(0, \lambda).$$

Therefore, by the theorem above,

$$\sqrt{n}(\sqrt{\bar{X}} - \sqrt{\lambda}) \overset{loi}{\to} N\Big(0, \frac{1}{4}\Big).$$

Indeed, here $\mathbb{R}^m = \mathbb{R}^p = \mathbb{R}$, $U_n = \bar{X}$, $U = \lambda$, $a_n = \sqrt{n}$, $V \sim N(0, \lambda)$, $\ell(u) = \sqrt{u}$, so $D\ell(U) = \ell'(\lambda) = \frac{1}{2\sqrt{\lambda}}$ and $D\ell(U)V \sim N(0, \frac{\lambda}{4\lambda})$.

1.5 Reminders on conditional expectation

Let X and Y be two random variables taking values in $\mathbb{R}^k$ and $\mathbb{R}^p$. For $x \in \mathbb{R}^k$, we denote by $\mathbb{P}_Y^{X=x}$ the conditional distribution of Y given $X = x$.

1.5.1 Computing conditional expectation

Recall that the conditional expectation of Y given X, denoted here by $\mathbb{E}(Y \mid X)$, is a random variable that can be written as a measurable function of X :

$$\mathbb{E}(Y \mid X) = g(X) \quad \text{where} \quad g(x) = \mathbb{E}(Y \mid X = x).$$

Exemple 1.6

1. Let U and V be two real-valued random variables. Recall the definition of conditional variance :

$$\mathbf{Var}(U \mid V) = \mathbb{E}\big[(U - \mathbb{E}(U \mid V))^2 \mid V\big] = \mathbb{E}[U^2 \mid V] - \big(\mathbb{E}[U \mid V]\big)^2.$$

We have

$$\mathbf{Var}(U \mid V) = g(V) \text{ where } g(v) = \mathbf{Var}(U \mid V = v),$$

i.e. $g(v)$ is the variance of U under the conditional distribution given $V = v$.
Indeed,

$$\mathbf{Var}(U \mid V) = \mathbb{E}[U^2 \mid V] - \big(\mathbb{E}[U \mid V]\big)^2 = \ell(V) - (h(V))^2$$

with $\ell(v) = \mathbb{E}[U^2 \mid V = v]$ and $h(v) = \mathbb{E}[U \mid V = v]$. Hence $\ell(v) - h(v)^2 = \mathbf{Var}(U \mid V = v)$.

2. Let r be a real-valued function defined on $\mathbb{R}$, and let $x \in \mathbb{R}$. Let $\hat{r}(x)$ be an estimator of $r(x)$ computed from an i.i.d. sample $(X_1, Y_1), \ldots, (X_n, Y_n)$ of pairs of real-valued random variables. Let E_Y denote the conditional expectation given $(X_1, \ldots, X_n)$. We have

$$E_Y\Big[(\hat{r}(x) - r(x))^2\Big] = \Big[E_Y(\hat{r}(x)) - r(x)\Big]^2 + E_Y\Big[\Big(\hat{r}(x) - E_Y[\hat{r}(x)]\Big)^2\Big].$$

Indeed,

$$E_Y\Big[(\hat{r}(x) - r(x))^2\Big] = g(X_1, \ldots, X_n)$$

with

$$g(x_1, \ldots, x_n) = E\Big[(\hat{r}(x) - r(x))^2 \mid X_1 = x_1, \ldots, X_n = x_n\Big].$$

For simplicity, we denote by E_n the expectation under the conditional distribution given $X_1 = x_1, \ldots, X_n = x_n$. We then have

$$\begin{aligned} g(x_1, \ldots, x_n) &= E_n\Big[(\hat{r}(x) - r(x))^2\Big] = \text{variance} + \text{squared bias} \\ &= E_n\Big[(\hat{r}(x) - E_n(\hat{r}(x))^2\Big] + \Big[E_n(\hat{r}(x)) - r(x)\Big]^2. \end{aligned}$$

Thus,

$$\begin{aligned} E_Y\Big[(\hat{r}(x) - r(x))^2\Big] &= g(X_1, \ldots, X_n) \\ &= \Big[E_Y(\hat{r}(x)) - r(x)\Big]^2 + E_Y\Big[\Big(\hat{r}(x) - E_Y[\hat{r}(x)]\Big)^2\Big]. \end{aligned}$$

1.5.2 Conditional transfer property

Let $f : \mathbb{R}^{p+k} \to \mathbb{R}^q$ be a Borel function. Then the conditional law of $f(X,Y)$ given $X = x$ satisfies

$$\mathbb{P}^{X=x}_{f(X,Y)} = \mathbb{P}^{X=x}_{f(x,Y)}$$

and therefore

$$\mathbb{E}\big[f(X,Y) \mid X = x\big] = \mathbb{E}\big[f(x,Y) \mid X = x\big].$$

In particular, if X and Y are independent, we have

$$\mathbb{P}^{X=x}_{f(X,Y)} = \mathbb{P}_{f(x,Y)}$$

and hence

$$\mathbb{E}\big[f(X,Y) \mid X = x\big] = \mathbb{E}\big[f(x,Y)\big].$$

Exemple 1.7

Assume that X and Y are independent real-valued random variables. We have

$$\mathbb{P}(Y \le X) = \mathbb{E}(\mathbb{1}_{Y \le X}) = \mathbb{E}\big(\mathbb{E}[\mathbb{1}_{Y \le X} \mid X]\big) = \mathbb{E}[g(X)]$$

where

$$g(x) = \mathbb{E}[\mathbb{1}_{Y \le X} \mid X = x] = \mathbb{E}[\mathbb{1}_{Y \le x}] = F_Y(x),$$

where F_Y denotes the cumulative distribution function (also called distribution function ; hereafter denoted CDF) of Y. Therefore,

$$\mathbb{P}(Y \le X) = \mathbb{E}[F_Y(X)]$$

as soon as X and Y are independent.
Another formulation of this result :

$$\begin{aligned}
\mathbb{P}(Y \le X) = \mathbb{E}[\mathbb{1}_{Y \le X}] &= \int \mathbb{1}_{y \le x}\, d\mathbb{P}_{X,Y}(x,y) \\
&= \int \mathbb{1}_{y \le x}\, d\mathbb{P}_Y \otimes d\mathbb{P}_X(y,x) \\
&= \int \Big[\int \mathbb{1}_{y \le x}\, d\mathbb{P}_Y(y)\Big] d\mathbb{P}_X(x) \\
&= \int F_Y(x)\, d\mathbb{P}_X(x) \\
&= \mathbb{E}[F_Y(X)].
\end{aligned}$$

If needed, you will find worked exercises on computing conditional expectations in the book by Yves Ouvrard, *Probabilités 2, master, agrégation.*

1.6 Review on quantiles and symmetric distributions

1.6.1 Quantiles

We only give here the definition in the simple case where the distribution has a CDF F that is continuous and strictly increasing.

Let X be a real-valued random variable with CDF F continuous and strictly increasing. For $\alpha \in (0,1)$, the α-quantile (quantile of order α) of the distribution F is the unique real number q^F_α such that

$$F(q^F_\alpha) = \mathbb{P}(X \le q^F_\alpha) = \alpha,$$

that is,

$$q_\alpha^F = F^{-1}(\alpha). \tag{1.1}$$

Warning : when the CDF is not continuous, the equation above does not always have a solution. Moreover, if the CDF is not strictly increasing, the equation may have infinitely many solutions. The general definition of a quantile appears in Chapter 2.

1.6.2 Symmetric distribution

Définition 1.8

A real-valued random variable X has a symmetric distribution (about 0) if $X \sim -X$.

— If the CDF F is continuous, this is equivalent to $F(x) = 1 - F(-x)$.

— If the CDF F is continuous and strictly increasing, this is equivalent to $q_{1-\alpha}^F = -q_\alpha^F$ for all $\alpha \in (0,1)$.

— If the distribution has a density f, this is equivalent to $f(-x) = f(x)$ for almost every $x \in \mathbb{R}$.

— A real-valued random variable X has a distribution symmetric about b iff $X - b$ has a distribution symmetric about 0, i.e. iff $X - b \sim -X + b$, equivalently

$$X \sim 2b - X.$$

— If X has a symmetric distribution then $\mathbb{P}(|X| > c) = \mathbb{P}(X > c) + \mathbb{P}(-X > c) = 2\mathbb{P}(X > c)$.

— If the distribution of X is symmetric and if $\mathbb{P}(X = 0) = 0$, then the random variable $|X|$ is independent of the random variable $\mathbb{1}_{X>0}$. Indeed, let A be measurable. The symmetry of the law of X implies

$$\mathbb{P}(|X| \in A, X > 0) = \mathbb{P}(|-X| \in A, -X > 0) \tag{1.2}$$

and

$$\mathbb{P}(X > 0) = \mathbb{P}(X < 0). \tag{1.3}$$

Equation (1.2) can be rewritten as

$$\mathbb{P}(|X| \in A, X > 0) = \mathbb{P}(|X| \in A, -X > 0),$$

which implies

$$\mathbb{P}(|X| \in A) = \mathbb{P}(|X| \in A, X > 0) + \mathbb{P}(|X| \in A, X < 0) = 2\mathbb{P}(|X| \in A, X > 0). \tag{1.4}$$

Equation (1.3) combined with the property $\mathbb{P}(X = 0) = 0$ implies

$$\mathbb{P}(X > 0) = \frac{1}{2}. \tag{1.5}$$

Equation (1.4) combined with (1.5) implies

$$\mathbb{P}(|X| \in A, X > 0) = \frac{1}{2}\mathbb{P}(|X| \in A) = \mathbb{P}(X > 0)\mathbb{P}(|X| \in A).$$

Exemple 1.9

The standard normal distribution and the Student t distribution are symmetric (about 0). The distribution $Be(1/2)$ is symmetric about $1/2$. The distribution $B(n, 1/2)$ is symmetric about $n/2$.

1.7 Review on (frequentist) parametric tests

Test and test error

Setting

We consider a statistical experiment generated by an observation X taking values in $(\mathcal{X}, \mathcal{A})$ and associated with the family of probability measures

$$\{\mathbb{P}_\theta,\ \theta \in \Theta\}.$$

The parameter set Θ is a subset of $\mathbb{R}^d$, with $d \geq 1$. Note that there is no probabilistic structure on Θ !

Principle of a statistical test

We want to "decide", based on the observation of X, whether or not a property of the distribution of X holds. This property is mathematically translated by a subset $\Theta_0 \subset \Theta$ of the parameter space, and the property means that $\theta \in \Theta_0$.

Définition 1.10

We test the "null hypothesis" denoted H_0

$$H_0 \ : \ \ \theta \in \Theta_0$$

against the "alternative" denoted H_1 or H_a

$$H_1 \ : \ \theta \in \Theta_1,$$

with $\Theta_0 \subset \Theta$, $\Theta_1 \subset \Theta$ and $\Theta_0 \cap \Theta_1 = \emptyset$. Constructing a test means constructing a procedure $\phi = \phi(X)$ of the form

$$\phi(X) = 1_{\{X \in \mathcal{R}\}} = \begin{cases} 0 & \text{if } X \notin \mathcal{R}. \quad \text{"we do not reject the null hypothesis"} \\ \\ 1 & \text{if } X \in \mathcal{R}. \quad \text{"we reject the null hypothesis"} \end{cases} \tag{1.6}$$

where $\mathcal{R}$ is measurable.

Mise en garde 1.11

Except for the trivial tests that always reject or never reject H_0, $\phi(X)$ takes two values, 0 and 1. Hence we should expect to be wrong sometimes, for example by rejecting H_0 incorrectly ; therefore one should be careful with the wording and, if possible, prefer saying "the data do not support H_0" rather than "H_0 is false" or even "we reject H_0" ; indeed, it is guaranteed that we will sometimes be wrong when rejecting H_0 (this is built into the definition of a nontrivial test).

Définition 1.12

We refer interchangeably to the set $\mathcal{R} \subset \mathcal{A}$ or to the event $\{X \in \mathcal{R}\}$ as the **rejection region**, or the **critical region**, of the test ϕ.

Définition 1.13

The hypothesis H_j ($j = 0$ or $j = 1$) is said to be **simple** if Θ_j is a singleton ; otherwise H_j is said to be **composite**.

For example, a test of the form $H_0 : \theta = 1$ versus $H_1 : \theta > 1$ has a simple null hypothesis and a composite alternative.

Test error

When performing a test, there are four possibilities. Two are unremarkable and correspond to a correct decision :

— Not rejecting the hypothesis H_0 while $\theta \in \Theta_0$ (i.e. H_0 is true).

— Rejecting the hypothesis H_0 while $\theta \in \Theta_1$ (i.e. H_0 is false).

The other two possibilities are the ones we care about, and correspond to an error of decision :

— Rejecting the hypothesis H_0 while $\theta \in \Theta_0$ (i.e. H_0 is true).

— Not rejecting the hypothesis H_0 while $\theta \in \Theta_1$ (i.e. H_0 is false).

Définition 1.14: Type I and Type II errors

The Type I error (first kind error) corresponds to the maximal probability of rejecting the hypothesis when it is true :

$$\sup_{\theta \in \Theta_0} \mathbb{E}_\theta[\phi(X)] = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta[X \in \mathcal{R}].$$

The Type II error (second kind error) corresponds to the maximal probability of not rejecting the null hypothesis H_0 when it is false :

$$\sup_{\theta \in \Theta_1} \mathbb{E}_\theta[1 - \phi(X)] = \sup_{\theta \in \Theta_1} \mathbb{P}_\theta[X \notin \mathcal{R}]. \tag{1.7}$$

Remarque 1.15

On Θ_0 and Θ_1, there is no preference (between parameters) expressed in the form of a probability distribution. All elements are equally important, which explains the “sup” (= “worst case”) in the definition.

Remarque 1.16

According to this terminology, the Type I error measures the (maximal) probability of rejecting incorrectly, and the Type II error the probability of accepting incorrectly. In everyday language, we have two notions : a “false positive” and a “false negative” . To match them with Type I or Type II errors, one must specify what H_0 is ; beware this can sometimes be counter-intuitive. For instance, in medicine, a test searches for the presence of a virus/bacterium/physiological or genetic anomaly/etc. In that case H_0 corresponds to the absence of the anomaly (negative test) and H_1 to its presence (positive test) ; thus making a Type I error amounts to a “false positive”, and making a Type II error amounts to a “false negative”.

Mise en garde 1.17

In most situations, Θ_0 is “smaller” than Θ_1, and controlling the Type II error (1.7) is difficult, especially if Θ_1 contains points “very close” to Θ_0. One may imagine that for points in Θ_1 converging to a point in Θ_0, the Type II error is 100% minus the Type I error. It then provides little new information about the test in question because it is too aggregated (because of the “sup”).
In the typical case where Θ_0 is a singleton and Θ_1 is its complement, the Type II error does not provide useful information for discriminating between statistical tests having the same Type I error.
For more precise information, one introduces the power function of a test, which measures its local performance (= at each point) over the alternative.

Définition 1.18

The power function of the test ϕ is the map

$$\beta : \Theta_1 \to [0, 1]$$

defined by

$$\theta \in \Theta_1 \rightsquigarrow \beta(\theta) = \mathbb{P}_\theta[X \in \mathcal{R}].$$

1.7.1 Comparing tests, Neyman's principle

Ideally, we would like both the Type I error and the Type II error to be simultaneously small. The two trivial tests

$$\phi_1 = 1_{\emptyset}, \quad \text{and} \quad \phi_2 = 1_{\mathcal{A}}$$

which respectively consist in never rejecting the hypothesis and in always rejecting it, without using the observation X, have respectively a null Type I error and a null Type II error. Unfortunately, the power of ϕ_1 is catastrophic : $\beta(\theta) = 0$ at every point θ of any alternative Θ_1. Similarly, the Type I error of ϕ_2 equals 1, even if the hypothesis is reduced to a single point, whatever the hypothesis.

A methodology, historically proposed by Neyman, consists in imposing an asymmetry in the testing problem : one decides that controlling the Type I error is crucial. The test construction procedure is then, among the tests whose Type I error is controlled, to choose the (one or more) most powerful test(s), i.e. those having the smallest possible Type II error.

Définition 1.19

Let $\alpha \in [0, 1]$ be a risk level. A test ϕ is said to be of **level** α if its Type I error is **less than or equal** to α.

Définition 1.20

A test is said to have **size** α if its Type I error is **equal** to α.

Remarque 1.21

A test aims to assess the adequacy of the hypothesis H_0 with the observations. To do so, it determines the typical values of X under H_0. If the realization x of X is not among the typical values, it rejects H_0. Otherwise, for lack of a better option, it retains H_0.
The level α can be seen as the maximal risk one is willing to take when rejecting H_0 incorrectly.
One takes as H_0 :

- a commonly accepted hypothesis ;
- a conservative hypothesis (cost, safety criterion, etc.) ;
- the only hypothesis under which one can work mathematically.

Exemple 1.22

In practice, two groups with different aims and interests will have reversed pairs (H_0, H_1) (e.g. industrials and consumers).
Let us give a concrete example of this situation : the legal limit for a pollutant contained in a factory's waste is 6 mg/kg. A measurement is performed on 20 samples, for which one observes an empirical mean of 7 mg/kg with an empirical standard deviation of 2.4 mg/kg. We assume the measurement distribution is Gaussian.
Thus we observe $X_1, \ldots, X_{20} \overset{iid}{\sim} N(\mu, \sigma^2)$ with μ and σ^2 unknown. For the factory manager, the most serious error would be to conclude that the pollutant level is too high when it is not. He therefore chooses the hypotheses

$$H_0 : \mu \leq 6 \quad \text{versus} \quad H_1 : \mu > 6.$$

Now consider the environmentalist's point of view. If the limit exceeds 8 mg/kg, there is danger. Unlike the factory manager, the environmentalist considers that the most serious error would be to conclude that the pollutant level is not too high when in reality it is. He therefore performs the following test :

$$H_0 : \mu \geq 8 \quad \text{versus} \quad \mu < 8.$$

1.8 Intuition and computations : L3-level examples

1.8.1 Intuition

Consider the simple testing example :

$$X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2), \quad \sigma \text{ known.}$$

$$H_0 : \mu = 3 \quad \text{versus } \mu \neq 3.$$

In practice, we observe $x_1, \dots, x_n$. Since we want to know whether the mean equals 3 or is greater than 3, we naturally look at the empirical mean $\bar{x}$. Suppose that $\bar{x} = 3.5$. What should we conclude ? This depends on several factors ; more precisely, it depends here on n and on σ.

Indeed, the issue is that, of course, we will never observe exactly 3. Suppose the true mean is 3. Since the X_i are random and we only have a finite number n of them, the sample never gives us exact information on μ, but only approximate and random information.

So if the empirical mean equals 3.5, the question is : is the true mean 3 and did we get 3.5 just because of randomness ? Or is it because the true mean is not 3 ?

To answer these questions, we must use tests, and in particular use all the information at our disposal (or that we can infer from the data), notably the sample size and the variance σ^2. Indeed these two pieces of information help us decide whether it is "normal" to obtain 3.5 when the true mean is 3, or whether it is "abnormal" (or "atypical").

Here, let us look at what happens under H_0, i.e. when $\mu = 3$ (this is always what one does in the frequentist approach : one examines what is supposed to happen under H_0, hence an asymmetry between the two hypotheses). If we are truly under H_0, the question is : what is a "normal" (or "usual" or "typical") value of $\bar{X}$ when $\mu = 3$? It suffices to standardize in order to reduce to a standard normal variable and use the quantiles of $N(0, 1)$. Indeed, if $\mu = 3$ then

$$\sqrt{n}\frac{\bar{X} - 3}{\sigma} \sim N(0, 1).$$

Thus, since the 97.5% quantile of the standard normal distribution is (approximately) 1.96, we have

$$\mathbb{P}_3\Big(\sqrt{n}\frac{|\bar{X} - 3|}{\sigma} \leq 1.96\Big) = 95\%.$$

In other words, we can say that, with very high probability—more precisely here with probability 95%—the random variable

$$T = \sqrt{n}\frac{\bar{X} - 3}{\sigma}$$

lies in the interval $[-1.96, 1.96]$. In other words, a "typical" value of the statistic T, if we are truly under H_0, is a value between -1.96 and 1.96.

Therefore, if we obtain a value outside this interval, we say that it is not a "normal" value for T under H_0 and we reject H_0.

Of course, it is always possible that, while being under H_0—i.e. here while having a true value of μ equal to 3—we obtain an observed value of T outside the interval $[-1.96, 1.96]$, since the normal distribution has support on $\mathbb{R}$. But this happens "rarely", and therefore the chance of making a mistake by incorrectly rejecting H_0 is small : here 5% (we always take α small). This is the Type I error.

Let us now illustrate the dependence on σ and n in our example (still assuming $\bar{x} = 3.5$) for our final decision :

1. First suppose that $\sigma = 1$ and $n = 100$. Then the observed value of T is

   $$t = \sqrt{100}\frac{3.5 - 3}{1} = 5.$$

   Since 5 is outside the interval $[-1.96, 1.96]$, we conclude that it is an "abnormal" value under H_0 and we reject H_0.

2. Suppose that $\sigma = 5$ and $n = 100$. Then the observed value of T is
$$t = \sqrt{100}\frac{3.5-3}{5} = 1.$$
Then we do not reject H_0. The idea is that it is quite possible that the true value of μ is 3 and that we observe a value as large as 3.5 here, because the data have a large variance.
3. Suppose that $\sigma = 1$ and $n = 9$. Then the observed value of T is
$$t = \sqrt{9}\frac{3.5-3}{1} = 1.5.$$
Then again we do not reject H_0. The idea is that a value $\bar{x} = 3.5$ is not "abnormal" under H_0 if we do not have much data (the result is imprecise if we have very little data, so it is not "abnormal" to have $\mu = 3$ while observing a sample mean $\bar{x}$ somewhat "far" from 3).

Another alternative H_1

In the previous example, we chose $H_1 : \mu \neq 3$. What if $H_1 : \mu > 3$?

We then simply modify the rejection region.

In fact, one must always look at H_1 to know when to reject. We start from the statistic T, which follows a standard normal distribution under H_0.

Under H_1, this statistic tends to take large values, because $\bar{X}$ is an estimator of μ and thus $\bar{X} - 3$ is close to $\mu - 3$, which is strictly positive under H_1. Then this quantity $\bar{X} - 3$, which will therefore be probably strictly positive if we are under H_1, is multiplied by $\sqrt{n}$ (and divided by σ) to obtain T. Hence we expect—at least if n is large enough and if μ is far enough from 3—that the statistic T will be "large" under H_1, so we reject H_0 when T is "too large". Thus the rejection region is of the form $T > c$ where c is a constant to be determined, again, from the typical behavior of T under H_0. Here we thus have a one-sided bound on T under H_0. The 95% quantile of $N(0, 1)$ is approximately 1.64. We can then say that
$$\mathbb{P}_3(T \leq 1.64) = 95\%,$$
i.e. with high probability—more precisely here 95%—and if we are truly under H_0, the statistic T should be less than 1.64. Therefore we reject H_0 if this is not the case.

Power function

A frequentist test is always based on a statistic whose behavior under H_0 is known, and the Type I error is always bounded by α. Therefore, by construction, we know that if we are truly under H_0 and we reject H_0 (thus incorrectly), the probability of making a mistake is small. In the construction, one does look at H_1, but only to determine the shape of the rejection region. (In fact, from the start one is supposed to choose a statistic whose behavior differs under H_0 and under H_1, so as to be able to distinguish the two hypotheses.)

Now, after constructing the test, we are interested in the Type II error and in the power function, i.e. we are interested in what happens under H_1. We want the probability of rejecting H_0 when we are under H_1 to be large, i.e. we want the power to be large. The power function allows us to compare different tests of the same level. One desired property is that, if we have enough data, we can correctly conclude we are under H_1 when we indeed are, with very high probability. This is the case when the test is "convergent" (or "consistent") : the power function tends to 1 as n tends to infinity.

Of course, as its name indicates, the power is a function, because it depends on the exact alternative. Indeed, in general, Θ_1 is a composite hypothesis, and one often has infinitely many possible cases (e.g. $\Theta_1 = \mathbb{R} \setminus \{3\}$ or $\Theta_1 =]3, +\infty[$)). It is obviously easier to detect that one is under H_1 when the true μ equals 10 than when it equals 3.5 (all else being equal). Moreover, the power also depends on the sample size and on σ.

1.8.2 Example of power computations

Let us return to the Gaussian-data example above :

$$X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2).$$

Let us compute the power in different cases. We denote by α the level in the three examples below.

1. $\underline{\sigma \text{ known}}$ and testing problem

$$H_0 : \mu = 3 \quad \text{versus} \quad H_1 : \mu \neq 3.$$

$\phi = \mathbb{1}_{|T|>q}$ with $q = q_{1-\frac{\alpha}{2}}^{N(0,1)}$ and $T = \sqrt{n}\frac{\bar{X}-3}{\sigma}$. The power function, for $\mu \neq 3$, is given by

$$\begin{aligned}
\underline{\beta}(\mu) &= \mathbb{P}_\mu(|T| > q) = \mathbb{P}_\mu\Big(\Big|\sqrt{n}\frac{\bar{X}-3}{\sigma}\Big| > q\Big) \\
&= \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-3}{\sigma} > q\Big) + \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-3}{\sigma} < -q\Big) \\
&= \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu+\mu-3}{\sigma} > q\Big) + \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu+\mu-3}{\sigma} < -q\Big) \\
&= \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} > q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) + \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} < -q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) \\
&= 1 - \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} \leq q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) + \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} < -q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) \\
&= 1 - \Phi\Big(q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) + \Phi\Big(-q + \sqrt{n}\frac{3-\mu}{\sigma}\Big).
\end{aligned}$$

A few numerical examples with $\alpha = 5\%$ (rounded to two decimals) :
Python code to compute α, β.

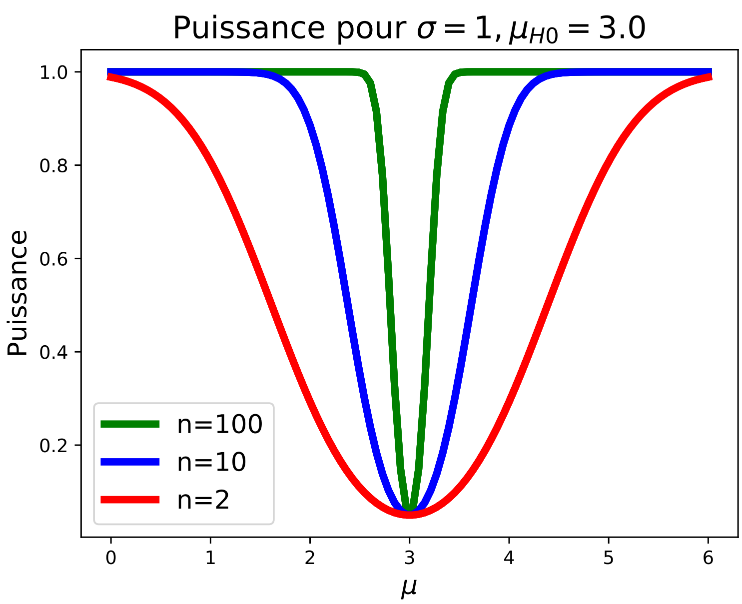


FIGURE 1.2 – Power function for the examples of the two-sided test $\mu = 3$ versus $\mu \neq 3$.

2. $\underline{\sigma \text{ known}}$ and testing problem

$$H_0 : \mu = 3 \quad \text{versus} \quad H_1 : \mu > 3.$$

$\phi = \mathbb{1}_{T>q}$ with $q = q_{1-\alpha}^{N(0,1)}$ and $T = \sqrt{n}\frac{\bar{X}-3}{\sigma}$. The power function, for $\mu > 3$, is given by

$$\underline{\beta}(\mu) = \mathbb{P}_\mu(T > q) = \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-3}{\sigma} > q\Big) = \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} > q + \sqrt{n}\frac{3-\mu}{\sigma}\Big)$$
$$= 1 - \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} \leq q + \sqrt{n}\frac{3-\mu}{\sigma}\Big) = 1 - \Phi\Big(q + \sqrt{n}\frac{3-\mu}{\sigma}\Big).$$

In these first two examples, one immediately sees that the power function tends to 1 as $n \to \infty$. This means that for any μ in the alternative and any $\epsilon > 0$, there exists a sample size n_0 such that the probability of failing to reject H_0 when one is under $\mathbb{P}_\mu$ for this particular μ is smaller than ϵ if $n \geq n_0$. However, in both cases, one can show that the power function does not converge to 1 uniformly, which means that this n_0 depends on μ (consider for instance the sequence $\mu_n = 3 + \frac{1}{n}$). The Type II error, which is defined via a supremum, does not tend to 0. See also Box 1.7.

Remarque 1.23

One must always **be cautious in one's conclusions when the sample size is small and the test result is acceptance (non-rejection) of** H_0.
Let us use the previous example to explain this. It is clear that, under H_1, the smaller $\mu - 3$ is, the smaller the probability of correctly rejecting H_0 is. Let us try to quantify this intuition. Fix a probability, say $1-\epsilon$ with $\epsilon > 0$, and consider the question : for which values of μ do we have $\beta(\mu) \geq 1-\epsilon$? Using the expression above, we find

$$\mu - 3 \geq \frac{\sigma}{\sqrt{n}}\Big(\Phi^{-1}(1-\alpha) - \Phi^{-1}(\epsilon)\Big).$$

We therefore see that, the smaller n is, the farther μ must be from 3 in order to satisfy this inequality.
In other words, when n is small and you are under H_1, it will be harder to detect it (unless the signal, i.e. here the difference $\mu - 3$, is truly strong). Therefore, when your sample size is small and your test does not reject H_0, always qualify the result by stating that there may have been a signal but that the sample size may not have allowed it to be detected.
By contrast, if your test rejects H_0 and your sample size is small, there is no concern : it means instead that the signal was particularly strong.
Moreover, the Type I error, unlike the Type II error or the power function, does not depend on n. Thus if your test is of level α, the Type I error is less than α whatever the value of n.

3. $\underline{\sigma \text{ unknown}}$ and testing problem

$$H_0 : \mu = 3 \qquad \textbf{versus} \qquad H_1 : \mu > 3.$$

If σ is unknown, we can no longer base our test on $T = \sqrt{n}\frac{\bar{X}-3}{\sigma}$ since T is no longer computable. We thus replace σ by an estimator ; here let

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

With this estimator $\hat{\sigma}$, we define

$$T = \sqrt{n}\frac{\bar{X}-3}{\hat{\sigma}}.$$

The distribution of this statistic under H_0 is the Student t distribution with $n-1$ degrees of freedom. Indeed,

$$T = \sqrt{n}\frac{\bar{X}-3}{\hat{\sigma}} = \frac{\sqrt{n}\frac{\bar{X}-3}{\sigma}}{\frac{\hat{\sigma}}{\sigma}} = \frac{\sqrt{n}\frac{\bar{X}-3}{\sigma}}{\sqrt{\frac{\hat{\sigma}^2}{\sigma^2}}}. \tag{1.8}$$

with

$$\sqrt{n}\frac{\bar{X}-3}{\sigma} \sim N(0,1) \quad \text{and} \quad \frac{\hat{\sigma}^2}{\sigma^2} = \frac{\sum_{i=1}^n \frac{(X_i-\bar{X})^2}{\sigma^2}}{n-1} \sim \frac{\chi^2(n-1)}{n-1},$$

and $\bar{X}$ is independent of $\hat{\sigma}^2$. We set $\phi = \mathbb{1}_{T>q}$ with $q = q_{1-\alpha}^{T(n-1)}$. This test is called Student's t-test.

The power function, for $\mu > 3$, is given by

$$\begin{aligned}
\underline{\beta}(\mu) = \mathbb{P}_\mu(T > q) &= \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-3}{\hat{\sigma}} > q\Big) = \mathbb{P}_\mu\Big(\sqrt{n}(\bar{X}-\mu+\mu-3) > q\hat{\sigma}\Big) \\
&= \mathbb{P}_\mu\Big(\sqrt{n}(\bar{X}-\mu) > q\hat{\sigma} + \sqrt{n}(3-\mu)\Big) = \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} > \frac{q\hat{\sigma}+\sqrt{n}(3-\mu)}{\sigma}\Big) \\
&= 1 - \mathbb{P}_\mu\Big(\sqrt{n}\frac{\bar{X}-\mu}{\sigma} \leq \frac{q\hat{\sigma}+\sqrt{n}(3-\mu)}{\sigma}\Big).
\end{aligned}$$

If we set $U = \sqrt{n}\frac{\bar{X}-\mu}{\sigma}$ and $V = \frac{q\hat{\sigma}+\sqrt{n}(3-\mu)}{\sigma}$, then $\underline{\beta}(\mu) = 1 - \mathbb{P}(U \leq V)$ with U and V independent, since $\bar{X}$ and $\hat{\sigma}$ are independent. Hence, by Example 2 in Section 1.5, $\underline{\beta}(\mu) = 1 - \mathbb{E}[F_U(V)]$, where F_U is the CDF of $U = \sqrt{n}\frac{\bar{X}-\mu}{\sigma} \sim N(0,1)$. We finally obtain

$$\underline{\beta}(\mu) = 1 - \mathbb{E}\Big(\Phi\Big(\frac{q\hat{\sigma}+\sqrt{n}(3-\mu)}{\sigma}\Big)\Big).$$

One can then check that the power does indeed converge pointwise to 1 as $n \to \infty$.
To do so one can use the expression above, but one can also deduce this property directly, without relying on it.
Under H_1 :

- $\hat{\sigma}^2 \underset{n\to\infty}{\overset{a.s.}{\to}} \sigma^2$.
- $\bar{X}_n \underset{n\to\infty}{\overset{a.s.}{\to}} \mu > 3$.
- $q_{1-\alpha}^{T(n-1)} \underset{n\to\infty}{\to} q_{1-\alpha}^{N(0,1)}$ since the Student t distribution with $n-1$ degrees of freedom converges to the standard normal distribution (see Theorem 2.22 and Example 2.23).

Thus $T_n \underset{n\to\infty}{\overset{a.s.}{\to}} +\infty$. Consequently,

$$P_\mu(T_n > q_{1-\alpha}^{T(n-1)}) \underset{n\to\infty}{\to} 1.$$

1.8.3 The p-value

Définition 1.24: p-value

Suppose we have constructed a family of tests $\phi_\alpha(X)$, each of level α, for $\alpha \in [0,1]$. The p-value associated with this family is the real-valued random variable defined by

$$p(X) = \inf\{\alpha \in [0,1] : \phi_\alpha(X) = 1\}.$$

Interpretation of the p-value : the smaller the observed p-value, the more inclined we are to reject H_0, because this means that the observed value of the statistic used for the test is atypical under H_0.

Remarque 1.25

We see that $p(X)$ is the level from which we start rejecting H_0. It is as if we carried out the test without knowing α and only drew the conclusion at the end once α is revealed. Hence :

— If $p(x) < \alpha$, then we reject H_0 at level α.

— If $p(x) > \alpha$, then we retain (do not reject) H_0 at level α.

Remarque 1.26

A small p-value does not mean that we are more likely to be under H_1 than under H_0 : this in fact depends on the behavior of the p-value under H_1. We know that, at least under certain conditions (see Chapter 2), the p-value has a uniform distribution under H_0, but we do not necessarily know the behavior of the p-value under H_1.
Moreover, the question of the probability of H_0 given the data is a Bayesian question, which can be answered if one has a prior on the alternative (and sometimes also a prior on H_0). Therefore, be careful with the interpretation of p-values : **do not say** "the p-value is small so the probability that H_0 is false is large".
Furthermore, a large p-value does not necessarily imply that H_0 is true. The test may simply be powerless. For example, consider the test $\phi(X) \equiv 0$: this test never rejects H_0 and therefore has a p-value equal to 1 [a]. A less trivial example : if the sample size is small, the p-value will not be particularly small if the signal is not particularly strong (not strong enough to compensate for the small sample size).

a. The set in the definition of the p-value is empty ; by convention, one takes its supremum to define the p-value, i.e. the p-value equals 1.

Exemple 1.27

It may seem difficult to work with the p-value since it involves all the tests in the family. This can create computational difficulties. However, solutions exist.
An example where computing the p-value is very simple [a] : suppose the test is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)>k_\alpha}$, that $\Theta_0 = \{\theta_0\}$, and that under $\mathbb{P}_{\theta_0}$ the statistic $T(X)$ has a distribution with CDF F_0 that is strictly increasing and continuous. Then $k_\alpha = F_0^{-1}(1-\alpha)$, and the observed p-value is given by

$$\begin{aligned} p(x) &= \inf\{\alpha \in]0,1[: T(x) > F_0^{-1}(1-\alpha)\} = \inf\{\alpha \in]0,1[: F_0(T(x)) > 1-\alpha\} \\ &= \inf\{\alpha \in]0,1[: \alpha > 1 - F_0(T(x))\} = 1 - F_0(T(x)). \end{aligned}$$

$$\mathbb{P}_{\theta_0}(T(X) > k_\alpha) = \alpha \Longleftrightarrow 1 - F_0(k_\alpha) = \alpha \Longleftrightarrow k_\alpha = F_0^{-1}(1-\alpha).$$

a. in this example one can also apply Wasserman's theorem directly

In this course, we will always assume that

— the test is designed so as to maximize the rejection region ;

— $\phi_\alpha(X)$ decreases as α decreases.

The first assumption is natural : it states that any larger rejection region (among those compatible with the structure of the test) yields a larger level α. In the definition of a level-α test, we require the Type I error to be less than α. This leads to infinitely many possible solutions : indeed, if $\mathbb{P}_{\theta_0}(T > c_1) \leq \alpha$, then $\mathbb{P}_{\theta_0}(T > c_2) \leq \alpha$ for all $c_2 > c_1$. If we want to minimize the Type II error, we must therefore maximize the rejection region (and thus take c as small as possible).
The second assumption is also very natural. It can be rewritten as

$$\alpha_1 \leq \alpha_2 \Longrightarrow \mathcal{R}_{\alpha_1} \subset \mathcal{R}_{\alpha_2},$$

that is, if we reject at level α_1, then we also reject at any level $\alpha_2 \geq \alpha_1$.

Théorème 1.28: "Wasserman's theorem"

We assume that, at a given level α, the tests we perform maximize the rejection region.

— Suppose a family of tests is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\leq k_\alpha}$, for $\alpha \in]0,1[$. Then, if the test has **size** α, the p-value is

$$p(x) = \sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T(X) \leq T(x)),$$

where x is the observed value of X.

— For a family of tests of size α of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\geq k_\alpha}$, we have

$$p(x) = \sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T(X) \geq T(x)).$$

— If the variable $T(X)$ has a discrete distribution with a fixed CDF F_0 under H_0 and if the family of tests is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\leq k_\alpha}$, then

$$p(x) = F_0(T(x)) = \mathbb{P}_{H_0}(T(X) \leq T(x)).$$

— If the variable $T(X)$ has a discrete distribution with a fixed CDF F_0 under H_0 and if the family of tests is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\geq k_\alpha}$, under the same assumptions, then

$$p(x) = \mathbb{P}_{H_0}(T(X) \geq T(x)).$$

Admitted. You can find a handwritten proof on Teams.

These formulas remain valid if there exists θ_0 such that for all t,

$$\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T(X) \leq t) = \mathbb{P}_{\theta_0}(T(X) \leq t)$$

when the test is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\leq k_\alpha}$, or

$$\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T(X) \geq t) = \mathbb{P}_{\theta_0}(T(X) \geq t)$$

when the test is of the form $\phi_\alpha(X) = \mathbb{1}_{T(X)\geq k_\alpha}$.

Remarque 1.29

We will often encounter tests based on discrete distributions written in the form $\phi = \mathbb{1}_{T>q}$ where q is a quantile of the distribution of the statistic T under H_0. The test is then not written directly as in the theorem above. But if the distribution of T is integer-valued (for instance; the same holds for any discrete set), then q is an integer (see L3 tutorial on Teams). Hence $\{T > q\} = \{T \geq q+1\}$.

Exemple 1.30

— Let $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with μ and σ^2 unknown. We want to test

$$H_0 : \mu = 2 \qquad \text{versus} \quad H_1 : \mu \leq 2.$$

The test used is Student's t-test (see "examples of power computations", item 3). The test is $\phi = \mathbb{1}_{T\leq q_\alpha^{T(n-1)}}$ with $T = \sqrt{n}\frac{\bar{X}-2}{\hat{\sigma}}$. We are then in the conditions of application of Wasserman's theorem, item 1 : indeed, since the Student distribution is continuous and since the null hypothesis is simple, the test has size α (not only level α). The observed p-value is therefore

$$p(x_1^n) = F_{T(n-1)}(T(x_1^n)),$$

where $T(x_1^n)$ is the observed value of the statistic $T(X_1^n)$.

Numerical application : $n = 20$, $\frac{1}{20}\sum_{i=1}^{20} x_i = 1.34$ and $\sqrt{\frac{1}{19}\sum_{i=1}^{20}(x_i - \bar{x})^2} = 1.06$, which gives the observed value

$$T(x_1^n) = \sqrt{20}(1.34 - 2)/1.06 = -2.78.$$

The observed p-value is $p(x_1^n) \approx 0.006$. One can find this value in R using the command `pt(-2.78,19)`. To obtain this result directly without doing any computation, one can use the R command `t.test(x,mu=2,alternative="less")` where $x = x_1^n$ is the observed sample (i.e. a vector).

— Let $X_1, \dots, X_n \overset{iid}{\sim} Be(p)$. We want to test

$$H_0 : p = 1/2 \quad \text{versus} \quad H_1 : p > 1/2.$$

We use the statistic $T = \sum_{i=1}^{n} X_i$, which under H_0 follows a binomial distribution $B(n, 1/2)$. In view of H_1, we reject when T is too large. Thus we set $\phi = \mathbb{1}_{T>c}$ where c is determined by the fact that the test is of level α :

$$\mathbb{P}_{1/2}(T > c) \leq \alpha. \tag{1.9}$$

Here the statistic T is discrete, so its CDF $F_{B(n,1/2)}$ under $\mathbb{P}_{1/2}$ is not continuous. Therefore we cannot always have equality.
We will see in Chapter 2 that the smallest integer c satisfying (1.9) is $c = q_{1-\alpha}^{B(n,1/2)}$, i.e. the $(1-\alpha)$-quantile of the binomial distribution $B(n, 1/2)$. This yields the following test :

$$\phi = \mathbb{1}_{T>q_{1-\alpha}^{B(n,1/2)}}.$$

Since T is almost surely integer-valued, we have

$$\phi = \mathbb{1}_{T>q_{1-\alpha}^{B(n,1/2)}} = \mathbb{1}_{\{T \geq q_{1-\alpha}^{B(n,1/2)}+1\}}.$$

Thus it indeed has one of the forms indicated in Wasserman's theorem. We are in the conditions of application of this theorem (discrete case), hence

$$p(x_1^n) = \mathbb{P}(Z \geq T(x_1^n)) = 1 - \mathbb{P}(Z < T(x_1^n)) = 1 - F_{B(n,1/2)}(T(x_1^n) - 1),$$

where Z denotes a random variable with distribution $B(n, 1/2)$.
For instance, if $n = 20$ and if the observed value of $\sum_{i=1}^{n} X_i$ is 11, then the p-value of the test is $p(x_1^n) = 0.41$. Hence we tend not to reject H_0 in view of the data.
One can obtain this value in R with the command `pbinom(10,20,1/2)`. One obtains this p-value directly using the command
`binom.test(11,20,1/2,alternative="greater")`
or in Python with

```
from scipy.stats import binom_test
binom_test(11,20,alternative='greater')
```

Remarque 1.31

We can make the same kind of remarks about the p-value as for the power function :

— Under H_1, the p-value depends, among other things, on the sample size.

— Under H_0, the p-value often follows a uniform distribution on $[0, 1]$ (or a distribution close to uniform) (see Chapter 2). Hence its behavior is known and "fixed".

For example, consider the testing problem $H_0 : \mu = 0$ versus $H_1 : \mu > 0$ for i.i.d. data with distribution $\mathcal{N}(\mu, 1)$, and denote by $\bar{x}$ the observed empirical mean. One obtains (easy exercise)

$$p(x_1^n) = 1 - \Phi(\sqrt{n}\bar{x}),$$

where Φ is the CDF of $\mathcal{N}(0, 1)$.
We clearly see that, for a given observed empirical mean, if it is positive, the p-value will be smaller when the sample size is larger.

Method for constructing a test

1. Choose H_0 and H_1.
2. Determine $T(X)$, the test statistic. One must know its distribution under H_0. Of course, one also wants this statistic to behave differently under H_0 and under H_1 in order to discriminate between the two hypotheses.
3. Determine the shape of the rejection region as a function of H_1 (i.e. as a function of the behavior of $T(X)$ under H_1).
4. Observe the realization $T(x)$ of $T(X)$.
5. Compute the associated p-value $p(x)$ and compare it to a threshold set by a non-statistician.
6. Retain or not retain H_0.

1.8.4 Interpretation of p-values

Since the p-value is defined as an infimum, it is not necessarily a "minimum", so we do not know a priori whether $p(x) \in \{\alpha \in]0, 1[: \phi_\alpha(x) = 1\}$, i.e. we do not know whether we reject H_0 at level $\alpha = p(x)$. Let us denote by α^* the p-value $p(x)$, so as to view it as a level. To fix ideas (this does not change the reasoning), assume that $\phi_{\alpha^*}(x) = 1$, i.e. at level α^* we reject H_0. Recall that the level α is chosen as the probability of incorrectly rejecting H_0 (or an upper bound on that probability if equality cannot be achieved for all α).

Thus, if we view the p-value as a level α^*, then we reject at level α^* and, if α^* is very small, the probability of rejecting incorrectly is very small. In other words, the smaller the observed p-value, the more inclined we are to reject H_0.

Suppose we observed a p-value $p = 0.001$, which is therefore very small. Then at level $\alpha = 5\%$ we reject H_0 since $0.001 < 0.05$. Moreover, knowing the p-value provides additional information : the fact that p is very small here gives us some confidence in our rejection. For instance, if we had obtained $p = 0.04$, we would also reject at level $\alpha = 5\%$, but with less confidence.

Statistical software always reports the p-value when asked to perform a test. Consider the example of Student's t-test. We have a vector x consisting of a sample of Gaussian observations with unknown mean and variance, and we want to test $H_0 : \mu = 1.5$ versus $\mu \neq 1.5$, where μ is the mean. One can use the following R command :

```
t.test(x,mu=1.5)
```

whose output is

```
One Sample t-test

data:  x
t = -1.9561, df = 19, p-value =
0.06532
alternative hypothesis: true mean is not equal to 1.5
```

```
95 percent confidence interval:
0.6181763 1.5298237
sample estimates:
mean of x
1.074
```

We obtain a p-value of about 0.06, so we do not reject H_0 at the 5% level. Again, since the p-value is close to the level, we do not have very high confidence in the final decision.

Interpretation using Wasserman's theorem

Let us return to one of the previous examples : the first item in "examples of p-value computations". In that example the p-value is $p(x_1^n) = F_{T(n-1)}(T(x_1^n))$. In the numerical application, the observed value of the statistic T is $t = -2.78$. If we are truly under H_0, then t is supposed to be an observed value of a statistic that follows a Student t distribution with 19 degrees of freedom, and the p-value then measures the probability that a Student t random variable with 19 degrees of freedom is less than -2.78, i.e. the probability of observing a value of T smaller than -2.78 **if we are truly under** H_0. Hence the p-value measures, in a sense, how atypical the observed value is relative to what is supposed to happen under H_0.

Here, if we were truly under H_0, there would be about a 1% probability of observing a value less than or equal to -2.78 for the statistic T. Therefore -2.78 is rather atypical under H_0, and we thus lean toward rejecting H_0.

Chapitre 2

Estimation of the distribution function

2.1 Consistency of empirical distribution functions

We consider $X_1^n = (X_1, \ldots, X_n)$ an i.i.d. n-sample with CDF F :

$$\forall x \in \mathbb{R}, \quad F(x) = \mathbb{P}(X_1 \leq x).$$

Recall that :

— F is nondecreasing,

— F is right-continuous,

— $\lim_{x\to+\infty} F(x) = 1$ and $\lim_{x\to-\infty} F(x) = 0$.

Since it is nondecreasing, F has a left limit at every point and it has at most countably many discontinuities (these are the points x such that $\mathbb{P}(X_j = x) \neq 0$).

There exists a natural estimator of F : the empirical distribution function.

Définition 2.1

The empirical distribution function associated with $X_1^n = (X_1, \ldots, X_n)$ is the random function defined by : $\hat{F}_n : \begin{array}{ccc} \mathbb{R} & \to & [0,1] \\ x & \to & \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{X_i \leq x} \end{array}$

Remarque 2.2

To emphasize the random nature of $\hat{F}_n$, one may sometimes write $\hat{F}_n(\omega, x)$ instead of $\hat{F}_n(x)$. Thus $\hat{F}_n(\omega, x)$ denotes the value of the CDF $\hat{F}_n$ at x when the outcome is ω.

Définition 2.3

Let $X_1, \ldots, X_n$ be n real-valued random variables. The order statistic $X^* = (X_{(1)}, \ldots, X_{(n)})$ is defined by $\{X_{(1)}, \ldots, X_{(n)}\} = \{X_1, \ldots, X_n\}$ and $X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$.

Remarque 2.4

One can easily construct $\hat{F}_n$ since it is a step function. Fix ω and write $(x_1, \ldots, x_n) = (X_1(\omega), \ldots, X_n(\omega))$. Then $\hat{F}_n(\omega, x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{x_i \leq x}$ is the distribution function of the random variable Z taking values in $\{x_1, \ldots, x_n\}$ and such that $\mathbb{P}(Z = x_i) = \frac{k}{n}$ if the value x_i appears k times in $\{x_1, \ldots, x_n\}$. For example, if all the x_i are distinct, then $\hat{F}_n(\omega, \cdot)$ is the CDF of the uniform distribution on $\{x_1, \ldots, x_n\}$.

Let $(X_{(1)}, \dots, X_{(n)})$ be the order statistic associated with X_1^n. The function $\hat{F}_n(\omega, \cdot)$ is discontinuous at the points $X_{(j)}(\omega)$. It has a jump equal to the number of times the value $X_i(\omega)$ appears in $\{X_1(\omega), \dots, X_n(\omega)\}$. In particular, if all the $X_i(\omega)$ are distinct, i.e. $X_{(j)}(\omega) < X_{(j+1)}(\omega)$ for all j, then $\hat{F}_n(\omega, x) = \frac{j}{n}$ for all $x \in [X_{(j)}(\omega), X_{(j+1)}(\omega)[$. In all cases, it equals 0 on $]-\infty, X_{(1)}(\omega)[$ and 1 on $[X_{(n)}(\omega), +\infty[$.

Proposition 2.5

Let $x \in \mathbb{R}$. Then $\hat{F}_n(x)$ is an unbiased estimator of $F(x)$ and $\lim_{n\to\infty} \hat{F}_n(x) = F(x)$ a.s. Moreover,

$$\sqrt{n}(\hat{F}_n(x) - F(x)) \xrightarrow{Loi} \mathcal{N}(0, F(x)(1 - F(x)))$$

Démonstration. $\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{X_i \le x}$ with $\mathbb{1}_{X_i \le x} \overset{iid}{\sim} Be(F(x))$, so $\lim_{n\to\infty} \hat{F}_n(x) = F(x)$ a.s. follows from the LLN. The second property follows from the CLT, noting that assumes $\mathbf{Var}(\mathbb{1}_{X_i \le x}) = F(x)(1 - F(x))$. □

This result is parametric in nature since x is fixed. One can go further.

Théorème 2.6: Glivenko-Cantelli

Let $(X_1, \dots, X_n)$ be an i.i.d. n-sample with distribution function F. Then the empirical distribution function is a strongly consistent estimator of F for the uniform convergence norm :

$$\lim_{n\to\infty} \|\hat{F}_n - F\|_\infty = \lim_{n\to\infty} \sup_{x\in\mathbb{R}} |\hat{F}_n(x) - F(x)| = 0 \quad \text{a.s. } \omega \in \Omega \tag{2.1}$$

We admit this theorem. You can find a proof in the 2015 midterm.

Définition 2.7: Quantile function

To any distribution function F one associates its generalized inverse $F^{(-1)}$ defined as follows :

$$\forall q \in [0,1] \quad F^{(-1)}(q) = \inf\{x \in \mathbb{R} : F(x) \ge q\}$$

$F^{(-1)}$ is also called the quantile function.

Proposition 2.8

We have $F^{(-1)} = F^{-1}$ when F is bijective. Moreover,

1. $F(F^{(-1)}(q)) \ge q$ for all $q \in [0,1]$.
2. $\forall x \in \mathbb{R},\ \forall q \in [0,1], \quad F(x) \ge q \Leftrightarrow x \ge F^{(-1)}(q)$.
3. If $U \sim U[0,1]$ then $F^{(-1)}(U)$ is a random variable with distribution function F.
4. If F is continuous then $F(F^{(-1)}(q)) = q$ for all $q \in]0,1[$.
5. If Z has continuous distribution function F, then $F(Z) \sim U[0,1]$.
6. $F^{(-1)}$ is nondecreasing.

Démonstration.
1. By definition of $F^{(-1)}(q)$, there exists a sequence $(u_n)_{n\ge 0}$ such that $F(u_n) \ge q$ and $u_n \underset{n\to\infty}{\to} F^{(-1)}(q)$ decreasing (so this is a right limit). Since F is right-continuous, $F(u_n) \to F(F^{(-1)}(q))$. Hence $F(F^{(-1)}(q)) \ge q$.
2. • If $F(x) \ge q$, then by definition $F^{(-1)}(q) \le x$.

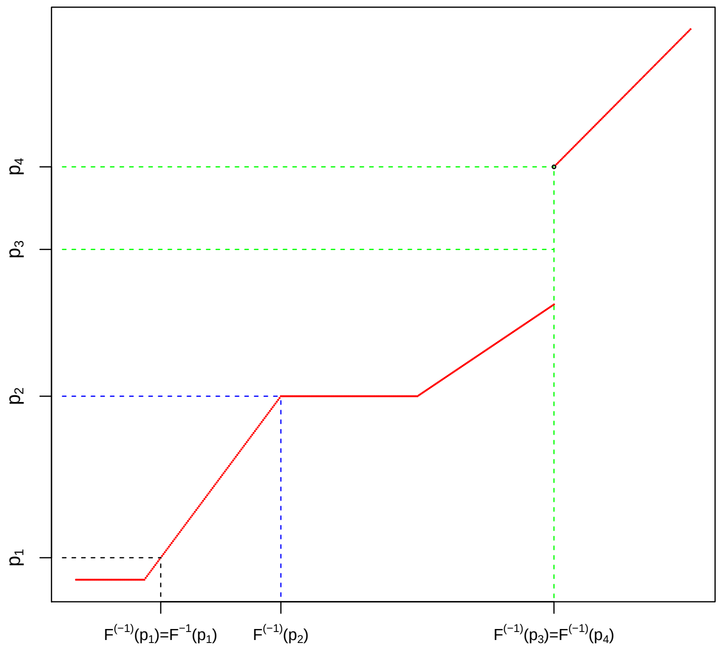


FIGURE 2.1 – Distribution function (in red) with a flat part and a jump

- If $x \geq F^{(-1)}(q)$, then by monotonicity of F we have $F(x) \geq F(F^{(-1)}(q))$, hence $F(x) \geq q$ by item 1.

3. By item 2, if $t \in [0,1]$, $\mathbb{P}(F^{(-1)}(U) \leq t) = \mathbb{P}(U \leq F(t))$, and $\mathbb{P}(U \leq F(t)) = F(t)$ since $F(t) \in [0,1]$.
4. By item 1, it suffices to show that $F(F^{(-1)}(q)) \leq q$. If F is continuous then $]0,1[\subset Im(F)$ by the intermediate value theorem. Hence there exists $x_q \in \mathbb{R}$ such that $F(x_q) = q$. By definition $F^{(-1)}(q) \leq x_q$. Therefore, by monotonicity of F, $F(F^{(-1)}(q)) \leq q$.
5. Let $t \in]0,1[$. We have

$$\begin{aligned}\mathbb{P}(F(Z) < t) &= 1 - \mathbb{P}(F(Z) \geq t) = 1 - \mathbb{P}(Z \geq F^{(-1)}(t)) = \mathbb{P}(Z < F^{(-1)}(t)) \\ &= F(F^{(-1)}(t)) = t\end{aligned}$$

   where we used item 2 for the second line, continuity of F for the fourth line, and item 4 for the last line. Since $]-\infty, x] = \cap_{t>x}]-\infty, t[$ we have $\mathbb{P}(F(Z) \leq x) = \lim_{t\to x, t>x} \mathbb{P}(F(Z) < t) = \lim_{t\to x, t>x} t = x$. Therefore $F(Z) \sim U[0,1]$.
6. Let $q_1, q_2 \in [0,1]$ with $q_1 \leq q_2$. Then $\{x \in \mathbb{R} : F(x) \geq q_2\} \subset \{x \in \mathbb{R} : F(x) \geq q_1\}$, hence $F^{(-1)}(q_1) \leq F^{(-1)}(q_2)$.

□

Remarque 2.9

Items 1 and 4 can be "derived" from a picture. One also "sees" that the flat parts of F correspond to a discontinuity point of $F^{(-1)}$, and that a jump of F corresponds to a flat part of $F^{(-1)}$.

Remarque 2.10

In a number of cases (see examples below), the p-value $p(X)$ of a test follows a uniform distribution under H_0.

Exemple 2.11

An i.i.d. sample $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ unknown. Testing problem $H_0 : \mu = \mu_0$ against $H_1 : \mu \leq \mu_0$. We use the Student test $\phi = \mathbb{1}_{T \leq q_\alpha^{T(n-1)}}$. Then, by Wasserman's theorem, the observed p-value is $p(x_1^n) = F_{T(n-1)}(T(x_1^n))$. Hence the p-value $p(X_1^n)$, as a random variable, satisfies $p(X_1^n) = F_{T(n-1)}(T(X_1^n))$. But $T(X_1^n) \sim T(n-1)$ under H_0. Therefore, by item 5 of the previous proposition, $p(X_1^n)$ follows a uniform distribution under H_0.

Exemple 2.12

Same setting but with $H_1 : \mu \geq \mu_0$. Then the test is $\phi = \mathbb{1}_{T \geq q_{1-\alpha}^{T(n-1)}}$. The observed p-value satisfies $p(x_1^n) = \mathbb{P}_{\mu_0}(T(X_1^n) \geq T(x_1^n)) = \mathbb{P}(Z \geq T(x_1^n))$ with $Z \sim T(n-1)$. Thus $p(x_1^n) = 1 - F_{T(n-1)}(T(x_1^n))$. Hence the p-value, as a random variable, satisfies $p(X_1^n) = 1 - F_{T(n-1)}(T(X_1^n))$. Again, under H_0, $F_{T(n-1)}(T(X_1^n)) \sim U[0,1]$ so $p(X_1^n) \sim U[0,1]$.

Exemple 2.13

Same setting but with $H_1 : \mu \neq \mu_0$. Then the test is $\phi_\alpha = \mathbb{1}_{|T| \geq q_{1-\frac{\alpha}{2}}^{T(n-1)}}$ and the observed p-value is

$$
\begin{aligned}
p(x_1^n) &= \inf \{\alpha \in]0,1[: |T(x_1^n)| \geq F_{T(n-1)}^{-1}(1 - \frac{\alpha}{2})\} \\
&= \inf \{\alpha \in]0,1[: F_{T(n-1)}(|T(x_1^n)|) \geq 1 - \frac{\alpha}{2}\} \\
&= \inf \{\alpha \in]0,1[: \alpha \geq 2[1 - F_{T(n-1)}(|T(x_1^n)|]\} \\
&= 2[1 - F_{T(n-1)}(|T(x_1^n)|].
\end{aligned}
$$

Thus

$$p(X_1^n) = 2[1 - F_{T(n-1)}(|T(X_1^n)|)].$$

For simplicity, we write $T(X_1^n) = T$ and $F_{T(n-1)} = F$.
We henceforth work under H_0.
We first prove that $2\Big(1 - F(|T|)\Big)$ indeed takes values in $[0,1]$. Then we will prove that $P\Big(2[1 - F(|T|)] \leq x\Big) = x$ for $x \in]0,1[$.
F is the CDF of $T(n-1)$, so F corresponds to a symmetric and continuous distribution. Hence $F(x) = 1 - F(-x)$. Thus $F(0) = 1/2$ and since F is nondecreasing, we have $u \geq 0 \Longrightarrow F(u) \geq 1/2$. Therefore $F(|T|) \geq 1/2$. Moreover $F(|T|) \leq 1$, so the first point is proved.
Next, if $x \in]0,1[$,

$$
\begin{aligned}
&\mathbb{P}(2[1 - F(|T|)] \leq x) = \mathbb{P}(F(|T|) \geq 1 - \frac{x}{2}) \\
&= \mathbb{P}(F(T) \geq 1 - \frac{x}{2}, T \geq 0) + \mathbb{P}(F(-T) \geq 1 - \frac{x}{2}, -T \geq 0) \\
&= 2\mathbb{P}(F(T) \geq 1 - \frac{x}{2}, T \geq 0) = 2\mathbb{P}(F(T) \geq 1 - \frac{x}{2}) = x
\end{aligned}
$$

We used :

- the symmetry of the distribution of T,
- $1 - \frac{x}{2} > 1/2$ when $x \in]0,1[$, and since $F(0) = 1/2$ and F is strictly increasing, $F(u) \geq \frac{1}{2} \Longrightarrow u \geq 0$,

— in the last equality : $1-F(T) \sim U[0,1]$ and $\frac{x}{2} \in]0,1[$.

Thus, again, $p(X_1^n) \sim U[0,1]$ under H_0.

Exemple 2.14: An example of computing a p-value with a discrete-statistic distribution

Reconsider Example 1.30 : $X_1, \dots, X_n \overset{iid}{\sim} Be(p)$. We want to test

$$H_0 : p = 1/2 \quad \text{against} \quad p > 1/2.$$

We use the statistic $T = \sum_{i=1}^n X_i$ which, under H_0, follows a binomial distribution $B(n, 1/2)$. In view of H_1, we set

$$\varphi = \mathbb{1}_{T > q_{1-\alpha}^{B(n,1/2)}}.$$

The quantiles of a random variable taking values in a discrete set $\{x_1, \dots, x_r\}$ also take values in this set. Hence the quantiles of the $B(n, 1/2)$ distribution are integers. Moreover, $T(x_1^n)$ is also an integer. Therefore $\{T > q_{1-\alpha}^{B(n,1/2)}\} = \{T \geq_{1-\alpha}^{B(n,1/2)} +1\}$. Thus we can use Wasserman's theorem. The observed p-value is therefore given by

$$p(x_1^n) = P_{p=1/2}(T(X_1^n) \geq T(x_1^n)).$$

The p-value, as a random variable, is therefore given by

$$1 - F_{B(n,1/2)}(T(X_1^n) - 1)$$

(It does not follow exactly a uniform distribution).

One can in fact recover directly the result of Wasserman's theorem :

$$\begin{aligned}
p(x_1^n) &= \inf\left\{\alpha \in]0,1[: T(x_1^n) > F_{B(n,1/2)}^{(-1)}(1-\alpha)\right\} \\
&= \inf\left\{\alpha \in]0,1[: T(x_1^n) \geq F_{B(n,1/2)}^{(-1)}(1-\alpha) + 1\right\} \\
&\overset{item\ 2\ prop\ 2.8}{=} \inf\left\{\alpha \in]0,1[: F_{B(n,1/2)}(T(x_1^n) - 1) \geq 1 - \alpha\right\} \\
&= \inf\left\{\alpha \in]0,1[: \alpha \geq 1 - F_{B(n,1/2)}(T(x_1^n) - 1)\right\} \\
&= 1 - F_{B(n,1/2)}(T(x_1^n) - 1) \\
&= P_{p=1/2}(T(X_1^n) \geq T(x_1^n))
\end{aligned}$$

2.2 Quantile estimation

For the construction of tests and confidence regions, we rely on the notion of quantiles. We recall the general definition of a quantile.

Définition 2.15

For $\beta \in]0,1[$, the quantile of order β of a probability distribution P supported on $\mathbb{R}$ is the quantity

$$q_\beta = \inf\{x \in \mathbb{R} : \mathbb{P}(]-\infty, x]) \geq \beta\}$$

In other words, using the generalized inverse function, if P has distribution function F,

$$\boxed{q_\beta = F^{(-1)}(\beta)}$$

Exemple 2.16

Consider the distribution $\frac{\delta_0+\delta_1+\delta_2}{3}$ [a]. The quantile of order 25% is 0 and the quantile of order 75% is 2.

a. This is the uniform distribution on $\{0,1,2\}$.

Proposition 2.17

1. When the distribution function F is invertible, the quantile of order β equals $F^{-1}(\beta)$ and then $F(q_\beta)=\beta$. Moreover, the quantile is the unique solution of this equation.
2. More generally, if F is continuous, we have $F(q_\beta)=\beta$ (but the solution is not unique).
3. We always have $F(q_\beta)\geq\beta$ and $F(q_\beta^-)\leq\beta$, i.e. $P(X<q_\beta)\leq\beta$. In other words,

$$P(X\leq q_\beta)\geq\beta \quad \text{and} \quad P(X\geq q_\beta)\geq 1-\beta.$$

Démonstration. 1. obvious.

2. $F(q_\beta)=\beta$ is item 4 of Proposition 2.8.
3. $F(x^-)\equiv\lim_{t\to x,t<x}F(t)=\lim_{t\to x,t<x}\mathbb{P}(]-\infty,t])=\mathbb{P}(]-\infty,x[)$. Moreover, if $t<q_\beta$, then by definition of q_β we have $F(t)<\beta$. Hence $F(q_b^-)\leq\beta$.

□

Exemple 2.18

Using point 3 of Proposition 2.17, it follows that the median m satisfies :

$$\mathbb{P}(X\leq m)\geq 1/2 \quad \text{and} \quad \mathbb{P}(X\geq m)\geq 1/2.$$

Moreover, when F is continuous or if $\mathbb{P}(X=m)=0$:

$$\mathbb{P}(X\leq m)=\mathbb{P}(X<m)=\mathbb{P}(X>m)=\mathbb{P}(X\geq m)=1/2. \tag{2.2}$$

Remarque 2.19

Other conventions exist for the definition of a quantile. One may also define a quantile in a non-unique way. Often, one calls a quantile of order β of the distribution F any number q_β such that

$$P(X\leq q_\beta)\geq\beta \quad \text{and } P(X\geq q_\beta)\geq 1-\beta. \tag{2.3}$$

Proposition 2.20

Let X be a real-valued random variable with CDF F, and $\alpha\in]0,1[$.
The smallest real number c such that $\mathbb{P}(X>c)\leq\alpha$ is equal to $q^F_{1-\alpha}$.

Démonstration. $\mathbb{P}(X>c)\leq\alpha\Leftrightarrow\mathbb{P}(X\leq c)\geq 1-\alpha$. By definition, the smallest real c satisfying this inequality is $F^{(-1)}(1-\alpha)$. □

Exemple 2.21

Let $X_1,\dots,X_n\overset{iid}{\sim}Be(p)$. We want to test at level α

$$H_0: p=1/2 \quad \text{against} \quad p>1/2.$$

We use a testing procedure $\varphi_\alpha = \mathbb{1}_{\sum_{i=1}^n X_i > c}$ with c chosen so that the level of the test is at most α and such that the rejection region is maximized. We therefore choose $c = q_{1-\alpha}^{B(n,1/2)}$.

We admit the following theorem. A proof, for students interested, can be found in the solutions to the 2018 exam.

Théorème 2.22

Let $(F_n)_{n\geq 0}$ be a sequence of distribution functions on $\mathbb{R}$ and F a distribution function on $\mathbb{R}$. Then F_n converges to F at every continuity point of F if and only if $F_n^{(-1)}$ converges to $F^{(-1)}$ at every continuity point of $F^{(-1)}$.

Exemple 2.23

The Student distribution with n degrees of freedom converges to the standard normal distribution. Φ^{-1} is continuous. Therefore, for all $\alpha \in]0,1[$, $q_\alpha^{T(n)} \underset{n\to\infty}{\to} q_\alpha^{N(0,1)}$.

We need quantiles for testing procedures as well as for confidence regions. Sometimes we do not know how to compute the quantiles of the distribution, but we can simulate from it. The empirical quantile can then be used in place of the true quantile.

We recall the following notation for order statistics :

$$X_{(1)} \leq \ldots \leq X_{(n)}$$

Définition 2.24

The empirical quantile of an i.i.d. n-sample $X = (X_1, \ldots, X_n)$ is defined, for $\beta \in]0,1]$, by

$$\hat{q}_{n,\beta} = \hat{F}_n^{(-1)}(\beta)$$

It is therefore the quantiles of the empirical distribution.

Proposition 2.25

$$\hat{F}_n^{(-1)}(\beta) = X_{(\lceil n\beta \rceil)}$$

where we denote $\lceil t \rceil = \min\{m \in \mathbb{N} : m \geq t\}$.

Thus, one easily obtains the empirical quantiles by sorting the x_i.

Démonstration. (not required) We use the following immediate property : for all x,

$$x \leq \lceil x \rceil < x + 1.$$

1. There are at least $\lceil n\beta \rceil$ indices $i \in [n]$ such that $X_i \leq X_{(\lceil n\beta \rceil)}$, hence

$$\hat{F}_n(X_{(\lceil n\beta \rceil)}) \geq \frac{\lceil n\beta \rceil}{n} \geq \beta. \tag{2.4}$$

2. Let $x < X_{(\lceil n\beta \rceil)}$. There are at most $\lceil n\beta \rceil - 1$ indices $i \in [n]$ such that $X_i \leq x$, hence

$$\hat{F}_n(x) \leq \frac{\lceil n\beta \rceil - 1}{n} < \beta. \tag{2.5}$$

Equations (2.4) and (2.5) give the result. □

Exemple 2.26

For $n = 101$ the first empirical quartile is given by $\hat{F}_{101}^{(-1)}(1/4) = \hat{q}_{101,25\%} = X_{(26)}$.

The Glivenko–Cantelli theorem ensures that $\|\hat{F}_n - F\|_\infty \to 0$ almost surely. We therefore expect $\hat{q}_{n,\beta}$ to be close to q_β when n is large.

Théorème 2.27

Let $\beta \in]0,1[$ such that $F^{(-1)}$ is continuous at β. Then

$$\lim_{n\to\infty} \hat{q}_{n,\beta} = q_\beta \quad \text{a.s.}$$

Démonstration. By the Glivenko–Cantelli theorem, there exists a measurable set A such that $\mathbb{P}(A) = 1$ and if $\omega \in A$, then $\|\hat{F}_n(\omega, \cdot) - F(\cdot)\|_\infty \underset{n\to\infty}{\to} 0$. Let $\omega \in A$. In particular, $\hat{F}_n(\omega, t) \underset{n\to\infty}{\to} F(t)$ for all $t \in \mathbb{R}$. Therefore, $\hat{F}_n^{(-1)}(\omega, t) \underset{n\to\infty}{\to} F^{(-1)}(t)$ at every continuity point t of $F^{(-1)}$ by Theorem 2.22. □

Remarque 2.28

A continuity point β for $F^{(-1)}$ corresponds to a point of strict increase q_β for F.

Remarque 2.29

We thus see that if one cannot easily compute the quantile of a distribution, but one can simulate from it, one can obtain an approximate value of its quantiles by simulating a sufficiently large sample and computing the empirical quantile. A related question is : what sample size is needed to achieve a given accuracy ? The following theorem partially answers this question. We admit this theorem.

Théorème 2.30

If F is differentiable at q_β with $F'(q_\beta) > 0$, then

$$\sqrt{n}(\hat{q}_{n,\beta} - q_\beta) \xrightarrow[n\to\infty]{Loi} \mathcal{N}\left(0, \frac{\beta(1-\beta)}{(F'(q_\beta))^2}\right)$$

Remarque 2.31

The conditions of the theorem are in particular satisfied if the distribution F has a density f that is strictly positive on $\mathbb{R}$. To construct a confidence interval for q_β, one then needs to know $f(q_\beta)$.
If the density is known, one can proceed as in Part B of the 2014 midterm.
One can nevertheless give a confidence interval for $q_\beta = F^{-1}(\beta)$ even without knowing the density (see, e.g., Arnaud Guyader's notes, Part I, pages 53–54 `https://perso.lpsm.paris/~aguyader/files/teaching/M1/PolycopiePartie1.pdf`).

2.3 Goodness-of-fit tests to a distribution or to a family of distributions

2.3.1 Fit to a given distribution

We fix a reference distribution with distribution function F_0, and we observe an i.i.d. n-sample $X_1^n = (X_1, \dots, X_n)$ whose common distribution function is denoted by F. We want to test

$$\boxed{H_0 : F = F_0 \quad \text{against} \quad H_1 : F \neq F_0}$$

We will naturally use the following test statistic

$$\boxed{h_n(X_1^n, F_0) = \|\hat{F}_n - F_0\|_\infty}$$

In what follows, this statistic is called the Kolmogorov–Smirnov (KS) statistic.

Remarque 2.32

This is indeed a statistic, i.e. h_n is measurable. Indeed, one can show (thanks to right-continuity) that

$$h_n(X_1^n, F_0) = \sup_{x\in\mathbb{Q}} \left|\hat{F}_n(x) - F_0(x)\right|.$$

Proposition 2.33

Assume that F_0 and F are continuous. Then (not required)

$$h_n(X_1^n, F_0) = \max_{1\le j\le n} \left\{ \max\{\frac{j}{n} - F_0(X_{(j)}),\, F_0(X_{(j)}) - \frac{j-1}{n}\}\right\}$$

Ce qu'il faut retenir 2.34

The specific form of the empirical CDF (piecewise constant, values j/n, see the illustration in Figure 2.2) allows a simple computation of the statistic associated with the KS test.

Démonstration. (not required) Since F is continuous, almost surely all the X_i are distinct (see Exercise 1 in TD1). Thus $X_{(1)} < X_{(2)} < \dots < X_{(n)}$. Hence we can describe $\hat{F}_n$ as follows :

$$\hat{F}_n(x) = \begin{cases} 0 & \text{if } x < X_{(1)} \\ \frac{j}{n} & \text{if } x \in [X_{(j)}, X_{(j+1)}[, \quad 1 \le j \le n-1 \\ 1 & \text{if } x \ge X_{(n)} \end{cases}$$

We will therefore use the following equality

$$h_n(X_1^n, F_0) = \sup_{0\le j\le n} M_j$$

where, for $1 \le j \le n-1$,

$$M_j = \sup_{x\in[X_{(j)}, X_{(j+1)}[} \left|\hat{F}_n(x) - F_0(x)\right|,$$

and

$$M_0 = \sup_{x<X_{(1)}} \left|\hat{F}_n(x) - F_0(x)\right| \quad \text{and} \quad M_n = \sup_{x\ge X_{(n)}} \left|\hat{F}_n(x) - F_0(x)\right|.$$

Using the monotonicity of F_0 we obtain

$$M_n = \sup_{x\ge X_{(n)}} \left|1 - F_0(x)\right| = \sup_{x\ge X_{(n)}} \{1 - F_0(x)\} = 1 - F_0(X_{(n)})$$

and

$$M_0 = \sup_{x<X_{(1)}} \left|0 - F_0(x)\right| = \sup_{x<X_{(1)}} F_0(x) = F_0(X_{(1)}^-)$$

And by continuity of F_0, $M_0 = F_0(X_{(1)})$. Now consider M_j for $1 \le j \le n-1$. We have

$$M_j = \sup_{x\in[X_{(j)}, X_{(j+1)}[} \left|\frac{j}{n} - F_0(x)\right|.$$

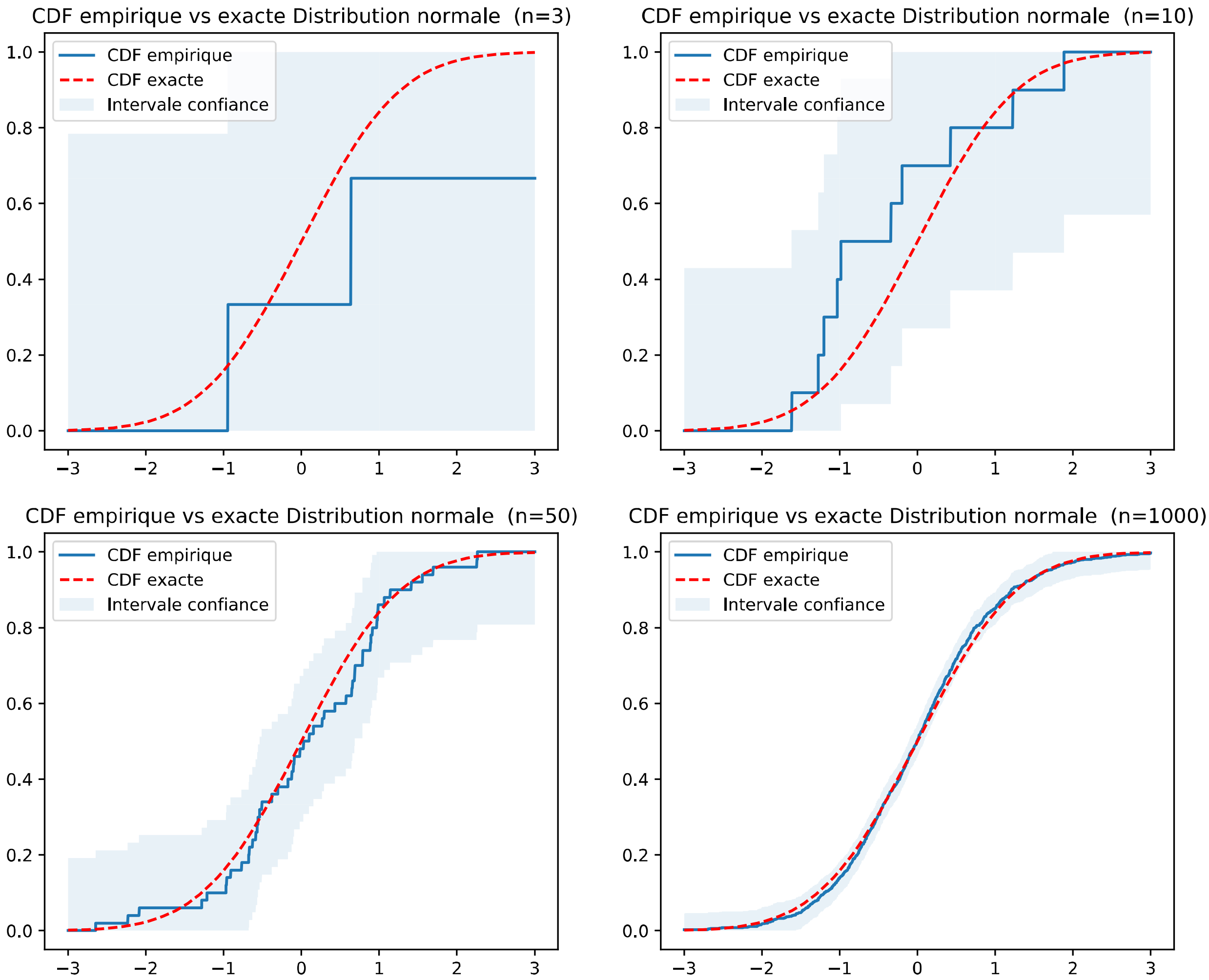


FIGURE 2.2 – Difference between the empirical and the true distribution functions for the normal distribution and for different values of n.

Let f be an increasing and continuous function on an interval $[a, b]$. Then

$$\sup_{a \le x < b} |f(x)| = \sup_{a \le x < b} \Big\{ \sup\{f(x), -f(x)\} \Big\} = \sup \Big\{ \sup_{a \le x < b} f(x), \sup_{a \le x < b} -f(x) \Big\}$$
$$= \sup \Big\{ \sup_{a \le x < b} f(x), - \inf_{a \le x < b} f(x) \Big\} = \max\{f(b), -f(a)\}.$$

Applying this property to the increasing and continuous function $F_0 - \frac{j}{n}$, we obtain

$$M_j = \max \Big\{ F_0(X_{(j+1)}) - \frac{j}{n}, \frac{j}{n} - F_0(X_{(j)}) \Big\}.$$

Collecting all results, we finally obtain

$$h_n(X_1^n, F_0) = \max \Big\{ \max_{1 \le j \le n-1} \Big(F_0(X_{(j+1)}) - \frac{j}{n} \Big), \max_{1 \le j \le n-1} \Big(\frac{j}{n} - F_0(X_{(j)}) \Big), F_0(X_{(1)}), 1 - F_0(X_{(n)}) \Big\}.$$

The final result follows by noting that

$$\max \Big\{ \max_{1 \le j \le n-1} \Big(\frac{j}{n} - F_0(X_{(j)}) \Big), 1 - F_0(X_{(n)}) \Big\} = \max_{1 \le j \le n} \Big(\frac{j}{n} - F_0(X_{(j)}) \Big)$$

and

$$\max \Big\{ \max_{1 \le j \le n-1} \Big(F_0(X_{(j+1)}) - \frac{j}{n} \Big), F_0(X_{(1)}) \Big\} = \max \Big\{ \max_{2 \le j \le n} \Big(F_0(X_{(j)}) - \frac{j-1}{n} \Big), F_0(X_{(1)}) \Big\}$$
$$= \max_{1 \le j \le n} \Big(F_0(X_{(j)}) - \frac{j-1}{n} \Big).$$

□

Définition 2.35

We say that a random variable Z is **diffuse** if its CDF is continuous, i.e. if

$$\forall x \in \mathbb{R} : \mathbb{P}(Z = x) = 0. \tag{2.6}$$

Définition 2.36

If the distribution of a statistic $\hat{g}(X_1, \ldots, X_n)$ does not depend on the sample CDF F_0, we say that this statistic is « distribution-free with respect to F_0 ».

We now make two remarks that are useful for the proof of the next proposition.

Remarque 2.37

If $F : \mathbb{R} \to [0, 1]$ is a distribution function, then

$$F \text{ is continuous } \Leftrightarrow \quad]0, 1[\subset F(\mathbb{R}).$$

Indeed :

— If F is continuous, we can apply the intermediate value theorem.

— If F is not continuous, then there is at least one jump at some $x \in \mathbb{R}$, and the values between $F(x)$ and $F(x^-)$ are not taken by F.

Remarque 2.38

If $Z = \max_{j=1,\ldots,k} X_j$ with diffuse random variables X_j, then Z is diffuse. Indeed, for all x,

$$\mathbb{P}(Z = x) \leq \mathbb{P}\Big(\cup_{j=1}^{k} \{X_j = x\} \Big) \leq \sum_{j=1}^{k} \mathbb{P}(X_j = x) = 0.$$

Proposition 2.39

Under H_0, if F_0 is continuous, then the statistic $h_n(X_1^n, F_0)$ of the K–S goodness-of-fit test is distribution-free with respect to F_0 and has a continuous distribution.

Démonstration. Let $U_1^n = (U_1, \ldots, U_n)$ be an i.i.d. n-sample from the uniform distribution on $]0, 1[$. Under H_0, since $X_i \overset{i.i.d.}{\sim} F_0$, we have $(X_1, \ldots, X_n) \sim (F_0^{(-1)}(U_1), \ldots, F_0^{(-1)}(U_n))$ by Proposition 2.8. Hence, under H_0,

$$h_n(X_1^n, F_0) \sim \sup_{x\in\mathbb{R}} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{F_0^{(-1)}(U_i)\leq x} - F_0(x)\Big|.$$

Using item 2 of Proposition 2.8, we obtain

$$\begin{aligned}\sup_{x\in\mathbb{R}} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{F_0^{(-1)}(U_i)\leq x} - F_0(x)\Big| &= \sup_{x\in\mathbb{R}} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq F_0(x)} - F_0(x)\Big| \\ &= \sup_{s\in Im(F_0)} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq s} - s\Big|.\end{aligned}$$

Using Remark 2.37, this gives, almost surely,

$$\sup_{s\in Im(F_0)} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq s} - s\Big| = \sup_{s\in]0,1[} \Big|\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq s} - s\Big|.$$

Indeed, $]0,1[\subset Im(F_0)\subset[0,1]$ and the value of the function $s\mapsto \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{U_i\le s}-s$ at $s=0$ and $s=1$ is equal to 0 almost surely. We have thus shown that, under H_0,

$$h_n(X_1^n,F_0)\sim h_n(U_1^n,G),$$

where G denotes the distribution function of the uniform distribution on $[0,1]$, i.e. the function defined by $G(s)=s$ for $s\in[0,1]$. This shows that the distribution under H_0 of $h_n(X_1^n,F_0)$ is distribution-free with respect to F_0.
We now prove that the distribution of $h_n(U_1^n,G)$ is continuous (proof not required). By Proposition 2.33, since G is continuous,

$$h_n(U_1^n,G)=\max_{1\le j\le n}\Big\{\max\{\frac{j}{n}-U_{(j)},\,U_{(j)}-\frac{j-1}{n}\}\Big\}.$$

Since the distribution of each U_j is absolutely continuous with respect to Lebesgue measure, so is that of $U_{(j)}$ (see also Exercise 1 in TD1 ; here we even have $U_{(j)}\sim Beta(j,n-j+1)$). Hence, by Remark 2.38, $h_n(U_1^n,G)$ indeed has a continuous distribution. □

Exemple 2.40

Let $X_1,\dots,X_n\overset{iid}{\sim}N(0,1)$ and $Y_1,\dots,Y_n\overset{iid}{\sim}exp(1)$. Then

$$\sup_{x\in\mathbb{R}}\Big|\frac{1}{n}\sum_{i=1}^n\mathbb{1}_{X_i\le x}-\Phi(x)\Big|\sim\sup_{x>0}\Big|\frac{1}{n}\sum_{i=1}^n\mathbb{1}_{Y_i\le x}-(1-\exp(-x))\Big|\sim\sup_{s\in]0,1[}\Big|\frac{1}{n}\sum_{i=1}^n\mathbb{1}_{U_i\le s}-s\Big|$$

(Φ is the CDF of $\mathcal{N}(0,1)$)

For all $\alpha\in]0,1[$, if we denote by $q_{n,\alpha}$ the quantile of order α of the distribution of the statistic $h_n(U_1^n,G)$, we thus have, by continuity of this distribution,

$$\mathbb{P}_{X_1,\dots,X_n\overset{iid}{\sim}F_0}(h_n(X_1^n,F_0)\le q_{n,\alpha})=\alpha.$$

For small values of n, the quantiles of this statistic have been tabulated.
We deduce a confidence band of level $1-\alpha$ by setting

$$\begin{aligned}B(n,\alpha)&=\{\text{distribution functions } G\mid \forall x\in\mathbb{R}\quad \hat F_n(x)-q_{n,1-\alpha}\le G(x)\le \hat F_n(x)+q_{n,1-\alpha}\}\\&=\{G: h_n(X_1^n,G)\le q_{n,1-\alpha}\}.\end{aligned}$$

To test $H_0: F=F_0$ against $H_1: F\neq F_0$, we set

$$\varphi_\alpha(X_1^n)=\mathbb{1}_{h_n(X_1^n,F_0)\ge q_{n,1-\alpha}}.$$

We have thus obtained the following result.

Théorème 2.41: Kolmogorov–Smirnov test

Let $(X_1,\dots,X_n)$ be an i.i.d. n-sample with distribution function F. The test $\varphi_\alpha(X_1^n)$ has size α for testing $H_0: F=F_0$ against $H_1: F\neq F_0$ when F_0 is continuous.

Remarque 2.42

When F_0 is not continuous, the test is no longer of size α but it remains of level α. Indeed, from the proof of Proposition 2.39,

$$h_n(X_1^n,F_0)\sim\sup_{s\in Im(F_0)}\Big|\frac{1}{n}\sum_{i=1}^n\mathbb{1}_{U_i\le s}-s\Big|.$$

(Continuity of F_0 is not needed to obtain this equality in distribution.) And since

$$\sup_{s\in Im(F_0)} |\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq s} - s| \leq \sup_{s\in]0,1[} |\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{U_i\leq s} - s|,$$

we have

$$\mathbb{P}_{X_1,\dots,X_n \overset{iid}{\sim} F_0}\big(h_n(X_1^n, F_0) \geq \xi_{n,1-\alpha}\big) \leq \mathbb{P}\big(h_n(U_1^n, G) \geq \xi_{n,1-\alpha}\big) = \alpha.$$

Remarque 2.43

When the number of observations n is large, one uses an asymptotic test.
One also has the Dvoretzky–Kiefer–Wolfowitz inequality [a], which holds without any condition on F_0. Under $H_0 : X_1, \dots, X_n \overset{iid}{\sim} F_0$ (not required),

$$\mathbb{P}(\|\hat{F}_n - F_0\|_\infty > \epsilon) \leq 2e^{-2n\epsilon^2} \text{ for all } n \in \mathbb{N} \text{ and all } \epsilon > 0.$$

a. More precisely, the inequality with constant 2 is due to Massart

Implémentation 2.44

1. R :
   - If we want to check that a sample `x` follows a Gaussian distribution with mean 3 and standard deviation 2 :
     `ks.test(x,"pnorm",3,2)`
   - If we want to check that a sample `x` follows a gamma distribution with shape parameter 3 and rate 2 :
     `ks.test(x,"pgamma",3,2)`

   Mise en garde 2.45

   Warning : the function `ks.test` behaves poorly in the presence of ties (in the two-sample equality-of-distributions test, one should not have a tie of the form $x_i = y_j$). In theory, one cannot have two identical values if the underlying distribution is continuous. But in practice, measurement imprecision may generate ties in the sample.

   Illustration of this behavior (here the value 4.25 appears twice in the sample) :

```
x= c(4.25, 7.79, 5.18, 8.43, 4.25, 5.71, 4.1 , 4.2)

ks.test(x,"pnorm",mean=5, sd=1)
```

   Output :

```
One-sample Kolmogorov-Smirnov test

data:  x
D = 0.27337, p-value = 0.5883
alternative hypothesis: two-sided

Warning message:
In ks.test(x, "pnorm", mean = 5, sd = 1) :
ties should not be present for the Kolmogorov-Smirnov test
```

One can avoid the warning message about ties by slightly modifying one of the tied values :

```
x= c(4.2500001, 7.79, 5.18, 8.43, 4.25, 5.71, 4.1 , 4.2)

ks.test(x,"pnorm",mean=5, sd=1)
```

Output :

```
One-sample Kolmogorov-Smirnov test

data:  x
D = 0.27337, p-value = 0.5045
alternative hypothesis: two-sided
```

We indeed observe a significance level value that is quite different from the previous one.

— In the case where we want to test that a sample x is normally distributed without specifying the mean or variance (see Section 2.3.2 and TD2), we can use the function `lillie.test` from the package `nortest` :

```
library(nortest)
lillie.test(x)
```

2. Python :

In Python there are two possibilities :

— The first, if one wants to stay within the R environment, is to call R commands. For example, one can do :

```
from rpy2 import robjects
rks=robjects.r('ks.test')
```

Then one uses the function that was called `rks` in the usual way, taking care to also convert the inputs. For example, if a sample is stored in the vector `x`, and we want to check that it is a standard Gaussian sample :

```
y=robjects.FloatVector(x)
z=rks(y,"pnorm")
```

— A variant is to use native functions such as `stats.kstest`; this function behaves differently from R in the presence of ties. Here we test whether the sample follows a $\mathcal{N}(5, 1)$ distribution :

```
from scipy import stats
x =[4.25, 7.79, 5.18, 8.43, 4.25, 5.71, 4.1, 4.2]
stats.kstest(x,'norm',args=(5.,1.))
Out[178]: KstestResult(statistic=0.2733726476231318, pvalue=0.504516077747122)

from scipy import stats
x =[4.2500001, 7.79, 5.18, 8.43, 4.25, 5.71, 4.1, 4.2]
stats.kstest(x,'norm',args=(5.,1.))
Out[180]: KstestResult(statistic=0.2733726175093874, pvalue=0.5045162143226165)
```

2.3.2 Fit to a parametric family of distributions : the case of exponential families

Let $(X_1, \ldots, X_n)$ be an i.i.d. n-sample of positive random variables with distribution function F. **We want to test whether the distribution of the X_i is exponential**, i.e. we want to test whether there exists $\lambda > 0$ such that $F = F_\lambda$ with $F_\lambda(x) = (1 - e^{-\lambda x})\mathbb{1}_{\mathbb{R}^+}(x)$ for all $x \in \mathbb{R}$. This hypothesis corresponds to H_0[1].

Under H_0 we estimate the parameter λ. The maximum likelihood estimator (MLE) is $\hat{\lambda} = 1/\bar{X}$. We then consider the statistic

$$\boxed{h'_n(X_1^n) = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F_{\hat{\lambda}}(x)| \text{ where } \hat{\lambda} = 1/\bar{X}}$$

Proposition 2.46

Under H_0, the distribution of $h'_n(X_1^n)$ is free of the parameter λ. Moreover, this distribution is continuous.

Démonstration. We work under H_0. Set $Y_i = \lambda X_i$, for $1 \le i \le n$. Then $Y_1, \ldots, Y_n \overset{iid}{\sim} exp(1)$.

$$\begin{aligned} h'_n(X_1^n) &= \sup_{x>0} \Big|\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{X_i \le x} - (1 - e^{-\frac{x}{\bar{X}}})\Big| = \sup_{x>0} \Big|\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{\frac{Y_i}{\lambda} \le x} - (1 - e^{-\frac{\lambda x}{\bar{Y}}})\Big| \\ &= \sup_{t>0} \Big|\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{Y_i \le t} - (1 - e^{-\frac{t}{\bar{Y}}})\Big|. \end{aligned} \tag{2.7}$$

The statistic $\sup_{t>0} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{Y_i \le t} - (1 - e^{-\frac{t}{\bar{Y}}})|$ has a distribution that does not depend on λ. We admit the continuity of this distribution. □

We deduce a test of size α by setting $\phi_\alpha(X_1^n) = \mathbb{1}_{h'_n(X_1^n) \ge q_{n,1-\alpha}}$, where $q_{n,1-\alpha}$ is the quantile of order $1-\alpha$ of the distribution of $\sup_{t>0} \left|\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{Y_i \le t} - (1 - e^{-\frac{t}{\bar{Y}_n}})\right|$ with $Y_1, \ldots, Y_n \overset{iid}{\sim} exp(1)$.

Remarque 2.47

One can perform the same type of test for a number of families of distributions (see TD2 for an example with normal distributions).

Mise en garde 2.48

Warning : this is a testing problem different from the Kolmogorov–Smirnov testing problem of the previous section. In the latter, we want to know whether the sample follows an exponential distribution with parameter λ_0, with λ_0 given. The test statistic is then $h_n = \|\hat{F}_n - F_{\lambda_0}\|_\infty$. The distributions under H_0 of these two test statistics h_n and h'_n are different.

2.4 Kolmogorov–Smirnov two-sample homogeneity test

We observe two i.i.d. samples $X_1^n = (X_1, \ldots, X_n)$ and $Y_1^m = (Y_1, \ldots, Y_m)$, independent of each other. The sample size m may differ from n. We want to test whether the two samples have the same distribution.

In other words, if we denote by F the CDF of the X_i and by G the CDF of the Y_i, we want to test

$$\boxed{H_0 : F = G \quad \text{against} \quad H_1 : F \neq G.}$$

As before, we denote by $\hat{F}_n$ and $\hat{G}_m$ the empirical distribution functions of the samples $X_1^n = (X_1, \ldots, X_n)$ and $Y_1^m = (Y_1, \ldots, Y_m)$, respectively, and we set

1. It is therefore a composite hypothesis.

$$\boxed{h_{n,m}(X_1^n, Y_1^m) = \sup_{t\in\mathbb{R}} |\hat{F}_n(t) - \hat{G}_m(t)|}$$

Proposition 2.49

Under $H_0 : F = G$ and if F is continuous, then the distribution of $h_{n,m}(X_1^n, Y_1^m)$ does not depend on F.

Démonstration. Under H_0, $X_1, \dots, X_n, Y_1, \dots, Y_m \overset{iid}{\sim} F$. Hence, if $U_1, \dots, U_n, V_1, \dots, V_m \overset{iid}{\sim} U[0,1]$, we have

$$(F^{(-1)}(U_1), \dots, F^{(-1)}(U_n), F^{(-1)}(V_1), \dots, F^{(-1)}(V_m)) \sim (X_1, \dots, X_n, Y_1, \dots, Y_m).$$

Thus, under H_0,

$$\begin{aligned}
h_{n,m}(X_1^n, Y_1^m) &= \sup_{t\in\mathbb{R}} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{X_i\le t} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{Y_i\le t}| \\
&\sim \sup_{t\in\mathbb{R}} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{F^{(-1)}(U_i)\le t} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{F^{(-1)}(V_i)\le t}| \\
&= \sup_{t\in\mathbb{R}} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{U_i\le F(t)} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{V_i\le F(t)}| \\
&= \sup_{s\in Im(F)} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{U_i\le s} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{V_i\le s}| \\
&\overset{p.s.}{=} \sup_{s\in]0,1[} |\frac{1}{n}\sum_{i=1}^n \mathbb{1}_{U_i\le s} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{V_i\le s}|.
\end{aligned}$$

We used Proposition 2.8 and continuity of F. Indeed, if F is continuous then $]0,1[\subset Im(F) \subset [0,1]$, and one checks immediately that, almost surely, the function $s \mapsto \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{U_i\le s} - \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{V_i\le s}$ equals 0 at $s = 0$ and $s = 1$. □

This distribution is tabulated. We set

$$\varphi_\alpha(X_1^n, Y_1^m) = \mathbb{1}_{h_{n,m}(X_1^n, Y_1^m) > x_{n,m,1-\alpha}},$$

where $x_{n,m,1-\alpha}$ is the quantile of order $1-\alpha$ of the distribution of the statistic $h_{n,m}(X_1^n, Y_1^m)$ under H_0.

Implémentation 2.50

To test whether two samples x and y have the same (continuous) distribution, one can use :

— R : `ks.test` from the `stats` package. The call is `ks.test(x,y)`.

— Python : (x and y are 1D arrays or lists)

```
from scipy import stats
stats.ks_2samp(x,y)
```

2.5 Chi-square goodness-of-fit tests

We recall two definitions related to asymptotic tests.

Définition 2.51

Let $\alpha \in]0,1[$. We say that a sequence of tests is asymptotically of size α if the Type I error converges to α as $n \to \infty$.
We say that a sequence of tests is consistent if the power function converges pointwise to 1 as $n \to \infty$.

2.5.1 Goodness-of-fit to a given distribution

Let $X_1, \ldots, X_n$ be an n-sample with the same distribution as a random variable X **taking values in a finite set** $\{x_1, \ldots, x_d\}$, for $d \in \mathbb{N}^*$.
Notation : for $1 \leq j \leq d$,

$$p_j = \mathbb{P}(X = x_j), \quad \hat{p}_j = \frac{1}{n}\sum_{i=1}^{d} \mathbb{1}_{X_i = x_j}.$$

We are given a reference distribution $p^0 = (p_1^0, \ldots, p_d^0)$ [a] such that $p_j^0 > 0$ for all $j \in \{1, \ldots, d\}$.
We want to test

$$H_0 : p = p^0 \quad \text{against} \quad H_1 : p \neq p^0.$$

We use a test based on the discrepancy between the empirical frequencies $\hat{p}_j$ and the theoretical frequencies p_j^0.

a. Thus $\sum_{j=1}^{d} p_j^0 = 1$.

More precisely, the test is based on Pearson's statistic (not required)

$$D^2 = n \sum_{j=1}^{d} \frac{(\hat{p}_j - p_j^0)^2}{p_j^0}.$$

Recall that, under H_0,

$$D^2 \underset{n\to\infty}{\overset{Loi}{\to}} \chi^2(d-1),$$

and that, under H_1, $D^2 \underset{n\to\infty}{\overset{p.s.}{\to}} +\infty$. We therefore have :

Théorème 2.52: Chi-square goodness-of-fit test to a given distribution

Let $\alpha \in]0,1[$. The test defined by $\varphi_\alpha(X_1^n) = \mathbb{1}_{D^2 > q_{1-\alpha}^{\chi^2(d-1)}}$ is asymptotically of size α and consistent.

Conditions for applying this test :

Recall that, in practice, the CLT-based approximation of a binomial distribution $B(n,p)$ by a normal distribution $\mathcal{N}(np, \sqrt{np(1-p)})$ is valid only under the following conditions :

- n must be sufficiently large.
- n must be larger when p is close to 0 or 1.

A common rule of thumb is : $np \geq 5$ and $n(1-p) \geq 5$ [2].

In the proof of the convergence in distribution of D^2 to the $\chi^2(d-1)$ distribution under H_0, one uses the CLT to approximate a multinomial vector by a normal distribution. Hence we need similar conditions.

2. of course, as always with such criteria, the thresholds are somewhat arbitrary

To fix ideas, we will consider the chi-square goodness-of-fit test to a given distribution to be valid if :

$$n \geq 30 \quad \text{and} \quad np_j^0 \geq 5 \quad \forall j \in \{1, \dots, d\}.$$

Exemple 2.53

We are given observations $x_1, \dots, x_n$ from an i.i.d. sample with unknown distribution. We suspect that it is a Poisson distribution with parameter $\lambda_0 = 1.5$. We can then use the chi-square goodness-of-fit test to a given distribution. Suppose that $n = 300$ and that we obtain the following results :

0	1	2	3	4	5	6
67	96	68	38	23	7	1

so that $\hat{p} = (\frac{67}{300}, \frac{96}{300}, \frac{68}{300}, \frac{38}{300}, \frac{23}{300}, \frac{7}{300}, \frac{1}{300})$

We then use the chi-square goodness-of-fit test to a given distribution for the testing problem

$$H_0 : p = p_0 \quad \text{against} \quad H_1 : p \neq p_0,$$

with $p^0 = (p_0^0, \dots, p_6^0)$ where, if $Y \sim \mathcal{P}(1.5)$,

$$p_i^0 = \begin{cases} \mathbb{P}(Y = i) & \text{if } i \in \{0, \dots, 5\} \\ \mathbb{P}(Y \geq 6) & \text{if } i = 6 \end{cases}$$

Indeed, one must reduce to a distribution with finite support if the reference distribution takes values in an infinite set.

One can check, using a computer, that the conditions in the previous box are not satisfied (np_6^0 is too small). Therefore we instead take $p^0 = (p_0^0, \dots, p_5^0)$ with $p_i^0 = \begin{cases} \mathbb{P}(X = i) & \text{if } i \in \{0, \dots, 4\} \\ \mathbb{P}(X \geq 5) & \text{if } i = 5 \end{cases}$ and $\hat{p} = (\frac{67}{300}, \frac{96}{300}, \frac{68}{300}, \frac{38}{300}, \frac{23}{300}, \frac{7+1}{300})$

Implémentation 2.54: implementation of this example with R

```
x=c(67,96,68,38,23,8)
p=rep(0,6)
p[1:5]=dpois(c(0,1,2,3,4),1.5)
p[6]=1-ppois(4,1.5)
chisq.test(x=x,p=p)
```

With the output :

```
##
##  Chi-squared test for given probabilities
##
## data:  x
## X-squared = 7.5475, df = 5, p-value = 0.183
```

Implémentation 2.55: implementation of this example with Python

```
from scipy.stats import poisson, chisquare
fobs=[67,96,68,38,23,8]
fexp=[300*poisson.pmf(k,1.5) for k in range(5)]
fexp.append(300*(1-poisson.cdf(4,1.5)))
chisquare(f_exp=fexp,f_obs=fobs)
```

Remarque 2.56: when the parameter is unknown

There is also a chi-square goodness-of-fit test to a parametric family of distributions with finite support in the case where the parameter is not specified (as in Section 2.3.2). One then replaces the unknown parameter in the chi-square goodness-of-fit test formula by the maximum likelihood estimator. The limiting distribution is again chi-square, but with a modified number of degrees of freedom (compared to the same testing problem where the parameter would be known).

2.5.2 Chi-square test of independence

We have n pairs of random variables $(X_1, Y_1), \ldots, (X_n, Y_n)$ taking finitely many values, independent and identically distributed according to a distribution denoted by ν. We want to test

$$H_0: \text{ the } X_i \text{ are independent of the } Y_i.$$

Let $\{x_1, \ldots, x_r\}$ and $\{y_1, \ldots, y_s\}$ be the sets of values taken by the X_i and the Y_i, respectively. The test statistic is as follows (not required)

$$D^2_{indep} = n \sum_{j=1}^{r} \sum_{\ell=1}^{s} \frac{\left(\hat{\nu}_{(j,\ell)} - \hat{p}_j \hat{q}_\ell\right)^2}{\hat{p}_j \hat{q}_\ell}$$

where

$$\hat{\nu}_{(j,\ell)} = \frac{\sum_{i=1}^n \mathbb{1}_{\{(X_i,Y_i)=(x_j,y_\ell)\}}}{n} \qquad \hat{p}_j = \frac{\sum_{i=1}^n \mathbb{1}_{X_i=x_j}}{n} \qquad \hat{q}_\ell = \frac{\sum_{i=1}^n \mathbb{1}_{Y_i=y_\ell}}{n}.$$

Recall that, under H_0,

$$D^2_{indep} \underset{n\to\infty}{\overset{Loi}{\to}} \chi^2((r-1)(s-1)),$$

and that, under H_1, $D^2_{indep} \underset{n\to\infty}{\overset{p.s.}{\to}} +\infty$. We therefore have :

Théorème 2.57: Chi-square test of independence

Let $\alpha \in]0,1[$. The test defined by $\phi(X_1^n, Y_1^n) = \mathbb{1}_{D^2_{indep} > q_{1-\alpha}^{\chi^2((r-1)(s-1))}}$ is asymptotically of size α and consistent.

Conditions for applying this test :

The convergence given in the previous theorem relies (among other things) on the CLT applied to a multinomial vector. Hence one also needs a validity condition of the form :

$$np_j q_\ell \text{ and } \nu_{j,\ell} \text{ must be sufficiently large, for all } (j,\ell).$$

Since we do not know p_j, q_ℓ, or $\nu_{j,\ell}$, we replace them by their estimators $\hat{p}_j$, $\hat{q}_\ell$ and $\hat{\nu}_{j,\ell}$ to check these conditions.

To fix ideas, we will consider the chi-square test of independence to be valid if :

$$n\hat{p}_j\hat{q}_\ell \geq 5 \text{ and } n\hat{\nu}_{j,\ell} \geq 5 \quad \forall (j,\ell) \in \{1,\ldots,r\} \times \{1,\ldots,s\}.$$

Exemple 2.58

For a certain disease, we want to know whether being ill depends on the person's sex. We take a sample and obtain the following table :

	M	F
ill	80	100
not ill	110	90

Here $r = s = 2$. If we denote by p_1 the probability of being ill and by q_1 the probability of being male, with $p_2 = 1 - p_1$ and $q_2 = 1 - q_1$, we then have

$$\hat{p}_1 = \frac{180}{380} \quad \hat{p}_2 = 1 - \hat{p}_1 = \frac{200}{380} \quad \hat{q}_1 = \frac{190}{380} \quad \hat{q}_2 = 1 - \hat{q}_1 = \frac{190}{380}$$

$$\hat{\nu}_{(1,1)} = \frac{80}{380} \quad \hat{\nu}_{(1,2)} = \frac{100}{380} \quad \hat{\nu}_{(2,1)} = \frac{110}{380} \quad \hat{\nu}_{(2,2)} = \frac{90}{380}$$

Checking the applicability conditions : one immediately checks that $n\hat{p}_1\hat{q}_1$, $n\hat{p}_1\hat{q}_2$, $n\hat{p}_2\hat{q}_1$, $n\hat{p}_2\hat{q}_2$, $n\hat{\nu}_{1,1}$, $n\hat{\nu}_{1,2}$, $n\hat{\nu}_{2,1}$ and $n\hat{\nu}_{2,2}$ are all greater than 5.

Implémentation 2.59: implementation of this example with R and Python

Implementation with R

```
M <- as.table(rbind(c(80, 100), c(110, 90)))
chisq.test(M)
##  Pearson's Chi-squared test with Yates' continuity correction##
## data:  M
## X-squared = 3.8106, df = 1, p-value = 0.05093
```

Implementation with Python

```
from scipy.stats import chi2_contingency
import numpy as np
table=np.array([[80,100],[110,90]])
chi2_contingency(table)
```

Remarque 2.60

Let us summarize the essential differences between the Kolmogorov–Smirnov test and the χ^2 goodness-of-fit test :

— The χ^2 test is better suited to discrete distributions. It can be used for continuous distributions as well, but one must discretize by choosing bins (which bins ? how many bins ?). The K–S test is designed for continuous distributions.

— The Kolmogorov–Smirnov test has the special feature of being exact for small samples : the distribution is distribution-free for fixed n. The χ^2 test is only asymptotic (one must check the condition $np_j^0 \gtrsim 5$).

Chapitre 3

Robust tests

The goal of this chapter is to present tests that require no assumptions about the underlying distributions, or only very weak assumptions. In that sense, these tests are nonparametric. They are also better suited to the presence of outlying observations in the sample : these are then called robust tests.

The R/Python code snippets are provided for illustration.

3.1 An example

An example of a question we would like to answer in this chapter is the following : do men earn more than women ? To answer this question, let us imagine that we have a sample

$$X_1^{n_1} = (X_1, \dots, X_{n_1})$$

of women's salaries and a sample

$$Y_1^{n_2} = (Y_1, \dots, Y_{n_2})$$

of men's salaries.

We will use different tests depending on whether

1. **the samples are i.i.d. and independent of each other**, that is,

$$X_1, \dots, X_{n_1} \text{ are i.i.d. with distribution } F_X, \qquad Y_1, \dots, Y_{n_2} \text{ are i.i.d. with distribution } F_Y, \qquad X_1^{n_1} \perp\!\!\!\perp Y_1^{n_2};$$

2. **the data are paired**. We will give a more precise definition of pairing later. For now, let us say that if the two samples have the same size, $n_1 = n_2 = n$, and if we have grouped the data into comparable pairs (for example by age : individuals with the same index have the same age), then the data are paired.

Let us now imagine that we are in the case of paired data (for example grouped by age). We can then consider the differences

$$D_i = Y_i - X_i, \qquad 1 \leq i \leq n,$$

and, to simplify, assume that $(D_i)_{1 \leq i \leq n}$ is an i.i.d. sample of size n.

The test we want to perform is therefore

$$H_0 : \text{ women earn as much as men} \qquad \text{versus} \qquad H_1 : \text{ men earn more than women.}$$

There are of course several ways to model the problem. Here, we work with the difference variable $D_1 = Y_1 - X_1$ and we wish to test a *location parameter*. Two usual choices are the mean and the median. We can translate "women earn as much as men" as :

— *mean version :* "the mean of D_1 is zero" ;

— *median version :* "the median of D_1 is zero".

In other words, if we choose the mean as the location parameter, we test

$$H_0 : \; \mathbb{E}[D_1] = 0 \qquad \text{versus} \qquad H_1 : \; \mathbb{E}[D_1] > 0.$$

If we choose the median as the location parameter, we instead test

$$H_0 : \; \mathrm{Med}(D_1) = 0 \qquad \text{versus} \qquad H_1 : \; \mathrm{Med}(D_1) > 0.$$

If we model the problem using the mean and assume Gaussian data, then we will naturally use a Student's t-test, which is a parametric test.

3.2 A parametric test : Student's t-test

3.2.1 One sample

Let $(X_1, \ldots, X_n)$ be an i.i.d. sample of size n from $\mathcal{N}(\mu, \sigma^2)$, with μ and σ^2 unknown. We want to test

$$H_0 : \; \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \; \mu \neq \mu_0.$$

(or $H_1 : \mu > \mu_0$, or $H_1 : \mu < \mu_0$.)

Student's t-test is based on the statistic

$$\widehat{T} = \sqrt{n}\,\frac{\overline{X} - \mu_0}{S}, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2,$$

and, under H_0, we have $\widehat{T} \sim T_{n-1}$.

At level α, Student's t-test is :

$$\varphi_\alpha(X_1^n) = \mathbf{1}\left\{|\widehat{T}| > q_{1-\alpha/2}\right\} \quad \text{for } H_1 : \mu \neq \mu_0,$$

$$\varphi_\alpha(X_1^n) = \mathbf{1}\left\{\widehat{T} > q_{1-\alpha}\right\} \quad \text{for } H_1 : \mu > \mu_0,$$

$$\varphi_\alpha(X_1^n) = \mathbf{1}\left\{\widehat{T} < q_{\alpha}\right\} \quad \text{for } H_1 : \mu < \mu_0,$$

where q_u denotes the quantile of order u of the Student t distribution with $(n-1)$ degrees of freedom.

Potential issues when carrying out this test in practice :

- the sample is not normally distributed ;
- the variables are Gaussian but not identically distributed (for example $X_i \sim \mathcal{N}(\mu, \sigma_i^2)$ with different σ_i^2) ;
- the sample is contaminated by outliers (atypical observations, "outliers") [a].

Roughly speaking, the non-normality issue is not necessarily serious if the sample size is large (see TD1 exo 5).

[a]. The notion of an outlier is subtle, especially for heavy-tailed distributions. A "low-probability" observation is not necessarily an outlier.

Remarque 3.1

However, if one wants to test the normality of a sample, it is recommended to start with graphical displays, in particular a *Q–Q plot* (or *QQ-plot*). One can then perform one of the many normality tests, for example the Shapiro–Wilk test (often considered one of the most powerful).

Implémentation 3.2

Student's t-test on one sample can be done in R with `t.test` and in Python with `scipy.stats.ttest_1samp`. The Shapiro–Wilk test can be done in R with `shapiro.test` and in Python with `scipy.stats.shapiro`.

3.2.2 Two independent samples

We have two independent samples $U_1, \dots, U_n$ and $V_1, \dots, V_p$, not necessarily of the same size, and we want to test equality of means. We assume that

$$U_1, \dots, U_n \text{ are i.i.d. with distribution } \mathcal{N}(\mu_1, \sigma^2), \qquad V_1, \dots, V_p \text{ are i.i.d. with distribution } \mathcal{N}(\mu_2, \sigma^2), \qquad U_1^n$$

$$\mu_1,\ \mu_2 \text{ and } \sigma^2 \text{ are unknown.}$$

and we want to test :

$$H_0 :\ \mu_1 = \mu_2 \qquad \text{versus} \qquad H_1 :\ \mu_1 \neq \mu_2.$$

(or $H_1 : \mu_1 < \mu_2$ or $H_1 : \mu_1 > \mu_2$).

We then use the statistic

$$T = \frac{\overline{V} - \overline{U}}{\widehat{\sigma}\sqrt{\frac{1}{n} + \frac{1}{p}}},$$

where the *pooled* estimator of the variance is defined by

$$\widehat{\sigma}^2 = \frac{1}{n+p-2}\left[\sum_{i=1}^{n}(U_i - \overline{U})^2 + \sum_{i=1}^{p}(V_i - \overline{V})^2\right].$$

Under H_0, we have

$$T \sim T(n+p-2).$$

Indeed :

- $\sum_{i=1}^{n} \frac{(U_i - \overline{U})^2}{\sigma^2} \sim \chi^2(n-1)$ and $\sum_{i=1}^{p} \frac{(V_i - \overline{V})^2}{\sigma^2} \sim \chi^2(p-1)$;
- these two variables are independent, hence

$$\sum_{i=1}^{n} \frac{(U_i - \overline{U})^2}{\sigma^2} + \sum_{i=1}^{p} \frac{(V_i - \overline{V})^2}{\sigma^2} \sim \chi^2(n+p-2);$$

- $\overline{U} \sim \mathcal{N}\left(\mu_1, \frac{\sigma^2}{n}\right)$ and $\overline{V} \sim \mathcal{N}\left(\mu_2, \frac{\sigma^2}{p}\right)$, and $\overline{U} \perp\!\!\!\perp \overline{V}$;
- therefore, under H_0,

$$\overline{V} - \overline{U} \sim \mathcal{N}\left(0, \sigma^2\left(\frac{1}{n} + \frac{1}{p}\right)\right).$$

Thus, under H_0,

$$\frac{\overline{V}-\overline{U}}{\sigma\sqrt{\frac{1}{n}+\frac{1}{p}}} \sim \mathcal{N}(0,1) \qquad \text{and} \qquad \frac{\widehat{\sigma}^2}{\sigma^2} \sim \frac{\chi^2(n+p-2)}{n+p-2}.$$

Moreover,

$$(U_1-\overline{U},\dots,U_n-\overline{U}) \perp\!\!\!\perp \overline{U}, \qquad (V_1-\overline{V},\dots,V_p-\overline{V}) \perp\!\!\!\perp \overline{V}, \qquad U_1^n \perp\!\!\!\perp V_1^p.$$

Since the vector

$$(U_1-\overline{U},\dots,U_n-\overline{U},\overline{U},\, V_1-\overline{V},\dots,V_p-\overline{V},\overline{V})$$

is Gaussian (as a linear transform of $(U_1,\dots,U_n,V_1,\dots,V_p)$), pairwise independence of its components implies mutual independence. Thus,

$$(U_1-\overline{U},\dots,U_n-\overline{U},\, V_1-\overline{V},\dots,V_p-\overline{V}) \perp\!\!\!\perp (\overline{U},\overline{V}),$$

so $\widehat{\sigma}^2 \perp\!\!\!\perp (\overline{V}-\overline{U})$, and finally

$$T = \frac{\dfrac{\overline{V}-\overline{U}}{\sigma\sqrt{\frac{1}{n}+\frac{1}{p}}}}{\dfrac{\widehat{\sigma}}{\sigma}} \sim T(n+p-2).$$

NB : for more details, see the solution to exercise 2 in TD3 of *statmaths* L3.

At level α, the two-sided test is

$$\varphi_\alpha = \mathbf{1}\left\{|T| > q_{1-\alpha/2}\right\}.$$

For one-sided tests :

$$\varphi_\alpha = \mathbf{1}\{T > q_{1-\alpha}\} \quad \text{for } H_1 : \mu_1 < \mu_2, \qquad \varphi_\alpha = \mathbf{1}\{T < q_\alpha\} \quad \text{for } H_1 : \mu_1 > \mu_2,$$

where q_u denotes the quantile of order u of the Student t distribution with $(n+p-2)$ degrees of freedom.

The same types of issues as for Student's t-test on a one-sample mean arise :

1. the data may not be Gaussian ;
2. the data may be Gaussian but with different variances ;
3. the data may be contaminated by outliers.

Remarque 3.3: Welch's test : $\sigma_1^2 \neq \sigma_2^2$

There exists a test adapted to Gaussian data, close to Student's t-test, but valid when we no longer assume equality of variances (in particular when $\sigma_1^2 \neq \sigma_2^2$ and/or when $n_1 \neq n_2$). This test is called *Welch's test*. The procedure is based on a statistic of the form

$$\frac{\overline{X}-\overline{Y}}{\widehat{\sigma}},$$

but the estimation of the standard deviation $\widehat{\sigma}$ is different since we no longer assume that the variances are equal. The statistic then no longer follows exactly a Student t distribution ; however, it is well approximated by a Student t distribution with a non-integer number of degrees of freedom, computed from s_X^2, s_Y^2, and the sample sizes.
A number of authors indicate that it is unnecessary to test equality of variances before choosing between Welch and Student, and that it is better to use Welch's test directly and systematically.

This is the majority view : on the one hand, Welch's test is more reliable when sample sizes and/or variances differ markedly ; on the other hand, it gives results very similar to Student's test when sizes and variances are close.
The issue of unequal variances is less important for Student's t-test when the sample sizes are approximately equal.

Implémentation 3.4

- R : to perform a Student test (variances assumed equal) or a Welch test with `t.test`, one must specify the argument `var.equal = TRUE` to obtain Student's test. By default (`var.equal = FALSE`), `t.test` performs Welch's test.
- Python : use `scipy.stats.ttest_ind` with the option `equal_var=True` (Student) or `equal_var=False` (Welch). In SciPy, `equal_var=True` is the default.

3.2.3 Paired samples (*paired data*)

"Definition" of pairing

For example, we want to compare the effects of two treatments on two populations of individuals that can be paired.

Let us first explain what pairing is. Concretely, we have two samples **of the same size** :

$$U_1, \ldots, U_n \qquad \text{and} \qquad V_1, \ldots, V_n.$$

We speak of *paired data* when "individual" i in the first sample is linked to "individual" i in the second sample.

Thus, for each i, U_i and V_i are not independent.

On the other hand, we assume independence between pairs (U_i, V_i) for different indices : for example (U_1, V_1) is independent of (U_2, V_2), but U_1 and V_1 are not independent.

Consider the example of a drug treatment. We want to compare the effectiveness of two drugs : U_i and V_i measure the respective effectiveness of drug 1 and drug 2 on two comparable individuals, for example two individuals of the same age. It could also be the same individual, to whom we administered two different treatments at two different times.

In general, when we consider paired samples, this means that
- either U_i and V_i correspond to two measurements on the same individual ;
- or the individuals are different but grouped according to covariates (sex, age, etc.).

Remarque 3.5

In general, pairing is used to improve the power of a test.

A test for paired data consists in testing a location parameter of the difference between the two measurements.

Student's t-test for paired data

We have two paired samples $U_1, \dots, U_n$ and $V_1, \dots, V_n$. We set

$$X_i = U_i - V_i, \qquad i = 1, \dots, n.$$

We assume that

$$X_1, \dots, X_n \text{ are i.i.d. with distribution } \mathcal{N}(\mu, \sigma^2).$$

We want to test

$$H_0 : \ \mu = 0 \qquad \text{versus} \qquad H_1 : \ \mu \neq 0$$

(or $H_1 : \mu > 0$ or $H_1 : \mu < 0$).

We then perform a Student t-test on the sample $\{X_i\}$. More precisely, at level α (two-sided test),

$$\varphi_\alpha(U_1^n, V_1^n) = \mathbf{1}\left\{|\widehat{T}| > q_{1-\alpha/2}\right\},$$

where

$$\widehat{T} = \sqrt{n}\,\frac{\overline{X}}{S_X} = \sqrt{n}\,\frac{\overline{U} - \overline{V}}{S_X}, \qquad S_X^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2.$$

Under H_0, we have $\widehat{T} \sim T(n-1)$.

Remarque 3.6

One of the assumptions is that the distribution of the X_i is the same for all i. In particular, the variance must be the same for all i. Some authors recommend a graphical check of this assumption using a Bland–Altman plot. Frequently, the dispersion is proportional to the level ; a logarithmic transformation can then be useful.

Implémentation 3.7

— R : one can use `t.test` specifying `paired = TRUE` : `t.test(x, y, paired = TRUE)`. Equivalently, one can use `t.test(x - y)`.

— Python : use `scipy.stats.ttest_1samp(x - y, popmean = 0)`.

3.2.4 Importance of the applicability conditions

Nonparametric tests apply in more general situations and are therefore more robust. They are generally used when the applicability conditions for parametric tests are not satisfied (or cannot be verified). However, a parametric test can become effective for large sample sizes, even if the theoretical conditions are not exactly satisfied. In particular, for Student's t-test comparing two populations, when sample sizes are large and under fairly weak conditions, Student's t-test is valid even if the samples are not Gaussian. This result is related to what happens in linear models when the errors are not Gaussian.
More generally, one may nevertheless prefer to use Wilcoxon nonparametric tests systematically (see the next section) when one does not know whether the samples are Gaussian. Indeed, the performance of Wilcoxon tests on Gaussian data is not much worse than that of Student's t-tests. Moreover, Wilcoxon tests often perform better than Student's t-test when the data are not Gaussian (even for large sample sizes).

3.2.5 Numerical illustration – (optional reading)

See the .rmd file in Teams.

3.3 Sign test

We have just seen a parametric test, Student's t-test, which can be used to compare two independent populations or to compare two treatments on paired data. This test relies on the Gaussian nature of the data.

We will now build tests based on much weaker assumptions.

We begin with the sign test and Wilcoxon's signed-rank test, which can be seen as nonparametric versions of Student's t-test for paired data.

We will quite often work with diffuse random variables ; see Definition 2.35 on page 34.

Intuition 3.8

Draw X at random from a continuous distribution with median 0. Then the sign of X is a symmetric variable : $\mathbb{P}(X > 0) = \mathbb{P}(X < 0) = 1/2$. If, on the contrary, repeated draws systematically yield positive (or negative) values, this would indicate that the median is probably not zero. This will be formalized via the sign test.

3.3.1 Sign test for one sample

The goal is to test a location parameter, which here is not the mean but the median. The interest of this choice is that the test does not even require the existence of a mean.

The assumptions on the data $X_1, \dots, X_n$ are :

$$\text{the } X_i \text{ are independent,} \tag{3.1}$$
$$\text{the } X_i \text{ share a common median } m, \tag{3.2}$$
$$\mathbb{P}(X_i = m) = 0,\ \forall i. \tag{3.3}$$

Note that the X_i are not necessarily identically distributed.
The null hypothesis is :

$$H_0 :\ m = 0.$$

Note that $m = 0$ implies here that $\mathbb{P}(X_i > 0) = \mathbb{P}(X_i < 0) = 1/2$ (see (2.2) in Example 2.18, page 29) ; this is where condition 3.3 comes in.

We set

$$Y_i = \mathbb{1}_{\{X_i > 0\}}, \qquad S = \sum_{i=1}^{n} Y_i.$$

One-sided test.

$$H_0 :\ m = 0 \quad \text{versus} \quad H_1 :\ m > 0,$$

1. Let us first, for simplicity of exposition, **assume that the X_i have the same distribution.** We then have

   $$Y_1, \dots, Y_n \overset{iid}{\sim} Be(p) \quad \text{with} \quad p = \mathbb{P}(X_i > 0).$$

   Thus we can rewrite the problem as follows :

   $$H_0 : p = 1/2 \quad \text{versus} \quad H_1 : p > 1/2.$$

   So we reduce to a test of equality for the parameter p of an i.i.d. sample of Bernoulli r.v.'s Y_i. We can therefore use

$$\varphi_\alpha(X_1,\ldots,X_n)=\mathbb{1}_{\left\{S>q_{1-\alpha}^{B(n,1/2)}\right\}}=\mathbb{1}_{\left\{\sum_{i=1}^n \mathbb{1}_{\{X_i>0\}}>q_{1-\alpha}^{B(n,1/2)}\right\}}.$$

2. Now, what happens if the X_i do not have the same distribution? The proposed test still works. Indeed, under H_0 we do have, since 0 is the common median, $Y_i \overset{iid}{\sim} Be(1/2)$, hence the test is indeed of level α.
Is it a test adapted to the problem? This amounts to asking whether the statistic $\sum_{i=1}^n \mathbb{1}_{X_i>0}$ takes large values under H_1. This is the case here because if $m>0$, the X_i tend to take positive values, and thus the number of positive X_i will be large.

Two-sided test.

$$H_0:\ m=0 \quad \text{versus} \quad H_1:\ m\neq 0,$$

Similarly, one can use

$$\varphi_\alpha(Y_1,\ldots,Y_n)=\mathbb{1}_{\left\{|S-n/2|>q_{1-\frac{\alpha}{2}}^{B(n,1/2)}-\frac{n}{2}\right\}}.$$

With the original sample, this gives

$$\varphi_\alpha(X_1,\ldots,X_n)=\mathbb{1}_{\left\{|\sum_{i=1}^n \mathbb{1}_{X_i>0}-n/2|>q_{1-\frac{\alpha}{2}}^{B(n,1/2)}-\frac{n}{2}\right\}}.$$

This test, which uses only the sign of the X_i, is called the *sign test*.

Implémentation 3.9

1. R : one can use `binom.test`. If the sample is in a vector `x` [a] and for the two-sided alternative :

```
binom.test(sum(x > 0), n = length(x), p = 0.5).
```

(By default the alternative is `"two.sided"`.)

2. Python : one can use `binomtest` from `scipy.stats` :

```
from scipy.stats import binomtest
binomtest((x > 0).sum(), n=len(x), p=0.5)   # two-sided by default
```

a. If there are ties, in practice one starts by removing zeros :

```
x <- x[x != 0].
```

3.3.2 Sign test for two samples

We have two paired samples $U_1,\ldots,U_n$ and $V_1,\ldots,V_n$. Since these are paired data, we reduce to a test on the sample of differences $X_i=V_i-U_i$.

To fix ideas, consider the case of two treatments that we want to compare. We take two populations of the same size n. We order them by age. We administer a first treatment to an individual i, whose effectiveness we measure by U_i, and we administer the other treatment to an individual of the same age, whose effectiveness we measure by V_i. For example, we want to know whether the second treatment is more effective than the first.

One can model the fact that the two treatments have the same effectiveness by the equality $\mathbb{P}(U_i\leq V_i)=\mathbb{P}(V_i\leq U_i)$, which gives, in terms of the X_i, $\mathbb{P}(X_i\leq 0)=\mathbb{P}(X_i\geq 0)$. Assuming that, almost surely, the X_i never take the value 0, this again translates into "the common median of the X_i is equal to 0" (cf. Subsection 3.3.1).

We will therefore perform a sign test on the difference sample X_1^n.

If the second treatment is more effective than the first, V_i will typically be larger than U_i, so $U_i < V_i$ (or equivalently $X_i > 0$).

To test "the second drug is more effective than the first" (one-sided alternative), one can use

$$\varphi_\alpha(U_1, \ldots, U_n, V_1, \ldots, V_n) = \mathbb{1}_{\left\{\sum_{i=1}^n \mathbb{1}_{\{U_i < V_i\}} > q_{1-\alpha}^{B(n,1/2)}\right\}}.$$

If, on the contrary, one thinks that the first is more effective, one swaps the roles :

$$\varphi_\alpha(U_1, \ldots, U_n, V_1, \ldots, V_n) = \mathbb{1}_{\left\{\sum_{i=1}^n \mathbb{1}_{\{V_i < U_i\}} > q_{1-\alpha}^{B(n,1/2)}\right\}}.$$

For the two-sided alternative, one uses

$$\varphi_\alpha(X_1, \ldots, X_n) = \mathbb{1}_{\left\{|\sum_{i=1}^n \mathbb{1}_{U_i < V_i} - n/2| > q_{1-\frac{\alpha}{2}}^{B(n,1/2)} - \frac{n}{2}\right\}}.$$

Remarque 3.10

The sign test uses very little information about the variables $U_i - V_i$ (only their sign, not their values). It is therefore a low-power test. Its interest is that it can be applied when only comparisons of the type "greater/smaller" are available.

Remarque 3.11

In practice, if there are ties ($U_i = V_i$), one removes them and replaces n by the number of nonzero pairs.

Implémentation 3.12

Since this is the sign test applied to the variables $X_i = U_i - V_i$, one uses the same procedure as for one sample : compute the differences (or the indicators $\mathbb{1}_{\{U_i < V_i\}}$), possibly remove ties, then apply `binom.test` (or `scipy.stats.binomtest`).

Remarque 3.13: Is the sign test a homogeneity test ?

One might be tempted to use this test to test that two samples have the same distribution (a homogeneity test). Suppose we are given two independent samples $U_1^n = (U_1, \ldots, U_n)$ and $V_1^n = (V_1, \ldots, V_n)$, with $U_i \overset{iid}{\sim} F$ (continuous) and $V_i \overset{iid}{\sim} G$ (continuous). We want to test

$$H_0: \ F = G \qquad \text{versus} \qquad H_1: \ F \neq G.$$

If $F = G$, then U and V are exchangeable : $(U, V) \sim (V, U)$. We deduce that $U - V \sim V - U$, hence $\mathbb{P}(U < V) = \mathbb{P}(V < U)$. Since F and G are continuous and U and V are independent, we have $\mathbb{P}(U = V) = 0$, and therefore

$$\mathbb{P}(U < V) = \mathbb{P}(V < U) = 1/2.$$

Thus, under H_0, the "null property" of the sign test is indeed satisfied.

But beware : the converse is false. One can have

$$\mathbb{P}(U < V) = \mathbb{P}(V < U) = 1/2$$

without U and V having the same distribution. In other words, equality of distributions is a much stronger property than "the median of $V-U$ is zero" (which is what the sign test actually tests).

For example, if U and V are independent, diffuse, and symmetric (about 0), then the median of $V-U$ is equal to 0, even if U and V do not have the same distribution. Thus, the sign test will not detect the difference between a $\mathcal{N}(0,1)$ sample and a Cauchy sample.

Indeed, if U and V are symmetric and independent, $(U,V) \sim (-U,-V)$, hence

$$U - V \sim -U - (-V) = V - U,$$

so $\mathbb{P}(U > V) = \mathbb{P}(V > U)$, and with $\mathbb{P}(U = V) = 0$ we obtain $\mathbb{P}(U > V) = \mathbb{P}(V > U) = 1/2$.

The sign test, viewed as a homogeneity test, is therefore an example of a particularly low-power test : it "does not see" certain alternatives.

3.4 Order and rank statistics

We will use order statistics (see Definition 2.1 on page 24).

Définition 3.14: Rank statistics

There exists a random permutation $\sigma_X \in \mathfrak{S}_n$ such that

$$(X_{(1)}, \dots, X_{(n)}) = (X_{\sigma_X(1)}, \dots, X_{\sigma_X(n)}).$$

We define the rank vector R_X as the inverse permutation of σ_X, that is,

$$R_X = \sigma_X^{-1}.$$

Of course, since one may have $X_i = X_j$ for $i \neq j$, there is not always uniqueness of σ_X nor of the rank vector. In theory, if the X_i are i.i.d. and have a continuous distribution, then there are almost surely no ties (cf. TD1, Exercise 1) and σ_X and R_X are then unique (a.s.).[1]

As its name indicates, the rank vector gives the rank of each observation in the sample.

Example. Let $x_1^7 = (4,2,1,5,3,0,6)$. Then a permutation σ_x and the rank vector R_x are given by :

number	1	2	3	4	5	6	7
σ_x	6	3	2	5	1	4	7

number	1	2	3	4	5	6	7
$R_x = \sigma_x^{-1}$	5	3	2	6	4	1	7

One can also find R_x directly from the observations :

x_1^7	4	2	1	5	3	0	6
R_x	5	3	2	6	4	1	7

1. In practice, due to limited measurement precision and rounding, there may be ties even for a sample drawn from a continuous distribution. One should pay attention to this issue : some procedures assumed valid for continuous data do not always handle ties correctly (and even when they do, the results may be less reliable ; see below).

3.5 Wilcoxon signed-rank test

Intuition 3.15

For a diffuse variable X with zero median, the sign test uses only very weak information : the sign of X. This gives it broad applicability but reduced power.
Often, additional usable information is available, for instance symmetry. Under symmetry about 0, the randomness of X decomposes into two parts : the absolute value $|X|$ and the sign of X. Formally, one can show that $|X|$ and $\mathbb{1}_{\{X>0\}}$ are independent. The absolute value is useful to establish an order (compare magnitudes). Wilcoxon's signed-rank test exploits exactly this idea.

3.5.1 For one sample

Once again we will test the median of a sample. We consider diffuse and independent random variables $(X_1, \dots, X_n)$, but not necessarily identically distributed.

Proposition 3.16

If the random variables $X_1, \dots, X_n$ are independent and diffuse, then

$$\mathbb{P}(\exists\, i \neq j : |X_i| = |X_j|) = 0.$$

Démonstration. For any $i \neq j$,

$$\begin{aligned}\mathbb{P}(|X_i| = |X_j|) &\leq \mathbb{P}(X_i = X_j) + \mathbb{P}(X_i = -X_j)\\ &= \int \mathbb{P}(X_i = x)\, dP_{X_j}(x) + \int \mathbb{P}(X_i = -x)\, dP_{X_j}(x) = 0,\end{aligned}$$

since X_i is diffuse and independent of X_j. Thus,

$$\mathbb{P}(\exists\, i \neq j : |X_i| = |X_j|) \leq \sum_{\substack{1 \leq i,j \leq n \\ i \neq j}} \mathbb{P}(|X_i| = |X_j|) = 0.$$

□

We assume that we have observations $(X_1, \dots, X_n)$ satisfying :

$$\text{the } X_i \text{ are mutually independent}\,; \tag{3.4}$$
$$\text{the } X_i \text{ are diffuse}\,; \tag{3.5}$$
$$\text{the } X_i \text{ share a common median } m; \tag{3.6}$$
$$\text{the distributions of the } X_i \text{ are symmetric about } m. \tag{3.7}$$

Note that we need an additional assumption compared to the sign test : symmetry in distribution around the common median m. In this course, we will assume these conditions hold without discussing how to check them.

The null hypothesis is

$$H_0 : \; m = 0.$$

We will again use the sign of the X_i, but also exploit the values of $|X_i|$. The idea is to assign a larger weight the larger an observation is in absolute value.

If one has access to the values of $|X_i|$, Wilcoxon's signed-rank test is preferable to the sign test, because it uses more information while having applicability conditions that are almost as broad (it requires symmetry).

We consider the ranks of the absolute values $\{|X_i|\}_{1\leq i\leq n}$. Since they are diffuse, we have a.s.

$$|X|_{(1)} < |X|_{(2)} < \cdots < |X|_{(n)}.$$

We denote by $R_{|X|}$ the associated rank vector. We set

$$W_n^+ = \sum_{i=1}^{n} R_{|X|}(i)\mathbb{1}_{\{X_i>0\}}, \qquad W_n^- = \sum_{i=1}^{n} R_{|X|}(i)\mathbb{1}_{\{X_i<0\}}.$$

Exemple 3.17

x_i	$-0{,}15$	$-0{,}42$	$0{,}22$	$0{,}60$	$-0{,}10$
$\lvert x_i\rvert$	$0{,}15$	$0{,}42$	$0{,}22$	$0{,}60$	$0{,}10$
$R_{\lvert X\rvert}(i)$	2	4	3	5	1

Then $w_n^+ = 3+5 = 8$ and $w_n^- = 2+4+1 = 7$.

Remarque 3.18

We have $0 \leq W_n^+ \leq \frac{n(n+1)}{2}$. The case $W_n^+ = 0$ corresponds to the case where $X_i < 0$ for all i; the case $W_n^+ = \frac{n(n+1)}{2}$ corresponds to the case where $X_i > 0$ for all i.
If $\mathbb{P}(X_i = 0) = 0$ (which is the case if the variables are diffuse), then

$$W_n^+ + W_n^- = \frac{n(n+1)}{2}.$$

Suppose, to fix ideas, that $H_1 : m > 0$. Under H_1, there are more positive X_i than negative X_i (same idea as for the sign test). But, due to symmetry, the positive X_i also tend to be larger in absolute value : this is the additional information exploited here. Under this alternative, W_n^+ therefore tends to be large. If the alternative is $H_1 : m < 0$, then on the contrary W_n^+ tends to be small.

Exemple 3.19

Let us take a concrete example. We simulate a sample of size $n = 30$ from the standard Cauchy distribution (i.e. with location parameter $m = 0$), with density $f(x) = \frac{1}{\pi(1+x^2)}$ (in black). We then simulate a sample of the same size from a Cauchy distribution with location parameter $m = 1$, i.e. with density $f(x) = \frac{1}{\pi(1+(x-1)^2)}$. These two samples are shown in Figure 3.1.
We note that :

- in the first sample, there are as many positive values as negative values; by symmetry around 0, the absolute values of the positive x_i do not tend to be larger than those of the negative x_i;
- in the second sample, there are more positive values than negative values; moreover, the positive values tend to receive the highest ranks among the $|x_i|$.

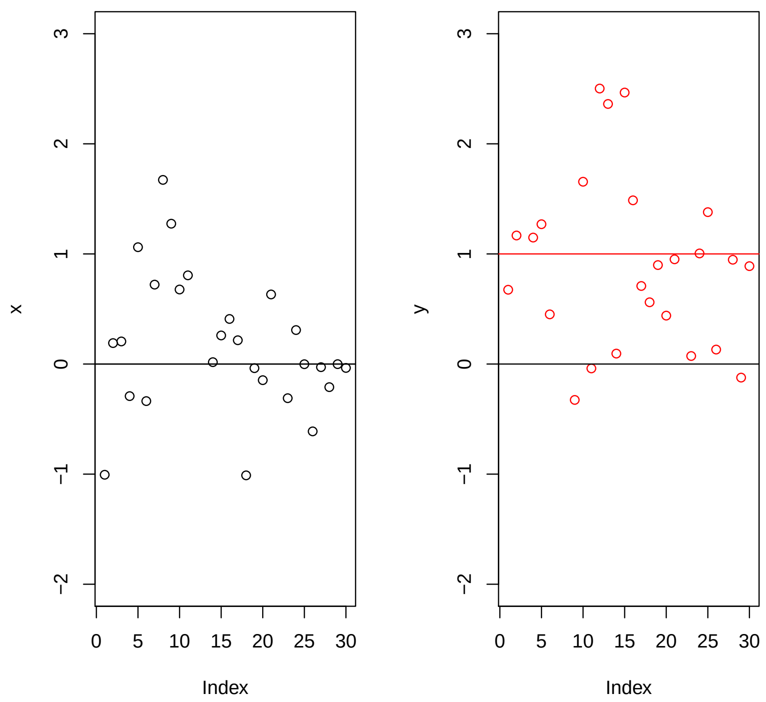


FIGURE 3.1 – Sample from a symmetric distribution centered at $m = 0$ (left) and at $m = 1$ (right).

Exemple 3.20: Necessity of the symmetry condition

We simulate a sample with density

$$f(x) = \frac{1}{2} e^{-x} \mathbb{1}_{\{x>0\}} + \frac{1}{2} \cdot 3e^{3x} \mathbb{1}_{\{x<0\}}.$$

This distribution has median 0 but is not symmetric. The sample is shown in Figure 3.2. We observe that there are roughly as many positive values as negative values, but that the positive x_i tend to take larger absolute values.

Théorème 3.21

Assume that conditions 3.4–3.7 hold. Under $H_0 : m = 0$:

1. $W_n^+ \sim W_n^-$;
2. $\mathbb{E}[W_n^+] = \frac{n(n+1)}{4}$;
3. the distribution of W_n^+ (and of W_n^-) does not depend on the distribution of X_1^n : we have

$$W_n^+ \sim \sum_{k=1}^{n} k\, Y_k, \qquad Y_1, \dots, Y_n \overset{iid}{\sim} Be(1/2);$$

4. $\mathbf{Var}(W_n^+) = \frac{n(n+1)(2n+1)}{24}$;
5. asymptotically,

$$\frac{W_n^+ - \mathbb{E}[W_n^+]}{\sqrt{\mathbf{Var}(W_n^+)}} \xrightarrow{\mathscr{L}} \mathcal{N}(0,1).$$

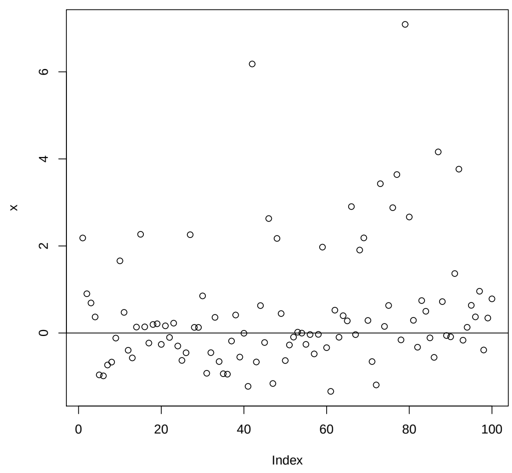


FIGURE 3.2 – Sample from a non-symmetric distribution with median equal to 0.

Démonstration. We admit the proof of item 5. We work under H_0.

1. We can write
$$W_n^+ = \sum_{i=1}^n R_{|X|}(i)\mathbb{1}_{\{X_i>0\}} = \sum_{j=1}^n j\,\mathbb{1}_{\{X_{\sigma_{|X|}(j)}>0\}},$$
where $\sigma_{|X|} = R_{|X|}^{-1}$. Similarly
$$W_n^- = \sum_{j=1}^n j\,\mathbb{1}_{\{X_{\sigma_{|X|}(j)}<0\}}.$$
Since the distributions of the X_i are symmetric and the X_i are independent, we have $X_1^n \sim -X_1^n$. Hence for any measurable function f, $f(X_1^n) \sim f(-X_1^n)$.
Now W_n^+ is a function of the vector X_1^n. Therefore,
$$W_n^+ = \sum_{j=1}^n j\mathbb{1}_{\{X_{\sigma_{|X|}(j)}>0\}} \sim \sum_{j=1}^n j\mathbb{1}_{\{-X_{\sigma_{|-X|}(j)}>0\}}.$$
But $\sigma_{|X|} = \sigma_{|-X|}$, hence
$$W_n^+ \sim \sum_{j=1}^n j\mathbb{1}_{\{-X_{\sigma_{|X|}(j)}>0\}} = \sum_{j=1}^n j\mathbb{1}_{\{X_{\sigma_{|X|}(j)}<0\}} = W_n^-,$$
i.e. W_n^+ and W_n^- have the same distribution.
2. Similarly, symmetry of the law of X_1^n implies that, for every $1 \leq j \leq n$, $X_{\sigma_{|X|}(j)} \sim -X_{\sigma_{|X|}(j)}$ and therefore
$$\mathbb{P}(X_{\sigma_{|X|}(j)} > 0) = \mathbb{P}(X_{\sigma_{|X|}(j)} < 0).$$
Thus, if we show that $\mathbb{P}(X_{\sigma_{|X|}(j)} = 0) = 0$, we will have $\mathbb{P}(X_{\sigma_{|X|}(j)} > 0) = 1/2$, and hence
$$\mathbb{E}(W_n^+) = \sum_{i=1}^n j\mathbb{P}(X_{\sigma_{|X|}(j)} > 0) = \frac{n(n+1)}{4}.$$
Let us show that $X_{\sigma_{|X|}(j)}$ is diffuse. For any $x \in \mathbb{R}$,
$$\mathbb{P}(X_{\sigma_{|X|}(j)} = x) = \sum_{i=1}^n \mathbb{P}(X_i = x, \sigma_{|X|}(j) = i) = 0$$
since the X_i are diffuse.

3. We use symmetry in the same way as in Subsection 1.6.2).
   We will show that
$$(\mathbb{1}_{X_{\sigma_{|X|}(1)}>0}, \dots, \mathbb{1}_{X_{\sigma_{|X|}(n)}>0}) \sim (Y_1, \dots, Y_n)$$
   where
$$Y_1, \dots, Y_n \overset{iid}{\sim} Be(1/2).$$
   Indeed, let $(\epsilon_1, \dots, \epsilon_n) \in \{0,1\}^n$. We have
$$\begin{aligned}&\mathbb{P}\Big((\mathbb{1}_{X_{\sigma_{|X|}(1)}>0}, \dots, \mathbb{1}_{X_{\sigma_{|X|}(n)}>0}) = (\epsilon_1, \dots, \epsilon_n)\Big)\\ &= \sum_{s\in\mathfrak{S}_n} \mathbb{P}\Big((\mathbb{1}_{X_{s(1)}>0}, \dots, \mathbb{1}_{X_{s(n)}>0}) = (\epsilon_1, \dots, \epsilon_n), \sigma_{|X|} = s\Big).\end{aligned}$$
   We will show that for $s \in \mathfrak{S}_n$,
$$\mathbb{P}\Big((\mathbb{1}_{X_{s(1)}>0}, \dots, \mathbb{1}_{X_{s(n)}>0}) = (\epsilon_1, \dots, \epsilon_n), \sigma_{|X|} = s\Big) = \frac{1}{2^n}\mathbb{P}(\sigma_{|X|} = s). \tag{3.8}$$
   Since $X_{s(i)} \sim -X_{s(i)}$ and the $X_{s(i)}$ are independent, we have
$$(X_{s(1)}, \dots, X_{s(n)}) \sim (\pm X_{s(1)}, \dots, \pm X_{s(n)}).$$
   Moreover, the vector $\sigma_{|X|}$ does not change if we change the signs of the $X_{s(i)}$, and since the variables are diffuse we have, a.s., $X_{s(i)} > 0$ or $X_{s(i)} < 0$ for all i.
   Thus, for any $(\tilde{\epsilon}_1, \dots, \tilde{\epsilon}_n) \in \{0,1\}^n$,
$$\begin{aligned}&\mathbb{P}\Big((\mathbb{1}_{X_{s(1)}>0}, \dots, \mathbb{1}_{X_{s(n)}>0}) = (\epsilon_1, \dots, \epsilon_n), \sigma_{|X|} = s\Big)\\ &= \mathbb{P}\Big((\mathbb{1}_{X_{s(1)}>0}, \dots, \mathbb{1}_{X_{s(n)}>0}) = (\tilde{\epsilon}_1, \dots, \tilde{\epsilon}_n), \sigma_{|X|} = s\Big).\end{aligned}$$
   Since
$$\mathbb{P}(\sigma_{|X|} = s) = \sum_{(\tilde{\epsilon}_1,\dots,\tilde{\epsilon}_n)\in\{0,1\}^n} \mathbb{P}\Big(\sigma_{|X|} = s, (\mathbb{1}_{X_{s(1)}>0}, \dots, \mathbb{1}_{X_{s(n)}>0}) = (\tilde{\epsilon}_1, \dots, \tilde{\epsilon}_n)\Big),$$
   we indeed obtain (3.8). Hence
$$\begin{aligned}&\mathbb{P}\Big((\mathbb{1}_{X_{\sigma_{|X|}(1)}>0}, \dots, \mathbb{1}_{X_{\sigma_{|X|}(n)}>0}) = (\epsilon_1, \dots, \epsilon_n)\Big)\\ &= \frac{1}{2^n}\sum_{s\in\mathfrak{S}_n} \mathbb{P}(\sigma_{|X|} = s) = \frac{1}{2^n} = \mathbb{P}((Y_1, \dots, Y_n) = (\epsilon_1, \dots, \epsilon_n)).\end{aligned}$$
   This proves item 3 : $W_n^+ \sim \sum_{j=1}^n jY_j$. It also allows us to compute the variance :
$$\mathbf{Var}\Big(\sum_{j=1}^n j\mathbb{1}_{\{X_{\sigma_{|X|}(j)}>0\}}\Big) = \sum_{j=1}^n j^2\mathbf{Var}(Y_j) = \sum_{j=1}^n \frac{j^2}{4} = \frac{n(n+1)(2n+1)}{24}.$$

□

Remarque 3.22

Under H_0, the statistic W_n^+ has a distribution symmetric about its mean $\frac{n(n+1)}{4}$. Indeed, under H_0, $W_n^+ \sim W_n^-$ and $W_n^+ + W_n^- = \frac{n(n+1)}{2}$, hence
$$W_n^+ \sim \frac{n(n+1)}{2} - W_n^+,$$

that is, $W_n^+ - \frac{n(n+1)}{4} \sim -\Big(W_n^+ - \frac{n(n+1)}{4}\Big)$.

Consequently, for the two-sided alternative $H_1 : m \neq 0$, we reject when $\Big|W_n^+ - \frac{n(n+1)}{4}\Big|$ is large. A level-α test can be written as

$$\varphi_\alpha(X_1, \ldots, X_n) = \mathbb{1}_{\{|W_n^+ - \frac{n(n+1)}{4}| > q_{1-\alpha/2} - \frac{n(n+1)}{4}\}},$$

where $q_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of the distribution of W_n^+ under H_0, i.e. of the distribution of $\sum_{j=1}^n jY_j$ with $Y_1, \ldots, Y_n \overset{iid}{\sim} Be(1/2)$.

Remarque 3.23: Exact or asymptotic test ?

We use the exact distribution of W_n^+ under H_0 when $n \leq 20$. For $n > 20$, we use an asymptotic test based on convergence in distribution.

Implémentation 3.24

To test $H_0 : m = 0$ versus $H_1 : m \neq 0$, if the sample is in a vector `x`, one can use :

```
wilcox.test(x, alternative="two.sided").
```

In Python, one can use :

```
scipy.stats.wilcoxon(x, alternative="two-sided").
```

(In practice one removes zeros : the function often does it automatically, but it is better to check explicitly.)

3.5.2 Paired samples

We have two paired samples of size n, $(U_1, \ldots, U_n)$ and $(V_1, \ldots, V_n)$. As for the sign test, we set $X_i = V_i - U_i$ and apply Wilcoxon's signed-rank test to the sample $(X_1, \ldots, X_n)$. We therefore assume that the X_i satisfy conditions 3.4–3.7.

The null hypothesis is $H_0 : m = 0$. If we think that the plausible alternative is $m > 0$ (i.e. V tends to be larger than U), we take $H_1 : m > 0$.

Remarque 3.25

The signed-rank test is more powerful than the sign test ; it is therefore recommended when one has access to the values of the differences (and when the symmetry assumption is plausible).

Implémentation 3.26

R. One can use `wilcox.test` with `paired=TRUE`. If the samples are stored in `x` and `y` and if $H_1 : m \neq 0$, one writes :

```
wilcox.test(x, y, paired=TRUE, alternative="two.sided").
```

Equivalent : `wilcox.test(x-y, alternative="two.sided")`.

Beware of ties. With ties in $|x_i|$, the exact p-value can no longer be computed and `wilcox.test` switches to a normal approximation ; one then typically gets : `Warning message: cannot compute exact p-value with ties`.
The normal approximation is reliable mainly for sufficiently large sample sizes ; for small n with many ties, the results should be interpreted with caution.

Python. One can use `scipy.stats.wilcoxon(x, y)` or `scipy.stats.wilcoxon(x-y)`. By default, ties are removed ; this behavior can be configured.

Remarque 3.27

To test the symmetry assumption for the X_i (in the i.i.d. case), one can start by plotting the data (histogram, density), or use a symmetry test (for example `symmetry.test` from the `lawstat` package, which can be somewhat slow). In case of strong asymmetry in the difference sample, it seems preferable to use the sign test.

3.6 Wilcoxon rank-sum test (Mann–Whitney)

Intuition 3.28

We saw in Intuition 3.15 (page 54) that Wilcoxon's signed-rank test uses the independence between absolute value and sign.
For an i.i.d. sample $(X_1, \ldots, X_n)$, another independence can be exploited : that between the order statistic $X^* = (X_{(1)}, \ldots, X_{(n)})$ and the rank vector R_X. Intuitively, one way to generate the sample $(X_1, \ldots, X_n)$ is to first draw the sorted values X^*, then draw a random permutation (the rank vector) that reassigns these values to indices. This independence is used in Wilcoxon's rank-sum test (also called the Mann–Whitney test, or Wilcoxon–Mann–Whitney test).

3.6.1 Preliminary results on the rank vector

We begin with a few results related to the rank vector in the i.i.d. case.

Théorème 3.29

Let $X_1, \ldots, X_n$ be i.i.d. random variables with a continuous distribution, with order statistic X^* and rank vector R_X. Then X^* and R_X are independent and, moreover, R_X is uniformly distributed over $\mathfrak{S}_n$.

Démonstration. The distribution is continuous, so a.s. there are no ties. The rank R_X takes values in $\mathfrak{S}_n$. Since R_X is the inverse permutation of σ_X, it actually suffices to show that σ_X is uniformly distributed over $\mathfrak{S}_n$ and that σ_X and X^* are independent.
Since the X_i are i.i.d., they are exchangeable. Hence, for any $s \in \mathfrak{S}_n$,

$$\mathbb{P}(\sigma_X = s) = \frac{1}{\mathbf{Card}(\mathfrak{S}_n)} = \frac{1}{n!}.$$

For example, for $n = 3$, we have

$$\begin{aligned}\mathbb{P}(X_1 < X_2 < X_3) &= \mathbb{P}(X_1 < X_3 < X_2) = \mathbb{P}(X_2 < X_1 < X_3) = \mathbb{P}(X_2 < X_3 < X_1)\\ &= \mathbb{P}(X_3 < X_2 < X_1) = \mathbb{P}(X_3 < X_1 < X_2) = 1/6.\end{aligned}$$

Now, the first term is nothing but $\mathbb{P}(\sigma_X = (1,2,3))$, the second is $\mathbb{P}(\sigma_X = (1,3,2))$, the third $\mathbb{P}(\sigma_X = (2,1,3))$, etc.
Let us prove independence between σ_X and X^*.
We want to show that, for any Borel set B of $\mathbb{R}^n$ and any permutation s in $\mathfrak{S}_n$,

$$\mathbb{P}(X^* \in B,\ \sigma_X = s) = \mathbb{P}(X^* \in B)\mathbb{P}(\sigma_X = s).$$

And since $\mathbb{P}(\sigma_X = s) = \frac{1}{n!}$, this amounts to showing that, for any $s \in \mathfrak{S}_n$,

$$\mathbb{P}(X^* \in B) = n!\mathbb{P}(X^* \in B,\ \sigma_X = s).$$

Since the X_i are independent and identically distributed, they are exchangeable, and thus, for any s and any B,

$$\begin{aligned}\mathbb{P}(X^* \in B,\ \sigma_X = s) &= \mathbb{P}\Big(X_{s(1)} < \cdots < X_{s(n)},\ (X_{s(1)}, \ldots, X_{s(n)}) \in B\Big) \\ &= \mathbb{P}(X_1 < \cdots < X_n,\ (X_1, \ldots, X_n) \in B).\end{aligned}$$

On the other hand,

$$\begin{aligned}\mathbb{P}(X^* \in B) &= \sum_{s \in \mathfrak{S}_n} \mathbb{P}(X^* \in B,\ \sigma_X = s) \\ &= \sum_{s \in \mathfrak{S}_n} \mathbb{P}(X_1 < \cdots < X_n,\ (X_1, \ldots, X_n) \in B) \\ &= n!\, \mathbb{P}(X_1 < \cdots < X_n,\ (X_1, \ldots, X_n) \in B) \\ &= n!\, \mathbb{P}(X^* \in B,\ \sigma_X = s),\end{aligned}$$

where s is any fixed permutation. □

The main consequence is that the distribution of R_X does not depend on the distribution of the X_i (as long as it is continuous). It follows that any statistic depending only on the ranks of i.i.d. continuous observations has a distribution that is independent of the underlying distribution.

Remarque 3.30

For any (non-random) s in $\mathfrak{S}_n$, we have

$$(X_{s(1)}, \ldots, X_{s(n)}) \sim (X_1, \ldots, X_n).$$

But this is not true if the permutation is random, unless it is independent of the X_i. For example,

$$X^* = (X_{(1)}, \ldots, X_{(n)}) \nsim (X_1, \ldots, X_n).$$

Proposition 3.31

Let $X_1, \ldots, X_n$ be i.i.d. random variables with a continuous distribution and rank vector $R_X = (R_1, \ldots, R_n)$. For any integer s such that $1 \le s \le n$, and for any sequence of distinct integers $(r_1, \ldots, r_s)$ in $\{1, \ldots, n\}$, we have

$$\mathbb{P}((R_1, \ldots, R_s) = (r_1, \ldots, r_s)) = \frac{1}{n(n-1)\cdots(n-s+1)}.$$

In particular, for any $i \in \{1, \ldots, n\}$, R_i is uniformly distributed on $\{1, \ldots, n\}$.

Démonstration. For simplicity, let us first consider three simple cases.

— $s = n$: $\mathbb{P}((R_1, \ldots, R_n) = (r_1, \ldots, r_n)) = \mathbb{P}(R = r)$ where $r = (r_1, \ldots, r_n) \in \mathfrak{S}_n$. Thus, by the previous theorem,
$$\mathbb{P}((R_1, \ldots, R_n) = (r_1, \ldots, r_n)) = \frac{1}{n!}.$$

— $s = 1$: $\mathbb{P}(R_1 = r) = \mathbb{P}(\text{“}X_1 \text{ is the } r\text{-th smallest element of the sample”}) = \frac{1}{n}$, again because the X_i are exchangeable.

— $s = 2$: then
$$\mathbb{P}((R_1, R_2) = (r_1, r_2)) = \mathbb{P}(R_1 = r_1)\mathbb{P}(R_2 = r_2 \mid R_1 = r_1) = \frac{1}{n}\frac{1}{n-1}.$$

The general case follows from the same idea (sequential construction). □

3.6.2 Definition and properties of the Mann–Whitney test

We are given two samples U_1^n and V_1^p such that :

1. $U_1, \dots, U_n \overset{iid}{\sim} U$ and $V_1, \dots, V_p \overset{iid}{\sim} V$;
2. U_1^n and V_1^p are independent ;
3. U and V are diffuse.

The samples need not have the same size.
Let F be the cdf of U and G that of V. The null hypothesis is

$$H_0 : \ F = G.$$

However, unlike the K–S homogeneity test, the alternative is **not** $F \neq G$.

We in fact assume that :

— either U and V have the same distribution (H_0),

— or one of the two variables tends to take larger values than the other (H_1).

Remarque 3.32

The test is often used against alternatives of the “location shift” type. This is why, in R, one reads `alternative hypothesis: true location shift is not equal to 0`.

Exemple 3.33

We want to test a new drug against an old one. We administer the first to a group of n people and the second to a group of p people, the two groups being independent. We want to see whether the new drug is more effective than the old one.

Remarque 3.34

Student’s test can be viewed as a test of equality in distribution when we assume the data are Gaussian and can differ only through their mean. In this sense, the Mann–Whitney test can be seen as a nonparametric (and more robust) version of the comparison of two independent populations.

We pool the two samples to form a single sample of size $N = n + p$:

$$(U_1, \dots, U_n, V_1, \dots, V_p).$$

We then rank all observations by their global rank in this pooled sample. We obtain a rank vector, denoted $R_{U,V}$. We denote by $R_1, \dots, R_n$ the ranks associated with the U_i and by $S_1, \dots, S_p$ the ranks associated with the V_j.

Exemple 3.35

Let $u_1 = 3.5$, $u_2 = 4.7$, $u_3 = 1.2$, $v_1 = 0.7$, $v_2 = 3.9$. Then $v_1 < u_3 < u_1 < v_2 < u_2$. Hence $r_1 = 3$, $r_2 = 5$, $r_3 = 2$, $s_1 = 1$, $s_2 = 4$.

We set

$$\Sigma_1 = R_1 + \cdots + R_n, \qquad \Sigma_2 = S_1 + \cdots + S_p.$$

Intuition 3.36

To fix ideas, first assume $n = p$. Under H_0, we expect Σ_1 and Σ_2 to be close. If the alternative corresponds to the fact that the U_i tend to be larger than the V_j, then the ranks R_i of the U_i will be larger on average than the ranks S_j of the V_j. Thus Σ_1 will tend to be "abnormally large" under H_1.
Even if $n \neq p$, the same phenomenon persists : under H_1, Σ_1 tends to be too large (or too small) relative to what happens under H_0.

More generally, if we have no prior on the direction of the effect, we expect Σ_1 to be abnormally large or abnormally small (relative to H_0).

The question becomes : which values of Σ_1 are "normal" under H_0 ? The following results answer this question.

Proposition 3.37

We have

$$\frac{n(n+1)}{2} \leq \Sigma_1 \leq np + \frac{n(n+1)}{2}, \qquad \frac{p(p+1)}{2} \leq \Sigma_2 \leq np + \frac{p(p+1)}{2}.$$

Under $H_0 : F = G$, and under conditions 1–3, we have, for all i and all j,

$$\mathbb{E}(R_i) = \mathbb{E}(S_j) = \frac{n+p+1}{2}, \qquad \mathbf{Var}(R_i) = \mathbf{Var}(S_j) = \frac{(n+p)^2 - 1}{12}.$$

Moreover,

$$\mathbb{E}(\Sigma_1) = \frac{n(n+p+1)}{2}, \qquad \mathbb{E}(\Sigma_2) = \frac{p(n+p+1)}{2},$$

and

$$\mathbf{Var}(\Sigma_1) = \mathbf{Var}(\Sigma_2) = \frac{np(n+p+1)}{12}.$$

In view of the proposition, we naturally consider :

$$M_U = \Sigma_1 - \frac{n(n+1)}{2}, \qquad M_V = \Sigma_2 - \frac{p(p+1)}{2},$$

both of which take values in $\{0, 1, \ldots, np\}$.

Proposition 3.38

Assume that conditions 1–3 hold. Then :

1. $M_U + M_V = np$ a.s. ;
2. under $H_0 : F = G$, the distribution of M_V is symmetric about $\frac{np}{2}$;
3. under $H_0 : F = G$, $M_U \sim M_V$;
4. M_V equals the number of pairs (U_i, V_j) such that $U_i < V_j$.

Démonstration. 1. The sum of the ranks of all $N = n+p$ observations equals $\sum_{k=1}^{N} k = \frac{N(N+1)}{2}$. Hence

$$M_U + M_V = \Sigma_1 - \frac{n(n+1)}{2} + \Sigma_2 - \frac{p(p+1)}{2} = \frac{(n+p)(n+p+1)}{2} - \frac{n(n+1)}{2} - \frac{p(p+1)}{2} = np.$$

2. Under H_0, consider the ranks $(S'_1, \ldots, S'_p)$ of the V_j when we order the pooled sample in

decreasing order. We have $(S'_1, \dots, S'_p) \sim (S_1, \dots, S_p)$ (same argument as in Proposition 3.31), hence $\Sigma'_2 \sim \Sigma_2$ where $\Sigma'_2 = \sum_{j=1}^p S'_j$. Now, for each j, $S'_j = N + 1 - S_j$, so

$$\Sigma'_2 = (N+1)p - \Sigma_2.$$

Thus,

$$\Sigma_2 \sim (N+1)p - \Sigma_2,$$

which implies

$$M_V = \Sigma_2 - \frac{p(p+1)}{2} \sim (N+1)p - \Sigma_2 - \frac{p(p+1)}{2} = np - M_V.$$

Therefore M_V is symmetric about $np/2$.

3. Combine the first two items.
4. Work under H_0. To interpret M_V, assume without loss of generality that $v_1 < \cdots < v_p$. For each j, there are $S_j - j$ elements of the first sample that are strictly smaller than v_j. Hence the total number of pairs (u_i, v_j) such that $u_i < v_j$ is

$$\sum_{j=1}^{p} (S_j - j) = \Sigma_2 - \frac{p(p+1)}{2} = M_V.$$

□

Théorème 3.39

Under conditions 1–3 :

— under H_0, the distribution of M_U (and hence of M_V) is distribution-free : it depends only on n and p ;

— asymptotically, as $n, p \to +\infty$,

$$\frac{M_U - \mathbb{E}(M_U)}{\sqrt{\mathbf{Var}(M_U)}} \xrightarrow{\mathscr{L}} \mathcal{N}(0,1),$$

and

$$\mathbb{E}(M_U) = \frac{np}{2}, \qquad \mathbf{Var}(M_U) = \frac{np(n+p+1)}{12}.$$

Démonstration. We admit the convergence in distribution.
The variable $M_U = \Sigma_1 - \frac{n(n+1)}{2}$ is a function of the ranks $(R_1, \dots, R_n)$. Under H_0, the distribution of these ranks is given by Proposition 3.31 : for any integer sequence $(r_1, \dots, r_n)$ with values in $[N]$, we have

$$\mathbb{P}_{F=G}\big((R_1, \dots, R_n) = (r_1, \dots, r_n)\big) = \frac{1}{N(N-1)\cdots(N-n+1)}.$$

It therefore does not depend on F but only on n and p.
The expectation and variance formulas follow from those of Σ_1 (Proposition 3.37). □

Remarque 3.40: Exact or asymptotic test

For small n and p, one can use the exact distribution (tabulated / exact algorithms). For larger sample sizes, one uses the Gaussian approximation.

Remarque 3.41: Continuity correction

Assume that the test statistic T_n takes discrete values (for instance integers) but that, for large n, its distribution is approximated by a Gaussian distribution, hence continuous. Then, for $p \in \mathbb{N}$ and $u \in [0,1)$, we have $\mathbb{P}(T_n \geq p) = \mathbb{P}(T_n \geq p-u)$. The continuity correction consists in replacing p by $p \pm 0.5$ in the Gaussian approximation. More precisely, if

$$a_n(T_n - t_n) \xrightarrow{\mathscr{L}} \mathcal{N}(0,1),$$

we approximate as follows :

$$\mathbb{P}(T_n \geq p) = \mathbb{P}(a_n(T_n - t_n) \geq a_n(p - t_n)) \approx 1 - \Phi(a_n(p - 0.5 - t_n))$$

$$\mathbb{P}(T_n \leq p) = \mathbb{P}(a_n(T_n - t_n) \leq a_n(p - t_n)) \approx \Phi(a_n(p + 0.5 - t_n))$$

Implémentation 3.42

— R : we use `wilcox.test` with `paired = FALSE` (the default). For instance :

```
wilcox.test(salaires$hommes, salaires$femmes, alternative = "greater")
```

the `exact` argument indicates whether we want the exact test or the Gaussian approximation. This argument defaults to `true` if one of the samples has size greater than 50, and to `false` otherwise. The `correct` argument indicates whether we want the continuity correction when using the Gaussian approximation. It defaults to `TRUE`.

— Python : `scipy.stats.mannwhitneyu(x, y, alternative='two-sided')`.

Remarque 3.43

In the same way that Student's test for comparing two means generalizes to more than two samples via one-way ANOVA, the nonparametric analogue of the Mann–Whitney test for more than two samples (possibly of different sizes) is the Kruskal–Wallis test.

Implémentation 3.44: Kruskal–Wallis

— R : `kruskal.test`

— Python : `scipy.stats.kruskal`

Remarque 3.45

Besides being applicable to a wider range of distributions, rank-based tests are more robust to the presence of outlying observations (*outliers*) in the sample.

Remarque 3.46: Parametric or nonparametric ?

Often, if the parametric model is correct, parametric tests are more powerful than nonparametric tests. However, they are also more demanding, since more assumptions/conditions of use must be checked. One will generally choose a nonparametric test when :

— the assumptions required for the parametric test are not satisfied ;

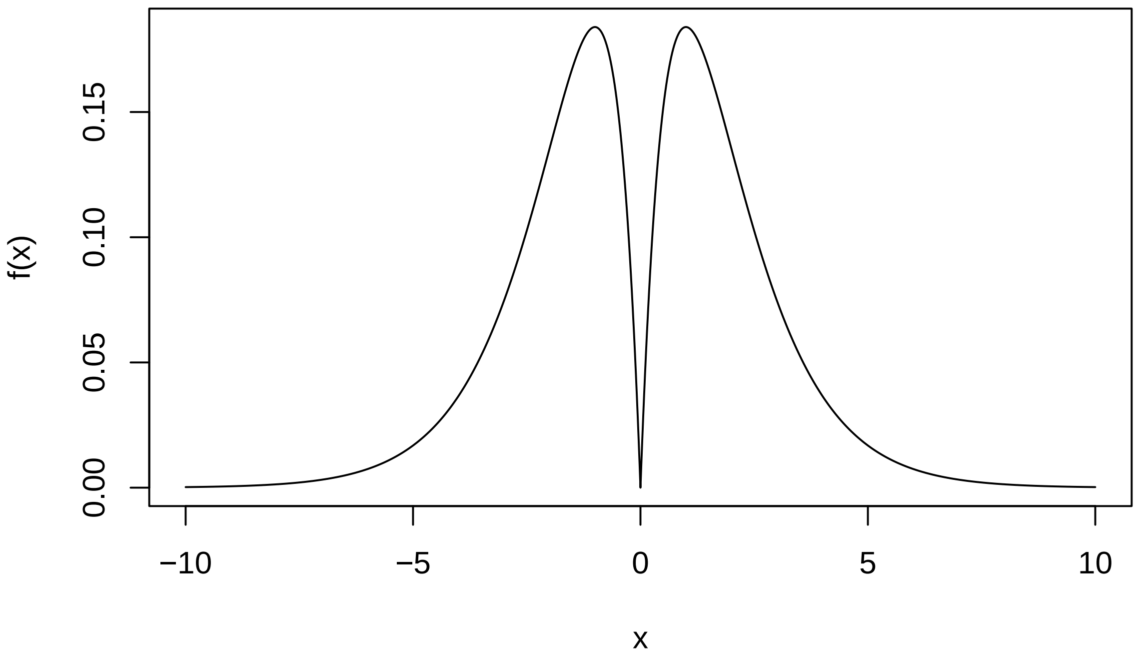
— or it is impossible to check these assumptions.

Remarque 3.47: Comparing the KS test and the Mann–Whitney (MW) test

The KS test is sensitive to any change between two distributions (shape, spread, location, etc.). In contrast, the MW test is mainly sensitive to a location shift (and more generally to a stochastic ordering), hence different behaviors (see the illustration).

Illustration: We compare below the respective performance of the Kolmogorov–Smirnov and Mann–Whitney tests in a particular case. More precisely, we use two samples that do not have the same distribution: one is Gaussian, the other is a mixture based on a (symmetrized) Gamma distribution, whose density is plotted below. Suppose we use KS and MW to test equality of distributions on these two samples. We look at each test's p-value as well as the empirical power for tests at level 95%.

```
set.seed(1)

n <- 100
B <- 1000
alpha <- 0.05

# density of the symmetrized Gamma(2,1)
f <- function(x) 0.5 * dgamma(abs(x), shape = 2, rate = 1)

x <- seq(-10, 10, by = 0.01)
plot(x, f(x), type = "l")
```



A function that simulates two samples from each distribution, performs the two tests on these samples, and stores the p-values:

```
one_run <- function() {
  z <- rnorm(n)
  t <- rgamma(n, shape = 2, rate = 1)
  rad <- sample(c(-1, 1), n, replace = TRUE)    # Rademacher signs
  y <- rad * t

  c(
    KS = ks.test(y, z)$p.value,
    MW = wilcox.test(y, z, exact = FALSE)$p.value
  )
}
```

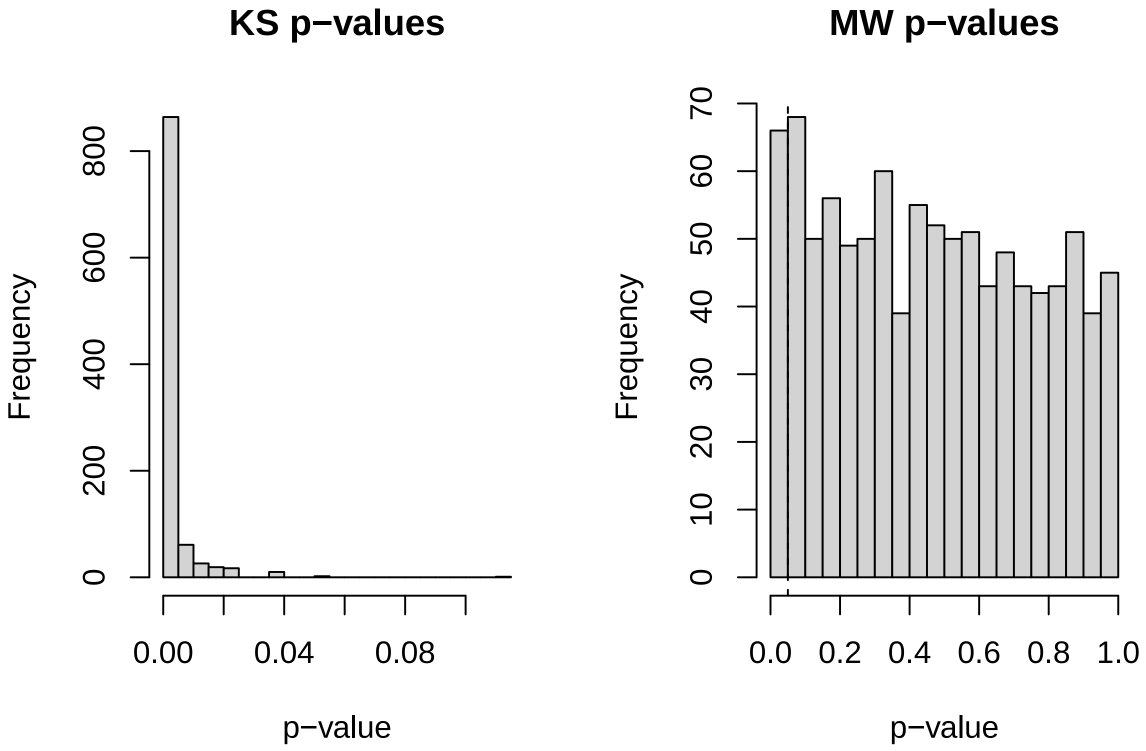
We repeat this experiment B=1000 times, compute the empirical power, and display histograms of the

p-values:

```
pvals <- replicate(B, one_run())
KS <- pvals["KS", ]
MW <- pvals["MW", ]

power_KS <- mean(KS <= alpha)
power_MW <- mean(MW <= alpha)


cat("Empirical power at alpha =", alpha,
    ": KS =", power_KS,
    ", MW =", power_MW, "\n")
```

```
## Empirical power at alpha = 0.05 : KS = 0.997 , MW = 0.066
```

```
par(mfrow=c(1,2))
hist(KS, breaks=30, main="KS p-values", xlab="p-value")
hist(MW, breaks=30, main="MW p-values", xlab="p-value")
abline(v=0.05, lty=2)
```



Chapitre 4

Density estimation via kernel estimators

4.1 Reminders

Definition of differentiability : let $\ell : \mathbb{R}^m \to \mathbb{R}^p$. The map ℓ is differentiable at u if there exists a linear map $D\ell(u) : \mathbb{R}^m \to \mathbb{R}^p$ (which can therefore be represented by a matrix in $M_{pm}(\mathbb{R})$) such that

$$\forall \epsilon > 0,\ \exists \delta > 0 :\ \|x-u\| \leq \delta \implies \|\ell(x) - \ell(u) - D\ell(u)(x-u)\| \leq \epsilon \|x-u\|.$$

Taylor–Lagrange formula : let $f : [a,b] \to \mathbb{R}$, where $[a,b]$ is an interval of $\mathbb{R}$. Assume that f is of class $\mathcal{C}^{n-1}$ on $[a,b]$ and n times differentiable on $]a,b[$. Then there exists $\eta \in]0,1[$ such that

$$f(b) = \sum_{k=0}^{n-1} f^{(k)}(a)\frac{(b-a)^k}{k!} + f^{(n)}(a+\eta(b-a))\frac{(b-a)^n}{n!}.$$

Taylor formula with integral remainder : assume that $f \in \mathcal{C}^n$ on $[a,b]$. Then

$$f(b) = \sum_{k=0}^{n-1} f^{(k)}(a)\frac{(b-a)^k}{k!} + \int_a^b \frac{(b-t)^{n-1}}{(n-1)!} f^{(n)}(t)\, dt.$$

4.2 Introduction

The goal of this chapter is to estimate a density f using an i.i.d. sample of size n, $X_1^n = (X_1, \ldots, X_n)$, where X_i has density f with respect to Lebesgue measure.

Measuring the quality of an estimator :

1. Definition of a distance on the space of functions :
   - L_p distance : $d(f,g) = \|f-g\|_p = \left(\int |f(x)-g(x)|^p\, dx\right)^{\frac{1}{p}}$.
     The usual cases are $p=2$ or $p=1$.
   - L_∞ distance : $d(f,g) = \|f-g\|_\infty = \sup_{x\in\mathbb{R}} |f(x)-g(x)|$.
   - Pointwise distance at x_0 : $d(f,g) = |f(x_0)-g(x_0)|$.
2. Definition of a loss function $\omega : \mathbb{R} \to \mathbb{R}^+$ such that ω is convex and $\omega(0)=0$.
   Example : $\omega(x) = x^2$.

3. Definition of the risk of an estimator $\hat{f}_n$:

$$R(\hat{f}_n, f) = \mathbb{E}[\omega(d(\hat{f}_n, f))],$$

where $\mathbb{E}$ denotes expectation under the law of the X_i.

Common examples :

— $d(f,g) = |f(x_0) - g(x_0)|, \quad \omega(x) = x^2$:

$$R(\hat{f}_n, f) = \mathbb{E}\big[|\hat{f}_n(x_0) - f(x_0)|^2\big].$$

— $d(f,g) = \|f - g\|_2, \quad \omega(x) = x^2$:

$$R(\hat{f}_n, f) = \mathbb{E}\big[\|\hat{f}_n - f\|_2^2\big].$$

We seek to determine $\hat{f}_n$ such that $R(\hat{f}_n, f)$ is minimal. We do not assume that the density f belongs to a parametric family. Instead, we make less restrictive assumptions : f belongs to a function class, denoted by $\mathcal{F}$. One can then define a risk, called the minimax risk of $\hat{f}_n$ over the class $\mathcal{F}$, by

$$R(\hat{f}_n, \mathcal{F}) = \sup_{f\in\mathcal{F}} R(\hat{f}_n, f).$$

We will therefore look for an estimator $\hat{f}_n$ such that the risk $R(\hat{f}_n, \mathcal{F})$ tends to 0 as fast as possible when $n \to \infty$.

Définition 4.1

Let $(r_n)_n$ be a sequence and C a constant such that

$$\forall n \quad R(\hat{f}_n, \mathcal{F}) \leq C r_n.$$

We say that the sequence of estimators $(\hat{f}_n)_n$ achieves the rate (or speed) r_n over the class $\mathcal{F}$, for the distance d and the loss ω.

We will see that the rate will be faster when the class $\mathcal{F}$ has higher regularity.
Examples of function classes : $\mathcal{C}^k$, the Hölder class (see the definition below).
If $\beta \in \mathbb{R}$, we denote by $\lfloor\beta\rfloor$ the natural integer equal to the greatest integer strictly less than β.
Example : if $\beta = 3.5$ then $\lfloor\beta\rfloor = 3$, and if $\beta = 4$ then $\lfloor\beta\rfloor = 3$.

Définition 4.2: (not required)

For any $\beta > 0$ and any $L > 0$, we define the Hölder class of regularity β and radius L by

$$\Sigma(\beta, L) = \{g : \mathbb{R} \to \mathbb{R} \text{ such that } g \text{ is } \lfloor\beta\rfloor \text{ times differentiable and}$$
$$\forall (x,y) \in \mathbb{R}^2 \quad |g^{(\lfloor\beta\rfloor)}(y) - g^{(\lfloor\beta\rfloor)}(x)| \leq L|x-y|^{\beta-\lfloor\beta\rfloor}\}.$$

We denote by $\Sigma_d(\beta, L)$ the intersection of $\Sigma(\beta, L)$ with the set of densities.

Remarque 4.3

— If $\beta = 1$ we obtain the set of Lipschitz functions.

— If $\beta > 1$, then $g' \in \Sigma(\beta - 1, L)$.

Proposition 4.4: (admitted and not required)

Let $\beta > 0$ and $L > 0$. There exists a constant $M(\beta, L)$ such that

$$\sup_{f \in \Sigma_d(\beta,L)} \|f\|_\infty = \sup_{x \in \mathbb{R}} \sup_{f \in \Sigma_d(\beta,L)} f(x) \leq M(\beta, L).$$

4.3 Nonparametric density estimation

The classical approach to estimating a density is to posit a parametric family. For instance, in dimension 1, one may start by visualizing the data using a histogram : if it has a bell shape and light tails, it is natural to consider a Gaussian model. One then only needs to estimate its parameters (μ, σ^2), that is, a vector of dimension 2. In other situations, prior knowledge (physical mechanisms, symmetry constraints, positivity, etc.) may also lead one to favor a specific parametric family.

This approach nevertheless has several limitations. In higher dimension, it becomes difficult to represent the data and to "visually recognize" a standard distribution. Moreover, one does not always have reliable prior information to guide model choice.

Above all, a poor choice of model can lead to a biased estimate of the density and, consequently, to a misleading interpretation of the data.

By contrast, nonparametric methods rely on weaker assumptions and offer greater flexibility : they aim to adapt to the shape of the density without imposing an overly restrictive structure a priori.

Finally, one should keep in mind that if the parametric model is correct—or, more generally, if it provides a good approximation of reality—then it is often more efficient than a nonparametric model.

4.3.1 A simple density estimator : the histogram

Assume, for simplicity, that we work in dimension 1 and that the observations $X_1, \ldots, X_n$ take values in $[0, 1]$. We therefore seek to estimate a density

$$f : [0, 1] \to \mathbb{R}_+.$$

We choose a partition of $[0, 1]$ into p bins

$$]a_1, a_2],\ ]a_2, a_3],\ \ldots,\ ]a_p, a_{p+1}],$$

and, for simplicity, we assume that these intervals all have the same length :

$$a_{j+1} - a_j = h, \qquad j = 1, \ldots, p,$$

where $h > 0$ is the bin width.

The histogram method consists in approximating f by a function *constant on each bin*, this constant being determined by the proportion of observations X_i that fall in the bin. More precisely, for $t \in]a_j, a_{j+1}]$, we define

$$\widehat{f}_n(t) = \frac{1}{nh} \#\{\, i \in \{1, \ldots, n\} : X_i \in]a_j, a_{j+1}]\, \}.$$

This formula is easily explained. If we assume that f is constant on $]a_j, a_{j+1}]$, equal to some value c_j, then

$$F(a_{j+1}) - F(a_j) = \int_{a_j}^{a_{j+1}} f(u)\, du = c_j\, h.$$

Now $F(a_{j+1}) - F(a_j) = \mathbb{P}(X \in]a_j, a_{j+1}])$ can be estimated by the empirical frequency

$$\frac{1}{n} \#\{\, i : X_i \in]a_j, a_{j+1}]\, \}.$$

Hence
$$c_j = \frac{F(a_{j+1}) - F(a_j)}{h} \;\approx\; \frac{1}{nh}\,\#\{\, i : X_i \in]a_j, a_{j+1}]\,\},$$
which leads exactly to the definition of $\widehat{f}_n$.

Finally, the performance of this estimator depends strongly on the choice of p (or, equivalently, of the width h) : too few bins yields an overly coarse estimate, whereas too many bins increases variability.

R code and graphical illustration of the choice of the number of bins

We illustrate the importance of choosing the number of bins (or, equivalently, the bin width h) for the histogram estimator, using an example of a bimodal density.

We simulate a mixture of two Gaussian distributions $\mathcal{N}(2,1)$ and $\mathcal{N}(6,1)$ with weights 1/2–1/2. The density is therefore

$$f(x) = \frac{1}{2}\,\varphi(x;2,1^2) + \frac{1}{2}\,\varphi(x;6,1^2), \qquad \varphi(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big).$$

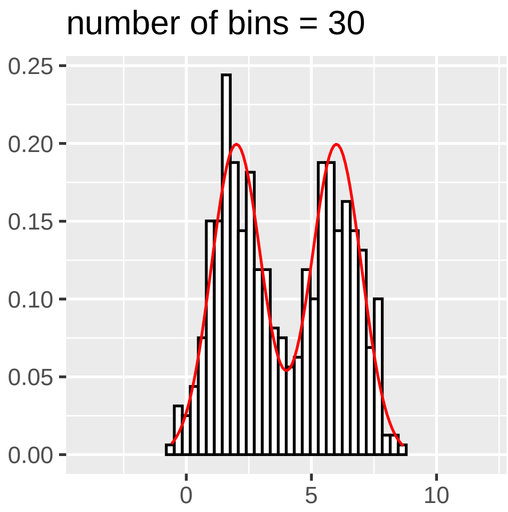
If the histogram approximation is satisfactory, we should observe two "bells" (standard deviation 1) centered at 2 and 6, with a slight overlap.

Simulating a sample $(n = 500)$

```
f <- function(x) {
  0.5 * dnorm(x, mean = 2, sd = 1) + 0.5 * dnorm(x, mean = 6, sd = 1)
}

sim <- function(n) {
  m <- sample(c(2, 6), size = n, replace = TRUE) # choose the center
  rnorm(n, mean = m, sd = 1)
}

n <- 500
Z <- sim(n)
df <- data.frame(x = Z)
```

We estimate the density using a histogram and overlay the true density f (in red). We explicitly set `bins = 30` (this is the default choice in `ggplot2`).

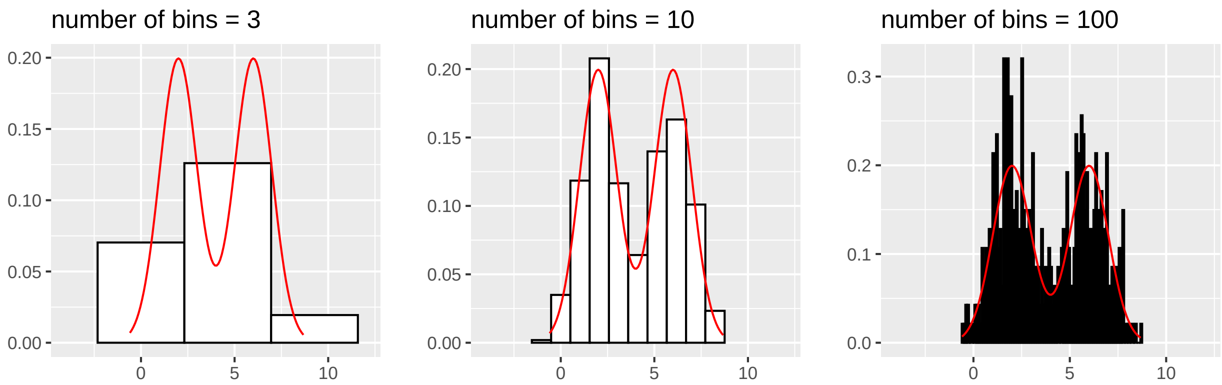
```
xlim <- c(-4, 12)

p_base <- ggplot(df, aes(x)) +
  labs(x = "", y = "") +
  coord_cartesian(xlim = xlim)

p_base +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30, color = "black", fill = "white") +
  stat_function(fun = f, colour = "red") +
  labs(title = "number of bins = 30")
```



Effect of the number of bins

The number of bins (or the width h) is a crucial parameter: it controls the bias–variance trade-off. Let us compare several choices.

```
plot_hist <- function(bins) {
  p_base +
    geom_histogram(aes(y = after_stat(density)),
                   bins = bins, color = "black", fill = "white") +
    stat_function(fun = f, colour = "red") +
    labs(title = paste0("number of bins = ", bins))
}

p1 <- plot_hist(3)
p2 <- plot_hist(10)
p3 <- plot_hist(100)

grid.arrange(p1, p2, p3, nrow = 1)
```



We observe that:

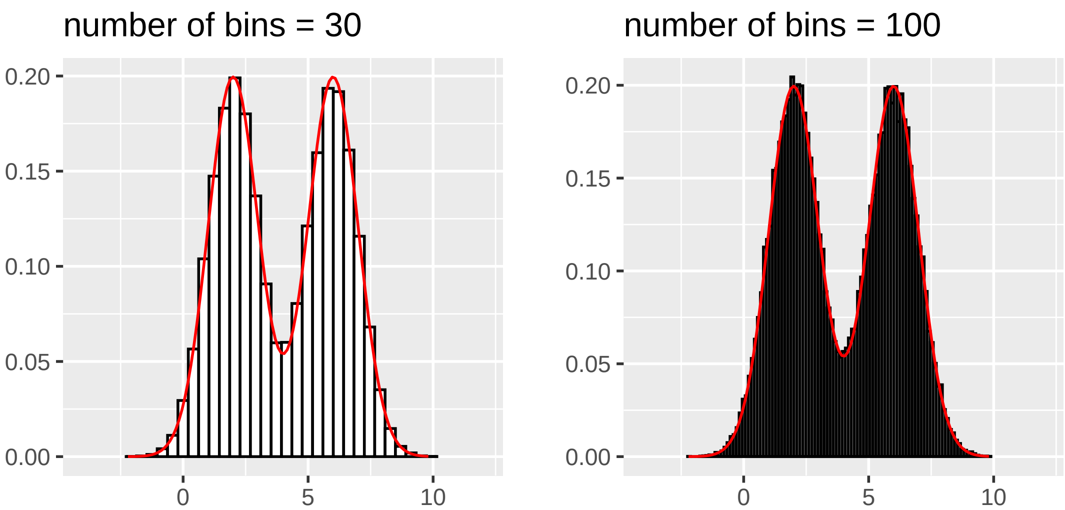
- if h is too small (hence too many bins), the histogram shows many unnecessary fluctuations (high variance);
- if h is too large (hence too few bins), the estimate is too coarse (high bias) and may hide the structure of the density (here, bimodality).

One can also specify the bin width h directly via the `binwidth` argument rather than the number of bins (`bins`).

Dependence on n

The optimal choice of h depends on the sample size. To illustrate this, we increase the sample size: we go from $n = 500$ to $n = 50\,000$.

```
n <- 50000
Z <- sim(n)
df <- data.frame(x = Z)

p_base <- ggplot(df, aes(x)) +
  labs(x = "", y = "") +
  coord_cartesian(xlim = xlim)

p30 <- p_base +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30, color = "black", fill = "white") +
  stat_function(fun = f, colour = "red") +
  labs(title = "number of bins = 30")

p100 <- p_base +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 100, color = "black", fill = "white") +
  stat_function(fun = f, colour = "red") +
  labs(title = "number of bins = 100")

grid.arrange(p30, p100, nrow = 1)
```



With more data, we can afford a finer histogram: the optimal number of bins generally increases with n (in other words, the optimal width h decreases with n).

The histogram estimator is presented here for illustrative purposes; more precise elements will be given for the estimator we are really interested in: the kernel estimator with bandwidth h.

Python illustrations are provided in a notebook on Teams.

4.3.2 Kernel estimators

A drawback of the histogram estimator is that the estimated density $\widehat{f}_n$ is a discontinuous function, whereas the density f we aim to estimate is in general more regular (at least continuous). Kernel estimators are designed precisely to *smooth* the estimation.

The starting point is the following observation : *if f is continuous at x*, then the distribution function F is differentiable at x and

$$f(x) = F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h} = \lim_{h\to 0} \frac{F(x+h) - F(x-h)}{2h}.$$

The idea is therefore to use, for small $h > 0$, the approximation

$$f(x) \approx \frac{F(x+h) - F(x-h)}{2h}.$$

To estimate f, we replace F by the empirical distribution function

$$\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \le x},$$

and we choose a window width $h > 0$. We then define

$$\widetilde{f}_{n,h}(x) = \frac{\widehat{F}_n(x+h) - \widehat{F}_n(x-h)}{2h} \overset{a.s.}{=} \frac{1}{n}\sum_{i=1}^{n} \frac{1}{2h} \mathbb{1}_{|X_i - x| \le h}.$$

If we set

$$K_0(u) = \frac{1}{2}\mathbb{1}_{[-1,1]}(u),$$

we can rewrite (a.s.) this estimator as

$$\widetilde{f}_{n,h}(x) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{h} K_0\left(\frac{x - X_i}{h}\right).$$

This estimator therefore corresponds to a "kernel" estimator with the uniform kernel K_0. However, since K_0 is not continuous, the function $x \mapsto \widetilde{f}_{n,h}(x)$ remains irregular. We therefore replace K_0 with a more regular function.

Définition 4.5

A function $K : \mathcal{R} \to \mathcal{R}$ is called a *kernel* if it is integrable and if

$$\int_{\mathcal{R}} K(u)\, du = 1.$$

Exemple 4.6: Examples of kernels

— Triangular kernel :

$$K(u) = (1 - |u|)\, \mathbb{1}_{[-1,1]}(u).$$

— Epanechnikov kernel :

$$K(u) = \frac{3}{4}(1 - u^2)\, \mathbb{1}_{[-1,1]}(u).$$

— Biweight kernel :

$$K(u) = \frac{15}{16}(1 - u^2)^2\, \mathbb{1}_{[-1,1]}(u).$$

— Gaussian kernel :

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right).$$

We then define the kernel estimator (or Parzen–Rosenblatt estimator) associated with a kernel K and a bandwidth $h > 0$.

Définition 4.7

Given a kernel K and $h > 0$, we set, for all $x \in \mathbb{R}$,

$$\widehat{f}_{n,h}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K\left(\frac{x - X_i}{h}\right).$$

The parameter h is called the *bandwidth*.

Remarque 4.8

1. Intuition : the estimator $\widehat{f}_{n,h}(x)$ is a sum of "bumps" centered at X_i (with shape K and scale h). In particular, if K is symmetric, nonnegative, and decreasing on $\mathcal{R}_+$ (as for the Gaussian kernel), then for fixed x, the closer X_i is to x, the larger
$$K\left(\frac{x - X_i}{h}\right)$$
is. Thus $\widehat{f}_{n,h}(x)$ increases when many observations lie near x.
2. If K is continuous, then $\widehat{f}_{n,h}$ is continuous (and more generally, the regularity of K is inherited by $\widehat{f}_{n,h}$).
3. The parameter h controls smoothing : a small h reduces the bias but increases the variance, whereas a large h yields more smoothing (lower variance) at the price of a larger bias. Choosing h is therefore crucial (bias–variance trade-off).
4. In practice, the choice of the kernel is generally less critical than that of the bandwidth h.
5. If $K \geq 0$, then $\widehat{f}_{n,h} \geq 0$ and, since
$$\int_{\mathcal{R}} \widehat{f}_{n,h}(x)\, dx = 1,$$
$\widehat{f}_{n,h}$ is then a density.

Remarque 4.9

When the kernel K is not nonnegative, the function $\widehat{f}_{n,h}$ is not necessarily nonnegative, hence not necessarily a density. If one nevertheless wants to obtain a density from such an estimator, one can take its positive part and then renormalize :

$$g \longmapsto \frac{\max\{g, 0\}}{\int_{\mathcal{R}} \max\{g(x), 0\}\, dx}.$$

Signed kernels can nevertheless be useful, for instance to construct kernels of order 2 (or higher) and reduce the bias (see below).

Graphical illustration and R commands: kernel density estimator

We consider the same bimodal distribution example as before. Kernel density estimation can be carried out using the function `density()` (from the base package `stats`), which estimates a one-dimensional density. For multivariate densities, one may use, for instance, the package `ks` (function `kde()`), which typically covers low to moderate dimensions (from 1 to 6 variables).

In `density()`, the default kernel is the Gaussian kernel. One can change the kernel via the `kernel` argument. In this document, we mainly use the **ggplot2** wrapper `geom_density()`, which also relies on `stats::density()`.

The smoothing parameter (the bandwidth h) is called `bw` (*bandwidth*). As for the histogram, the choice of `bw` is crucial.

Simulated data

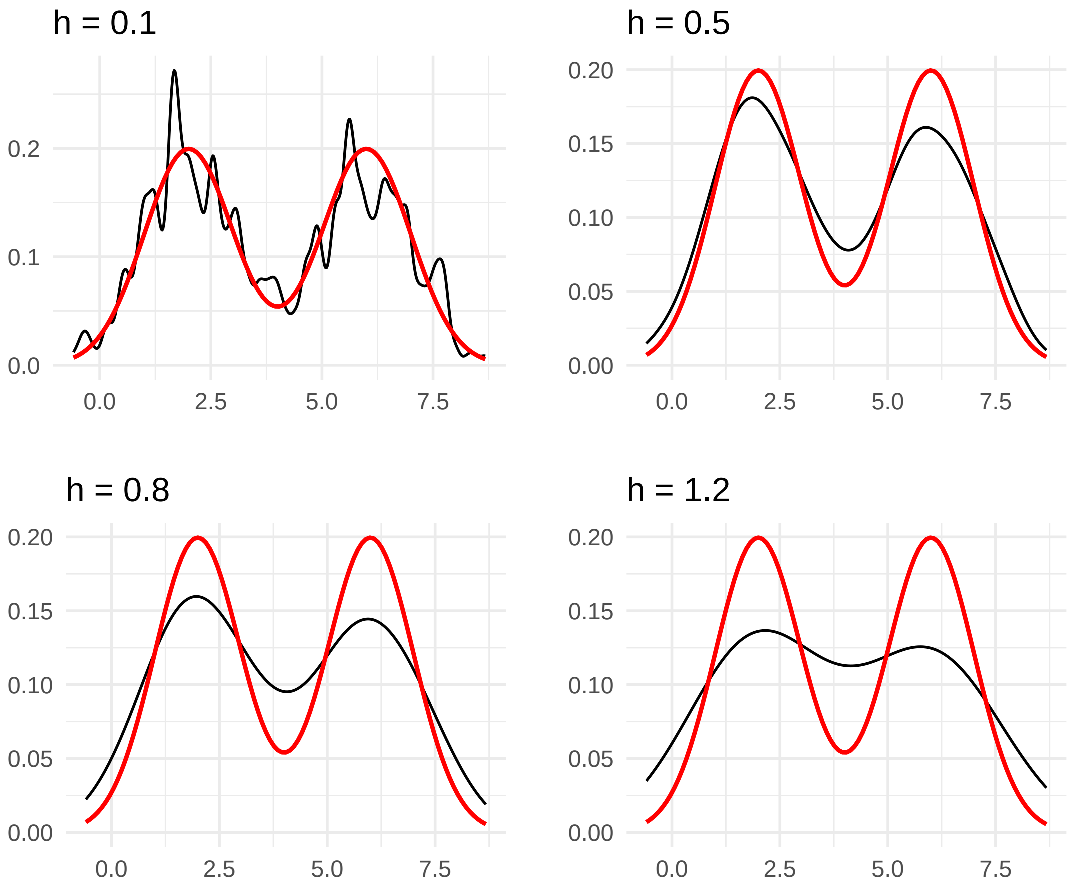
We simulate a mixture of two Gaussian distributions $\mathcal{N}(2,1)$ and $\mathcal{N}(6,1)$ with weights 1/2–1/2.
The true density is therefore

$$f(x) = \tfrac{1}{2}\varphi(x;2,1) + \tfrac{1}{2}\varphi(x;6,1), \qquad \varphi(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Influence of the bandwidth choice

We illustrate the influence of the choice of `bw`. We recover the same phenomenon as for the histogram: - `bw` too small $\Rightarrow$ very irregular estimate (high variance); - `bw` too large $\Rightarrow$ over-smoothing (high bias), which may hide bimodality.

```
p0 <- ggplot(df, aes(x)) + labs(x = "", y = "") + theme_minimal()

p1 <- p0 + geom_density(bw = 0.1) + stat_function(fun = f, col = "red", size = 0.8) + ggtitle("h = 0.1"
p2 <- p0 + geom_density(bw = 0.5) + stat_function(fun = f, col = "red", size = 0.8) + ggtitle("h = 0.5"
p3 <- p0 + geom_density(bw = 0.8) + stat_function(fun = f, col = "red", size = 0.8) + ggtitle("h = 0.8"
p4 <- p0 + geom_density(bw = 1.2) + stat_function(fun = f, col = "red", size = 0.8) + ggtitle("h = 1.2"

grid.arrange(p1, p2, p3, p4, nrow = 2, ncol = 2)
```



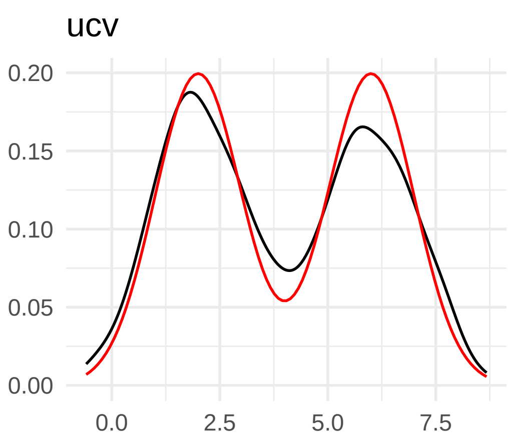
Data-driven bandwidths

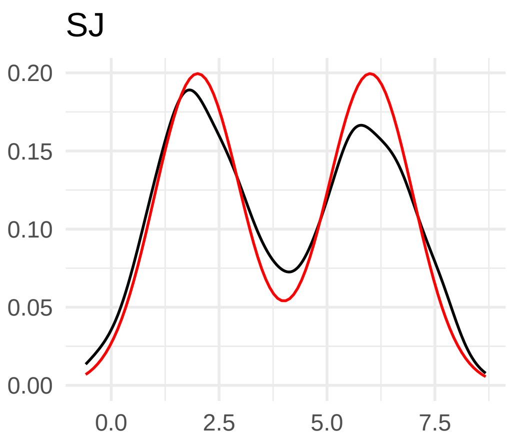
We now illustrate two bandwidths computed from the data:

- **SJ**: Sheather–Jones method;
- **ucv**: unbiased cross-validation, discussed later in the course.

For other methods, see the documentation of `bw.*` and `density()` (see also Silverman's rule in Exercise 2 of the 2019 exam).

```
p5 <- p0 + geom_density(bw = "ucv") + stat_function(fun = f, col = "red") + ggtitle("ucv")
bwSJ <- bw.SJ(Z)
p6 <- p0 + geom_density(bw = bwSJ) + stat_function(fun = f, col = "red") + ggtitle("SJ")

grid.arrange(p5, p6, ncol = 2)
```

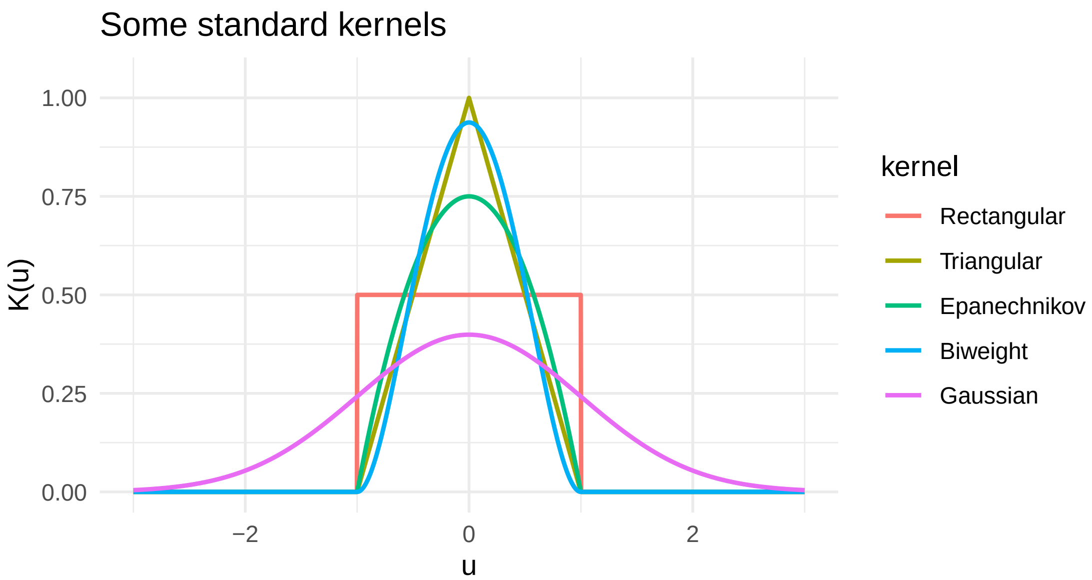




Some standard kernels

We plot below a few classical kernels (on the standardized scale u).
Compactly supported kernels (rectangular, triangular, Epanechnikov, biweight) vanish outside $[-1, 1]$, whereas the Gaussian kernel has unbounded support.

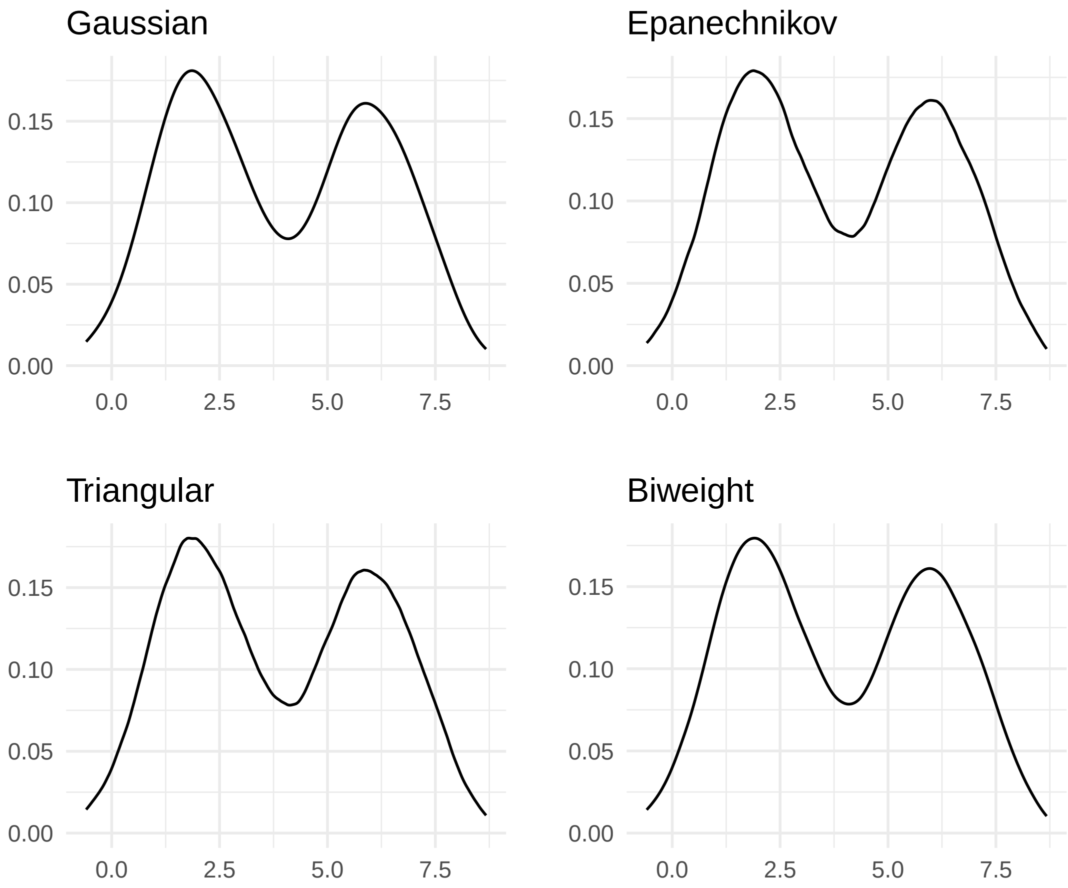
```
u <- seq(-3, 3, length.out = 2001)

K_rect <- 0.5 * (abs(u) <= 1)
K_tri  <- (1 - abs(u)) * (abs(u) <= 1)
K_epa  <- 0.75 * (1 - u^2) * (abs(u) <= 1)
K_biw  <- (15/16) * (1 - u^2)^2 * (abs(u) <= 1)
K_gau  <- dnorm(u)

kern_df <- data.frame(
  u = rep(u, 5),
  K = c(K_rect, K_tri, K_epa, K_biw, K_gau),
  kernel = factor(
    rep(c("Rectangular", "Triangular", "Epanechnikov", "Biweight", "Gaussian"), each = length(u)),
    levels = c("Rectangular", "Triangular", "Epanechnikov", "Biweight", "Gaussian")
  )
)

ggplot(kern_df, aes(x = u, y = K, colour = kernel)) +
  geom_line(size = 0.8) +
  labs(x = "u", y = "K(u)", title = "Some standard kernels") +
  theme_minimal() +
  coord_cartesian(xlim = c(-3, 3), ylim = c(0, max(kern_df$K) * 1.05))
```



Effect of the kernel (fixed bandwidth)

In general, the choice of the kernel has a smaller effect than the choice of the bandwidth.
To illustrate this, we fix the same bandwidth `bw = 0.5` and compare several kernels available in `stats::density()`.

```
bw0 <- 0.5

q0 <- ggplot(df, aes(x)) + labs(x = "", y = "") + theme_minimal()

q1 <- q0 + geom_density(bw = bw0, kernel = "gaussian")     + ggtitle("Gaussian")
q2 <- q0 + geom_density(bw = bw0, kernel = "epanechnikov") + ggtitle("Epanechnikov")
q3 <- q0 + geom_density(bw = bw0, kernel = "triangular")   + ggtitle("Triangular")
q4 <- q0 + geom_density(bw = bw0, kernel = "biweight")     + ggtitle("Biweight")

grid.arrange(q1, q2, q3, q4, nrow = 2, ncol = 2)
```



Remark: the two-dimensional version

`ggplot2` provides a two-dimensional version via `geom_density_2d()` (and also `geom_density_2d_filled()`), which allows one to display contour lines of the estimated density.

4.4 Pointwise quadratic risk of kernel estimators over the class of Hölder spaces

In this section, we study the pointwise quadratic risk of $\hat{f}_n$, i.e. given $x_0 \in \mathbb{R}$

$$R(\hat{f}_n, f) = \mathbb{E}\Big[|\hat{f}_n(x_0) - f(x_0)|^2\Big]$$

Recall the "bias squared + variance" decomposition of the quadratic risk :

$$\mathbb{E}\Big[|\hat{f}_n(x_0) - f(x_0)|^2\Big] = \Big(\mathbb{E}[\hat{f}_n(x_0)] - f(x_0)\Big)^2 + \mathbf{Var}(\hat{f}_n(x_0))$$

Définition 4.10

Soit $\ell \in \mathbb{N}^*$. We say that the kernel K is of order ℓ if $\forall j \in \{1, \dots, \ell\}$, the function $u \to u^j K(u)$ is integrable and $\int u^j K(u) du = 0$.

Proposition 4.11: Bias

If $f \in \Sigma(\beta, L)$ with $\beta > 0$ and $L > 0$ and if K is a kernel of order $\ell = \lfloor \beta \rfloor$ such that $\int |u|^\beta |K(u)| du < \infty$ then for every $x_0 \in \mathbb{R}$, and for every $h > 0$, the bias can be bounded as follows :

$$|\mathbb{E}[\hat{f}_n(x_0)] - f(x_0)| \leq C_1 h^\beta$$

where $C_1 = \frac{L}{\ell!} \int |u|^\beta |K(u)| du$.

Démonstration. We have

$$\begin{aligned}
\mathbb{E}[\hat{f}_n(x_0)] &= \mathbb{E}[\frac{1}{n}\sum_{i=1}^n \frac{1}{h} K(\frac{x_0 - X_i}{h})] = \mathbb{E}[\frac{1}{h} K(\frac{x_0 - X_i}{h})] \\
&= \frac{1}{h}\int K(\frac{x_0 - X_i}{h}) f(u) du = \int K(v) f(x_0 - hv) dv.
\end{aligned}$$

Aside : technical detail : $\hat{f}_n(x_0)$ is indeed integrable since

$$\begin{aligned}
\mathbb{E}[|\hat{f}_n(x_0)|] &\leq \mathbb{E}[\frac{1}{n}\sum_{i=1}^n \frac{1}{h}|K|(\frac{X_i - x_0}{h})] = \mathbb{E}[\frac{1}{h}|K|(\frac{X_1 - x_0}{h})] \\
&= \frac{1}{h}\int |K|(\frac{u - x_0}{h}) f(u) du = \int |K|(v) f(x_0 + hv) dv \leq M(\beta, L) \int |K| < \infty.
\end{aligned}$$

End of aside.
Moreover,

$$f(x_0) = f(x_0) \times 1 = f(x_0) \int K(v) dv.$$

Hence

$$|\mathbb{E}[\hat{f}_n(x_0)] - f(x_0)| = |\int K(v)[f(x_0 - hv) - f(x_0)] dv|$$

As $f \in \Sigma(\beta, L)$, f admits $\lfloor \beta \rfloor := \ell$ derivatives and, by a Taylor–Lagrange expansion, for every $x \in \mathbb{R}$,

$$f(x) = \sum_{k=0}^{\ell-1} \frac{(x - x_0)^k}{k!} f^{(k)}(x_0) + \frac{(x - x_0)^\ell}{\ell!} f^{(\ell)}(x_0 + \xi(x - x_0))$$

with $\xi \in]0, 1[$. In other words, with $x = x_0 - hv$, we have

$$f(x_0 - hv) - f(x_0) = \sum_{k=1}^{\ell-1} \frac{(-hv)^k}{k!} f^{(k)}(x_0) + f^{(\ell)}(x_0 - hv\xi) \frac{(hv)^\ell}{\ell!}$$

for some $\xi \in]0,1[$. Therefore

$$\int K(v)[f(x_0 - hv) - f(x_0)]dv = \int K(v)[\sum_{k=1}^{\ell-1} \frac{(-hv)^k}{k!} f^{(k)}(x_0) + f^{(\ell)}(x_0 - hv\xi)\frac{(-hv)^\ell}{\ell!}]dv$$
$$= \frac{(-h)^\ell}{\ell!} \int K(v)v^\ell f^{(\ell)}(x_0 - hv\xi)dv$$

As K is of order ℓ, we also have $\int K(v)v^\ell f^{(\ell)}(x_0)dv = 0$. Hence

$$\int K(v)[f(x_0 - hv) - f(x_0)]dv = \frac{(-h)^\ell}{\ell!} \int K(v)v^\ell [f^{(\ell)}(x_0 - hv\xi) - f^{(\ell)}(x_0)]dv$$

Now, since $f \in \Sigma(\beta, L)$, we have $|f^{(\ell)}(x_0 - hv\xi) - f^{(\ell)}(x_0)| \leq L|hv|^{\beta-\ell}$. Finally,

$$\Big|\int K(v)[f(x_0 - hv) - f(x_0)]dv\Big| \leq \frac{|h|^\ell}{\ell!} \int |K(v)||v|^\ell L|hv|^{\beta-\ell}dv$$

which means that

$$\Big|\mathbb{E}[\hat{f}_n(x_0)] - f(x_0)\Big| \leq \frac{L|h|^\beta}{\ell!} \int |K(v)||v|^\beta dv.$$

□

Thus, the smoother the function f, the smaller the bias (provided, of course, that the order of the kernel is sufficiently large).

Proposition 4.12: Variance

If f is bounded and if K is square-integrable, then

$$\mathbf{Var}(\hat{f}_n(x_0)) \leq \frac{\|f\|_\infty \|K\|_2^2}{nh}$$

In particular, if $f \in \Sigma(\beta, L)$, then

$$\mathbf{Var}(\hat{f}_n(x_0)) \leq \frac{C_2}{nh}$$

where $C_2 = M(\beta, L)\|K\|_2^2$.

Démonstration.

$$\begin{aligned}
\mathbf{Var}(\hat{f}_n(x_0)) &= \mathbf{Var}\Big(\frac{1}{nh}\sum_{i=1}^n K\Big(\frac{X_i - x_0}{h}\Big)\Big) \\
&\overset{indep}{=} \sum_{i=1}^n \mathbf{Var}\Big(\frac{1}{nh} K\Big(\frac{X_i - x_0}{h}\Big)\Big) \\
&= \sum_{i=1}^n \frac{1}{n^2h^2}\mathbf{Var}\Big(K\Big(\frac{X_i - x_0}{h}\Big)\Big) \\
&\overset{i.d.}{=} \frac{1}{nh^2}\mathbf{Var}\Big(K\Big(\frac{X_1 - x_0}{h}\Big)\Big) \\
&\leq \frac{1}{nh^2}\mathbb{E}\Big(K^2\Big(\frac{X_1 - x_0}{h}\Big)\Big) \\
&= \frac{1}{nh^2}\int K^2\Big(\frac{u - x_0}{h}\Big) f(u)du \\
&= \frac{1}{nh}\int K^2(v) f(x_0 + vh)dv
\end{aligned}$$

And finally, we use Proposition 4.4 : there exists a positive constant $M(\beta, L)$ such that $\|f\|_\infty \leq M(\beta, L)$. This implies that

$$\mathbf{Var}(\hat{f}_n(x_0)) \leq \frac{1}{nh} M(\beta, L) \int K^2(v)dv.$$

□

For the variance to tend to zero, it is necessary that nh tends to infinity.
In particular, for fixed n, the variance is a decreasing function of h, unlike the bias, which is an increasing function of h. Therefore there is an optimal value of h that must strike a balance between the squared bias and the variance.

We can now give a bound on the pointwise quadratic risk.

Théorème 4.13

Let $\beta > 0$ and $L > 0$, and let K be a square-integrable kernel of order $\lfloor \beta \rfloor$ such that $\int |u|^\beta |K(u)|du < \infty$. Then, by choosing a bandwidth of the form $h = cn^{-\frac{1}{2\beta+1}}$ with a constant $c > 0$, we obtain

$$\forall x_0 \in \mathbb{R}, \quad R(\hat{f}_n(x_0), \Sigma_d(\beta, L)) := \sup_{f \in \Sigma_d(\beta, L)} \mathbb{E}[|\hat{f}_n(x_0) - f(x_0)|^2] \leq Cn^{-\frac{2\beta}{2\beta+1}}$$

where C is a constant depending on L, β, c, and K.

Démonstration. We have

$$R(\hat{f}_n(x_0), f(x_0)) = \text{ Squared bias + Variance}$$

The bias term was treated in Proposition 4.11, and the variance term was treated in Proposition 4.12. We find

$$R(\hat{f}_n(x_0), f(x_0)) \leq \Big(\frac{h^\beta L}{\ell!}\int |u|^\beta |K(u)|du\Big)^2 + \frac{M(\beta, L)\|K\|_2^2}{nh}.$$

We look for the bandwidth h that minimizes this upper bound. This can be formulated as an optimization problem with respect to h and admits an easy solution ; below we give a more intuitive interpretation.
Here we will not worry about the exact constants, so we use the notation $c_1 = \Big(\frac{L}{\ell!}\int |u|^\beta |K(u)|du\Big)^2$

and $c_2 = M(\beta, L)\|K\|_2^2$. We then have to minimize with respect to h the quantity

$$c_1 h^{2\beta} + \frac{c_2}{nh}$$

We have one quantity that increases and one quantity that decreases with h. Since we are not concerned with the constants, we only seek a bandwidth h corresponding to the minimal order of the risk. When h is too large, the bias is too large, and when h is too small, it is the variance that is too large. We therefore seek the bandwidth h that balances the squared bias and the variance :

$$h^{2\beta} \approx \frac{1}{nh}$$

where the symbol $\approx$ here means "of the same order as". This gives

$$h \approx n^{-\frac{1}{2\beta+1}}$$

In other words, for a bandwidth h of order $n^{-\frac{1}{2\beta+1}}$, the squared bias and the variance are of the same order. More precisely, if we choose the bandwidth $h_* = cn^{-\frac{1}{2\beta+1}}$, with c a positive constant, then

$$\text{Squared bias} \approx h_*^{2\beta} \approx \text{ variance } \approx \frac{1}{nh_*}$$

Moreover, we then have $h_*^{2\beta} \approx n^{\frac{-2\beta}{2\beta+1}}$. In other words, there exists some constant C (depending on β, L, K, and c) such that, for this bandwidth h_*, we have

$$R\Big(\hat{f}_n(x_0), \Sigma_d(\beta, L)\Big) \leq Cn^{\frac{-2\beta}{2\beta+1}}.$$

This bandwidth is therefore optimal up to a constant (if we change c, this modifies C but does not change the rate, which is $n^{\frac{-2\beta}{2\beta+1}}$). $\square$

Remarque 4.14

1. The estimator depends on β through the bandwidth h. Therefore, **without prior knowledge of the smoothness of the function f, one cannot use this estimator**. One then tries to find a choice of bandwidth depending only on the data and performing as well as the estimator using this optimal bandwidth.
2. The theorem is an asymptotic result, that is, valid when n is large. But when n is small, the choice of K and h becomes an issue.
3. The larger β is, the faster the rate. In the limit $\beta \to \infty$, one obtains a parametric rate.
4. **One can generalize the concept of kernel estimators to a multivariate density. But beware : in high dimension, the problem of the "curse of dimensionality" often arises.** In fact, the kernel estimator in dimension d yields a rate of $n^{-\frac{2\beta}{2\beta+d}}$. Therefore this rate deteriorates very quickly with the dimension. (More details on this topic in Chapter 5.)

4.5 Choosing the bandwidth h by cross-validation

The choice of the bandwidth in the previous section can be criticized : it depends on the smoothness, which is generally unknown. One may therefore try to replace this unknown bandwidth by an estimator $\hat{h}$. To emphasize the dependence on the bandwidth h, we write $\hat{f}_{n,h}$ for the estimator associated with a choice of bandwidth h. The final estimator will be $\hat{f}_{n,\hat{h}}$, once the choice of $\hat{h}$ has been made.

We seek to minimize with respect to h the quadratic risk for the L^2 distance :

$$R(\hat{f}_{n,h}, f) = \mathbb{E}[\|\hat{f}_{n,h} - f\|_2^2]$$

But since the function f is unknown, this risk cannot be computed from the data. We therefore seek to estimate this risk using only the data.
Minimizing with respect to h the quantity $R(\hat{f}_{n,h}, f)$ is equivalent to minimizing with respect to h the quantity $R(\hat{f}_{n,h}, f) - \|f\|_2^2$. In fact, we will replace the minimization of the unknown quantity $R(\hat{f}_{n,h}, f) - \|f\|_2^2$ by the minimization of an estimator $\hat{R}(h)$ of this quantity. More precisely, we will look for an unbiased estimator of $R(\hat{f}_{n,h}, f) - \|f\|_2^2$ [a].

a. same idea as Mallows' C_p, cf. the MLG course

Throughout this section, we assume that the density f and the kernel K are square-integrable. Let us check that this implies that the integrated quadratic risk is indeed finite : we have

$$\mathbb{E}[\hat{f}_n^2(x)] \leq \frac{1}{h} \int_{\mathcal{R}} K^2(v) f(x - hv) dv.$$

Thus [1]

$$\int_{\mathcal{R}} \mathbb{E}[\hat{f}_n^2(x)]dx \leq \frac{1}{h} \int_{\mathcal{R}^2} K^2(v) f(x-hv) dv dx = \frac{1}{h} \int_{\mathcal{R}} K^2(v) (\int_{\mathcal{R}} f(x-hv) dx) dv = \frac{1}{h} \int_{\mathcal{R}} K^2(v) dv < +\infty$$

Théorème 4.15: Unbiased cross-validation (UCV)/ Least-squares cross validation (LSCV)

If we define

$$\hat{R}(h) = \|\hat{f}_{n,h}\|_2^2 - \frac{2}{n(n-1)} \sum_{i,j=1, j\neq i}^{n} \frac{1}{h} K(\frac{X_j - X_i}{h})$$

then $\hat{R}(h)$ is an unbiased estimator of $R(\hat{f}_{n,h}, f) - \|f\|_2^2$.

Démonstration. Two technical remarks :

— $\int E[|\hat{f}_n(x)|] f(x) dx < \infty$ because $\int \mathbb{E}[\hat{f}_n^2(x)]dx < \infty$ and $\int f^2(x)dx < \infty$.

— $\sum_{i,j=1, j\neq i}^{n} \frac{1}{h} K(\frac{X_j - X_i}{h})$ is indeed integrable : indeed, if we denote by K_h the function defined by $K_h(x) = \frac{1}{h} K(\frac{x}{h})$, then

$$\begin{aligned} E\Big[\frac{1}{h}|K|(\frac{X_1 - X_2}{h})\Big] &= \int\int |K_h|(u-v) f(u) f(v) du dv \\ &\leq \sqrt{\int \Big[\int |K_h|(u-v) f(u) du\Big]^2 dv} \sqrt{\int f^2(v) dv} \end{aligned}$$

Moreover $\left(\int K_h(u-v) f(u) du\right)^2 \leq \int K_h^2(u-v) f(u) du$ by .... And therefore

$$\int (\int K_h(u-v) f(u) du)^2 dv \leq \int\int K_h^2(u-v) f(u) du dv = \int K_h^2(x) dx < \infty$$

We want to show that

$$\mathbb{E}\hat{R}(h) = R(\hat{f}_{n,h}, f) - \|f\|_2^2$$

1. Of course this can also be justified using a very well-known result on convolution

Now

$$\begin{aligned} R(\hat{f}_{n,h}, f) - \|f\|_2^2 &= \mathbb{E}\Big(\|\hat{f}_{n,h}\|_2^2 - 2\int \hat{f}_{n,h}(x)f(x)dx\Big) \\ &= \mathbb{E}\|\hat{f}_{n,h}\|_2^2 - 2\int \mathbb{E}\hat{f}_{n,h}(x)f(x)dx \end{aligned}$$

It is therefore enough to show that

$$\int \mathbb{E}\hat{f}_{n,h}(x)f(x)dx = \frac{1}{n(n-1)}\mathbb{E}\Big[\sum_{i,j=1,j\neq i}^{n} \frac{1}{h}K(\frac{X_j - X_i}{h})\Big]$$

The left-hand side gives, according to the calculation made in Proposition 4.11,

$$\begin{aligned} \int [\mathbb{E}\hat{f}_{n,h}(x)]f(x)dx &= \int \Big[\int \frac{1}{h}K(\frac{u-x}{h})f(u)du\Big]f(x)dx \\ &\overset{Fubini}{=} \int\int \frac{1}{h}K(\frac{u-v}{h})f(u)f(v)dudv \end{aligned}$$

The right-hand side gives

$$\begin{aligned} \frac{1}{n(n-1)}\mathbb{E}\Big[\sum_{i,j=1,j\neq i}^{n} \frac{1}{h}K(\frac{X_j - X_i}{h})\Big] &= \mathbb{E}\Big[\frac{1}{h}K(\frac{X_1 - X_2}{h})\Big] \\ &= \int\int \frac{1}{h}K(\frac{u-v}{h})f(u)f(v)dudv \end{aligned}$$

□

We then define

$$\hat{h} = \arg\min_{h\in H} \hat{R}(h)$$

if this minimum is attained. H is an interval $H = [h_{\min}, h_{\max}]$, where $h_{\min}$ and $h_{\max}$ depend on n.

Remarque 4.16

This criterion is unbiased but may have high variance, in particular for small values of h, and it may tend to choose a bandwidth that is too small (undersmoothing). There are many other methods for choosing the bandwidth from the data.

Remarque 4.17: Likelihood cross-validation LCV/MLCV

In the `model_selection` submodule of `sklearn`, the function `GridSearchCV` performs cross-validation (cf. Chapter 5 or the learning course) using the log-likelihood as the default scoring function.

— we choose a number K of folds.

— We divide the sample into K blocks of equal length (or approximately equal length) : $B_1, B_2, \ldots B_K$.

— We denote by ℓ_k the cardinality of B_k (thus $\ell_k \approx n/K$).

— for each $k \in \{1, \ldots, K\}$, we compute the estimator $\hat{f}_h^k$ based only on the data X_i with $i \in \{1, \ldots, n\} \cap B_k^c$, i.e. we use all the data except those corresponding to block B_k to build $\hat{f}_h^k$.

Then we compute

$$CV(h) := \frac{1}{K}\sum_{k=1}^{K}\frac{1}{\ell_k}\sum_{i\in B_k}\log(\hat{f}_h^k(X_i))$$

This is the quantity computed by default by `GridSearchCV` when this function is used with the `KernelDensity` estimator from the `neighbors` submodule of `sklearn`. One then looks for the bandwidth that **maximizes** this quantity over a grid of values.

Remarque 4.18: Choice of criterion

One tends to choose Likelihood-CV when the density is really used as a predictive probabilistic model and in particular when one wants to avoid estimators that yield an almost zero density where data may actually fall. The KL divergence penalizes much more severely than the quadratic error in situations where $\hat{f}_h(x)$ becomes very small whereas $f(x)$ is not, because of the logarithm. Likelihood-CV is, for instance, particularly natural in anomaly detection using a density score.
One tends to choose a criterion based on quadratic loss (e.g. Least-Squares CV) if one wants a good global reconstruction of the density as a function, or a visually stable and smooth estimator, but also for example in the presence of heavy tails or outliers.

Remarque 4.19

One can use nonparametric density estimates for classification in a direct way by applying Bayes' theorem [a]. Suppose we have a binary classification problem. Let us reuse the same notation as in Exercise 5, TD1, from the statistical learning course. Suppose that we fit nonparametric density estimates $\hat{f}_0$ and $\hat{f}_1$, and that we also have estimates of the prior class probabilities $\hat{p}_0$ and $\hat{p}_1$. Then to estimate $\eta(x) = P(Y = 1 \mid X = x)$, one can use

$$\hat{\eta}(x) = \frac{\hat{p}_1\hat{f}_1(x)}{\hat{p}_0\hat{f}_0(x) + \hat{p}_1\hat{f}_1(x)}$$

In the same way as for Naive Bayes in the learning course, we can then propose the following classifier :

$$\hat{g}(x) = \begin{cases} 1 & \text{if} \quad \hat{p}_1\hat{f}_1(x) > \hat{p}_0\hat{f}_0(x) \\ 0 & \text{otherwise} \end{cases}$$

This can be extended to a K-class problem, for any $K \geq 3$, in a straightforward way :

$$\hat{g}(x) = \arg\max_{j\in\{1,\ldots,K\}} \hat{p}_j\hat{f}_j(x)$$

An example is provided in the notebook on Teams.

a. especially interesting if you care about estimating η, and not only the final classification

Chapitre 5

Nonparametric Regression

5.1 Introduction

In this chapter, we aim to estimate a regression function.
The random variables $Y_1, \ldots, Y_n$ are real-valued.

5.1.1 Random design

We seek to explain the values that a real-valued variable Y can take from the values that a (generally multidimensional) variable X can take.

Exemple 5.1 (medicine)**.** *The variable to be explained, Y, is the insulin level in the blood, which we explain (or predict) using X = (BMI, blood pressure, concentration of molecules).*

Exemple 5.2 (sociology)**.** *The variable to be explained, Y, is the level of education attained, which we explain using X = (age, parents' income, parents' occupation).*

We assume that the variable Y is integrable, $\mathbb{E}|Y| < \infty$, and we denote by r the regression function of Y on X :

$$r(x) = \mathbb{E}(Y|X = x)$$

The goal is to estimate the function r in order to explain and predict Y from X. To do so, we observe n i.i.d. pairs of variables $(X_1, Y_1), \ldots, (X_n, Y_n)$ having the same distribution as (X, Y).

- Y is called the variable to be explained, or the response variable, or the output variable (target, output variable).
- the vector X represents the explanatory variables, also called covariates, or input variables (features, input variables, explanatory variables).

If we set $\epsilon = Y - \mathbb{E}(Y \mid X)$, then we have

$$Y = r(X) + \epsilon \quad \text{with} \quad \mathbb{E}(\epsilon|X) = 0. \tag{5.1}$$

If we assume that $\mathbb{E}(Y^2) < \infty$, then r is the Bayes estimator for the quadratic loss.
Recall indeed that $r(X) = \mathbb{E}(Y \mid X)$ is such that

$$r(X) = \arg\min_{g \in \mathcal{G}} \mathbb{E}(Y - g(X))^2$$

(where $\mathcal{G}$ is the set of real-valued measurable functions such that $\mathbb{E}(g^2(X)) < \infty$). This regression function is therefore the best function, in the sense of quadratic loss, for predicting Y from X.
We set

$$\sigma^2 = \mathbb{E}(\epsilon^2) = \mathbb{E}(Y - E(Y \mid X))^2$$

Thus, σ^2 corresponds to the Bayes error R^* encountered in the learning course (corresponding here to the case of regression and quadratic loss ($\ell(x, y) = (y - g(x))^2$)).

Interpretation : σ^2 tends to be large when the features in the vector X do not predict Y well enough.

Thus, for the sample, we have

$$Y_i = r(X_i) + \epsilon_i, \quad i = 1, \ldots, n, \quad \mathbb{E}(\epsilon_i | X_i) = 0.$$

In particular, we therefore have $\mathbb{E}(\epsilon_i) = 0$.

Since the (X_i, Y_i) are i.i.d., the ϵ_i are as well.

Quite often, to simplify the calculations, one also assumes that the ϵ_i are independent of the X_i.

5.1.2 Deterministic design

The design may also be deterministic.

Example : a scientist wants to study crop yield as a function of the amount of a certain fertilizer. The scientist then fixes the experiment : they have a certain number n of fields [1] and carry out the experiment of applying different fertilizer amounts denoted $x_1, \ldots, x_n$. Thus it is the scientist who decides the amounts applied to the different fields. They then obtain observed yields $(y_1, \ldots, y_n)$.

They could apply the same amount of fertilizer to different fields (this is of course not what they would do ; this is just to explain the modeling) : in general, the yield in those fields would be fairly close, but it would not be exactly the same. The presence of these fluctuations is translated by saying that the observed yields $y_1, \ldots, y_n$ are the observations of random variables $Y_1, \ldots, Y_n$. For a fertilizer amount x, $r(x)$ then denotes the mean yield.

If we model this mathematically, this means that we observe n independent random variables $Y_1, \ldots, Y_n$ associated with the values $x_1, \ldots, x_n$ (fixed by the experimenter). And we have

$$Y_i = r(x_i) + \epsilon_i, \quad i = 1, \ldots, n$$

with

$$\mathbb{E}(\epsilon_i) = 0.$$

(which expresses the fact that $E(Y_i) = r(x_i)$)

Remark (notation) : In the deterministic design case, one often writes x_i instead of X_i.

Since the Y_i are independent [2], the ϵ_i are as well.

In what follows, we make the following assumption :

$$\mathbf{Var}(\epsilon_i) = \sigma^2 < \infty \quad \text{finite variance and independent of } i$$

5.2 Nonparametric Least Squares (OLS) Estimator

5.2.1 Linear model : reminders

The linear model consists in assuming that r can be written, if $x = (x_1, \ldots, x_p) \in \mathbb{R}^p$,

$$r(x) = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$$

1. which must be approximately identical according to obvious criteria such as exposure to sunlight, wind, initial soil composition, etc.

2. the Y_i are no longer identically distributed, even if the ϵ_i are

We denote $\boldsymbol{X} = \begin{pmatrix} 1 & X_{11} & \dots & X_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{np} \end{pmatrix}$, $\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}$ and $\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}$

In this case, estimating r amounts to estimating the vector $\boldsymbol{\beta}$. This is a parametric problem. One usually uses ordinary least squares :

$$\hat{\boldsymbol{\beta}} = \arg\min_{\beta \in \mathbb{R}^p} \|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = \arg\min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p X_{ij}\beta_j)^2.$$

If $\boldsymbol{X}$ has full column rank, then $\boldsymbol{X}^T\boldsymbol{X}$ is invertible and $\hat{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}$, and $\hat{Y} = \boldsymbol{X}\hat{\beta} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y} = A\boldsymbol{Y}$ where $A = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$. Finally, the estimator of the regression function is

$$\hat{r}(x) = (1, x^T)\hat{\beta}$$

for $x \in \mathbb{R}^p$.

Recall that the matrix A (often called the hat matrix) is the orthogonal projection matrix onto the subspace spanned by the column vectors of the design matrix $\boldsymbol{X}$.

In the case of a deterministic design, one shows that if the noise vector $(\epsilon_1, \dots, \epsilon_n)$ is made of independent random variables, centered and with variance σ^2, then one has [3]

$$\mathbb{E}(\hat{\beta}) = \beta \quad \text{and} \quad \Sigma_{\hat{\beta}} = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

where $\Sigma_{\hat{\beta}}$ is the covariance operator of $\hat{\beta}$.

Remark : the theoretical results obtained for a deterministic design are not without interest when one has a random design. A certain number of results are still valid, under some assumptions, by conditioning on the design.

For example : in the case of a random design, with i.i.d. observations $(X_i, Y_i)_{1 \leq i \leq n}$, if the regression function is indeed affine and if the distribution of the X_i is such that almost surely the columns of the matrix $\boldsymbol{X}$ are linearly independent, then we recover the fact that the bias of the OLS estimator $\hat{\beta}$ is zero :

$$\mathbb{E}(\hat{\beta} \mid X_1, \dots, X_n) = \beta$$

and therefore

$$E(\hat{\beta}) = \beta.$$

If moreover one assumes that $\mathbb{E}(\epsilon_i^2 \mid X_i)$ is constant equal to σ^2 (which is true if one further assumes that the ϵ_i are independent of the X_i), then one has ("covariance matrix of $\hat{\beta}$ given the design")

$$\mathbb{E}[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T \mid X_1, \dots, X_n] = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$$

5.2.2 Bias-variance tradeoff : reminders

Assume that the design is deterministic. We have

$$Y_i = r(x_i) + \epsilon_i, \quad i = 1, \dots, n$$

We assume that r is well approximated by a linear function of the variables. Let m be a subset of $\{1, \dots, p\}$. The set m represents a collection of covariates. If $m = \{1, \dots, p\}$, this is the OLS estimator on the full model. For simplicity, we assume that there is no intercept. The OLS prediction on the i-th training data point is then

$$\hat{r}(x_i) = x_i^T\hat{\beta}$$

Thus the OLS predictions on the training data $(x_1, \dots, x_n)$ are given by

$$\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\beta}$$

3. and one does not need the noise to be Gaussian for these results...

If one uses an OLS estimator computed only with the features whose indices belong to m, one obtains an estimator denoted by $\hat{r}^m$. The predictions of this estimator on the training data are

$$\hat{\boldsymbol{Y}}^m = \boldsymbol{X}^m \hat{\beta}^m$$

where $\boldsymbol{X}^m$ is the matrix $\boldsymbol{X}$ after removing the columns whose indices do not belong to m, and where $\hat{\beta}^m$ is the corresponding OLS estimator, i.e.

$$\hat{\beta}^m = (\boldsymbol{X}^{mT} \boldsymbol{X}^m)^{-1} \boldsymbol{X}^{mT} \boldsymbol{Y}$$

We are interested in the following risk :

$$R_m := \mathbb{E}\|\hat{Y}^m - r^*\|_2^2$$

where $r^* = (r(x_1), \ldots, r(x_n))^T$. We have

$$\boldsymbol{Y} = r^* + \xi$$

where $\xi = (\epsilon_1, \ldots, \epsilon_n)^T$

$$R_m = \mathbb{E}\|P_m(r^* + \xi) - r^*\|_2^2$$

where P_m is the orthogonal projection matrix onto the subspace spanned by the columns of the matrix $\boldsymbol{X}^m$. Hence

$$\begin{aligned} R_m &= \|(I - P_m)r^*\|^2 + \mathbb{E}\|P_m \xi\|_2^2 \\ &= \|(I - P_m)r^*\|^2 + \sigma^2 |m| \end{aligned}$$

where $|m|$ is the cardinality of the set m.

This risk therefore decomposes into two terms :

— a bias term $\|(I - P_m)r^*\|^2$ which reflects the quality of the approximation of r^* by the features whose indices are in m,

— a variance term $\sigma^2 |m|$ which increases with the number of features.

The best model strikes a compromise between this bias term and this variance term.